

# Master Progressive Pull-Request Blueprint — μOS++ Ille SMP Migration

**Status 2026-10-06.** This file-by-step map belongs to the earlier 30-step, script-driven plan; the executed per-commit file lists are the commits themselves (pr/<ID>). The migration from xpack-development to smp was then executed with the commit-only procedure of [xpack-dev-smp.md](#) Part II: no scripts/smp/, no chunk recipes, one subject per commit, only added or modified files. The result is recorded in its Part III: 343 commits over micro-os-plus-iii, -posix-arch, -cortexm, -aarch32 and -aarch64; micro-os-plus-iii-devices dissolved into the ports' drivers/ and soc/<chip>/ (no UOS\_DEVICES\_DIR); every commit built; the full xpm test run (native, QEMU Cortex-M, test-smp-cmake) green. Where this document disagrees with it (step or PR numbering, scripts, the devices repository, test counts), xpack-dev-smp.md applies.

*Files Modified Per Step, In-Depth Technical Rationale ("WHY"), and Lifting Recipes (Part 0 → Part D, Steps 1–31)*

**Source:** The verified multi-core smp branch across all component repositories, compared against the uniprocessor upstream baseline (origin/xpack-development).

**Scope:** Complete, exhaustive blueprint for the progressive upstream integration across all **7 Pull Requests** and all **31 Steps**, spanning:

- **Part A (Steps 1–13):** Uniprocessor correctness, ISO C/C++17 conformance, and toolchain hardening.
- **Part B (Steps 14–23 collapsed into Step 14):** SMP multi-core kernel, port contracts, and genuine host dual-core verification (native-smp).
- **Part C1 (Step 24):** Port package semantic version releases (posix-arch v1.1.0, cortexm v1.2.0).
- **Part 0:** History-preserving dissolution of micro-os-plus-iii-devices via git subtree.
- **Part C2 (Steps 25–26):** New silicon support (Cortex-M33, RP2350) and modular add-only CMake architecture.
- **Part C3 (Steps 27–28):** New test suites (fp-switch, smp-support), multi-arch harness integration (AArch32, AArch64), and add-only test platforms.
- **Part D (Steps 29–30):** Documentation synchronization suite and final merge reconciliation.
- **Step 31:** Dedicated Raspberry Pi Pico 2 (cortexm-pico2) hardware platform wiring.

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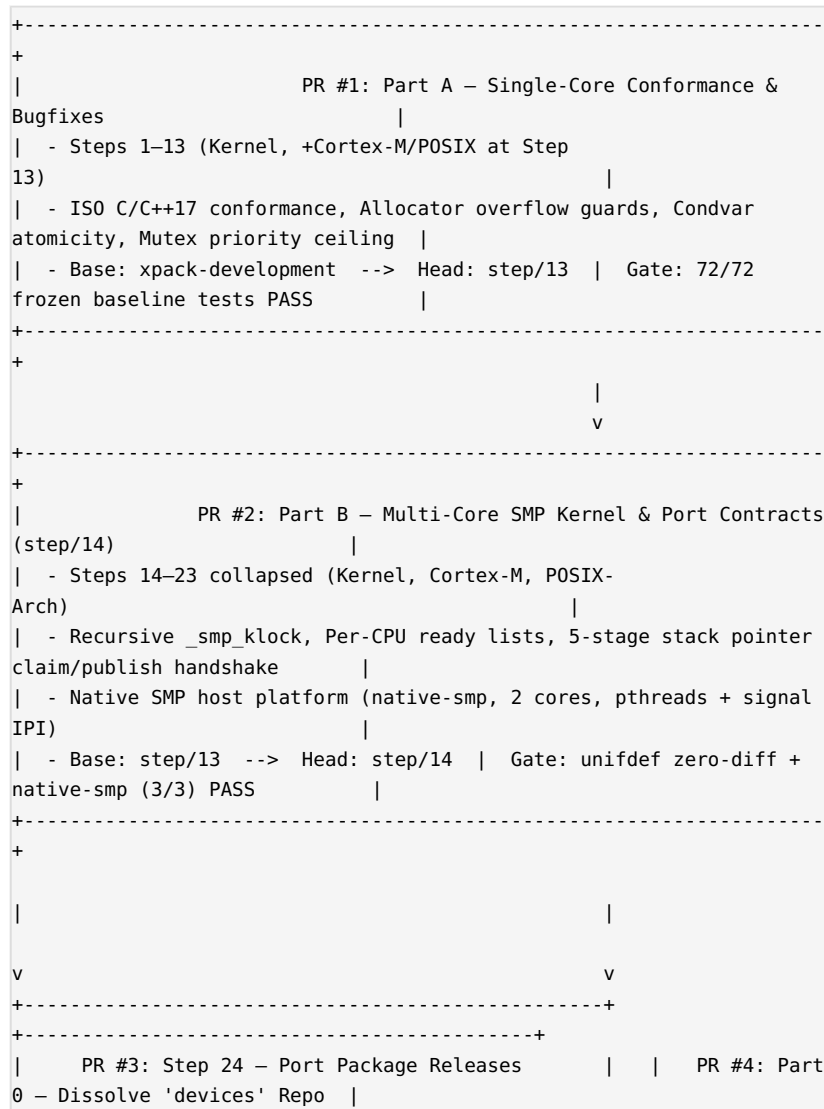
## Legend and Repository Key

- **Change Types:**
  - **A** = Added file or directory
  - **M** = Modified file
  - **D** = Deleted file
  - **K** = Baseline Kept (deliberately preserved to avoid upstream regressions)
- **Lifting Methods:**
  - **whole:** Unmodified file checkout from origin/smp (git checkout origin/smp -- <path>).

- `sc-project.py`: Single-core projection filter stripping `#if defined(OS_USE_SMP_SCHEDULER)` and `OS_INTEGER_RTOS_PORT_NCPU > 1` blocks.
- `_edit.py`: Surgical hunk insertion/substitution with strict anchor matching and occurrence assertions.
- `additive`: Add-only CMake target or JSON entry without modifying existing baseline configurations.
- **Repository Codes:**
  - **K** = `micro-os-plus-iii` (Kernel core and shared test harness)
  - **C** = `micro-os-plus-iii-cortexm` (ARM Cortex-M port and CMSIS-M HAL)
  - **P** = `micro-os-plus-iii-posix-arch` (POSIX host port and multi-threaded CPU emulation)
  - **A32** = `micro-os-plus-iii-aarch32` (ARMv7-A / Cortex-A7/A53 32-bit port)
  - **A64** = `micro-os-plus-iii-aarch64` (ARMv8-A / Cortex-A53 64-bit port)
  - **DEV** = `micro-os-plus-iii-devices` (Source repository dissolved in Part 0)

## 1. The 7 Progressive Pull Requests (Architecture & Dependency Flow)

To ensure seamless upstream review and maintain an unbroken bisectable history, the integration is decomposed into seven strictly ordered pull requests. Each PR builds upon its predecessors and satisfies rigorous acceptance invariants.



```

| - posix-arch v1.1.0, cortexm v1.2.0 | | - History-
preserving git subtree add |
| - Python package.json bump (no npm push) | | - soc/
native, drivers/*, soc/rp2350, etc. |
| - Base: step/14 --> Head: step/24 | | - Base: Port
branches -> Head: part0-dev |
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PR #5: Steps 25–26 (Part C2) – Silicon Cores & Modular
Add-Only CMake |
| - Cortex-M33 (MPS2 AN521, SSE-200 MHU IPI), RP2350 (SIO Hardware
Spinlock 0 + FIFO IRQ 25) |
| - Modular toolchains, uos-app.cmake, 2xcortex-m33 target (links
fat ::iii library) |
| - Base: step/24 (+ PR #4) --> Head: step/26 | Gate: 2xcortex-
m33 (4/4) PASS |
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|
v
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|
PR #6: Steps 27–28 (Part C3) – Multi-Core Test Suites &
Multi-Arch Add-Only Platforms |
| - Hardware FPU context switch test (fp-switch), tests/smp-support/
relocation |
| - 4-core QEMU raspi3b platforms (AArch32 & AArch64), composite
test-smp-cmake action |
| - Base: step/26 --> Head: step/28 | Gate: 72/72 baseline +
test-smp-cmake (100% green) |
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|
|
v
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|
PR #7: Steps 29–30 (Part D) – Documentation Suite & Final
Merge Reconciliation |
| - Synchronized Markdown & PDF runbooks, pristine CI metadata
(.github/**, README.md, LICENSE) |
| - Dev-tooling removal (scripts/smp/), finalize.sh acceptance
validation |
| - Base: step/28 --> Head: step/30 | Gate: Byte-identical CI
metadata + full suite PASS |
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```

## 1.1 Pull Request Summary Matrix

| PR #  | Phase  | Steps Included | Repositories       | Core Focus & Artifacts                                            | Acceptance Gate & Invariants                                      |
|-------|--------|----------------|--------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|
| PR #1 | Part A | Steps 1–13     | K, +C+P at Step 13 | Single-core bugfixes, ISO C/C++ conformance, allocator hardening, | <b>72/72 frozen baseline tests PASS</b> ; zero SMP macro leakage. |

| PR #         | Phase          | Steps Included                     | Repositories          | Core Focus & Artifacts                                                                                                                                                     | Acceptance Gate & Invariants                                                                                              |
|--------------|----------------|------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|              |                |                                    |                       | condvar<br>atomicity,<br>mutex ceiling,<br>highres clock<br>sync.                                                                                                          |                                                                                                                           |
| <b>PR #2</b> | <b>Part B</b>  | Steps 14-23<br>(collapsed step/14) | <b>K, C, P</b>        | Portable SMP scheduler,<br>recursive<br>_smp_klock,<br>per-CPU ready lists, deferred stack switch,<br>IPI dispatch,<br>port fixes A-J,<br><b>native-smp host platform.</b> | unifdef<br>-UOS_USE_SMP_SCHEDULER<br>zero-diff vs Step 13;<br>72/72 tests pass;<br>native-smp dual-core suite (3/3) PASS. |
| <b>PR #3</b> | <b>Part C1</b> | Step 24                            | <b>P, C</b>           | Version releases<br>publishing SMP port contract:<br>posix-arch v1.1.0,<br>cortexm v1.2.0.                                                                                 | package.json version bump only; local git tags created without pushing.                                                   |
| <b>PR #4</b> | <b>Part 0</b>  | Part 0                             | <b>P, C, A32, A64</b> | History-preserving git subtree migration of devices into architecture repositories;<br>legacy target kept resolvable.                                                      | All existing board targets linking micro-os-plus::devices resolve and build.                                              |
| <b>PR #5</b> | <b>Part C2</b> | Steps 25-26                        | <b>K, C, P, Arch</b>  | Silicon additions (Cortex-M33, RP2350 SIO spinlocks),<br>modular CMake (uos-app.cmake, toolchains),<br>2xcortex-m33 QEMU target.                                           | 72/72 baseline pass;<br>2xcortex-m33 dual-core suite (4/4) PASS.                                                          |
| <b>PR #6</b> | <b>Part C3</b> | Steps 27-28                        | <b>K, C, P, Arch</b>  | fp-switch suite,<br>test_smpl absorption into tests/smp-support/,<br>AArch32/64 raspi3b harness wiring,<br>test-smp-cmake action.                                          | 72/72 baseline pass;<br>test-smp-cmake (M33 + Native-SMP + AArch) 100% green.                                             |
| <b>PR #7</b> | <b>Part D</b>  | Steps 29-30                        | <b>All</b>            | Documentation PDF build,<br>.github/ CI workflow restoration,<br>scripts/smp/                                                                                              | Upstream CI metadata byte-identical to baseline;<br>full test matrix green.                                               |

| PR # | Phase | Steps Included | Repositories | Core Focus & Artifacts        | Acceptance Gate & Invariants |
|------|-------|----------------|--------------|-------------------------------|------------------------------|
|      |       |                |              | removal, master branch merge. |                              |

### 1.2 Executive Resume of All Pull Request Actions Across Repositories (PR #1 through PR #32)

This resume catalogs all 32 pure single-theme pull requests, providing a direct 1:1 cross-reference between `GITHUB-PROGRESSIVE-PR-GUIDE.md` and `files-modif-by-step.md`:

| Pull Request ID                                  | Repositories Touched | Primary Target Files & Artifacts                                                                                           | Rationale & Failure Mode Averted                                                                                                                                             | Gate Verification                                          |
|--------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>PR #1: POSIX dirent.h Conformance</b>         | K                    | include/cmsis-plus/posix/dirent.h, timegm.c                                                                                | Fixes ISO C empty struct constraint. Baseline timegm.c preserved.                                                                                                            | 72/72 tests pass; -Wextra-semi clean                       |
| <b>PR #2: List Iterator Member Access</b>        | K                    | include/cmsis-plus/utils/lists.h                                                                                           | Corrects iterator postfix/prefix access to call member functions <code>node_-&gt;next()</code> .                                                                             | 72/72 tests pass; template concepts clean                  |
| <b>PR #3: Newlib Syscall Return Types</b>        | K                    | include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h                                                                  | Harmonizes read/write syscall returns with newlib <code>_READ_WRITE_RETURN_TYPE</code> .                                                                                     | 72/72 tests pass; newlib reentrant tests pass              |
| <b>PR #4: Export Thread Suspend Symbol</b>       | K                    | src/rtos/os-thread.cpp                                                                                                     | Removes out-of-line inline from <code>this_thread::suspend</code> so linker exports symbol.                                                                                  | 72/72 tests pass; C wrapper resolves symbol                |
| <b>PR #5: Allocator Usable Size &amp; Calloc</b> | K                    | os-memory.h, first-fit-top.h, os-memory.cpp, first-fit-top.cpp, lifo.cpp, block-pool.cpp, malloc.cpp, estd/memory_resource | Adds <code>do_usable_size()</code> virtual API; fixes integer multiplication overflow in <code>calloc()</code> ; corrects <code>polymorphic_allocator</code> copy semantics. | 72/72 tests pass; <code>malloc_usable_size</code> verified |
| <b>PR #6: C++17 Aligned Allocation</b>           | K                    | src/libcpp/new.cpp                                                                                                         | Adds standard C++17 <code>operator new/delete(std::size_t, std::align_val_t)</code> overloads.                                                                               | 72/72 tests pass; aligned new overloads clean              |
| <b>PR #7: Meyers Error Singletons</b>            | K                    | src/libcpp/system-error.cpp                                                                                                | Thread-safe Meyers singletons for error categories with Clang destructor pragma wraps.                                                                                       | 72/72 tests pass; -Wexit-time destructors clean            |
| <b>PR #8: Chrono 64-Bit Duration Overflow</b>    | K                    | src/libcpp/chrono.cpp                                                                                                      | Rewrites steady clock <code>highres</code> multiplication to avoid 64-bit integer overflow.                                                                                  | 72/72 tests pass; chrono math overflow-free                |
| <b>PR #9: CMSIS-RTOS C Wrapper Cleanups</b>      | K                    | src/rtos/os-c-wrapper.cpp                                                                                                  | Corrects one-shot timer default, polymorphic deletion, and 64-bit timeout casts.                                                                                             | 72/72 tests pass; cmsis-os-validator (60/60)               |
| <b>PR #10: POSIX I/O FD Table Mutexing</b>       | K                    | src/posix-io/file-descriptors-manager.cpp                                                                                  | Wraps file descriptor table under critical section locks to prevent concurrent races.                                                                                        | 72/72 tests pass; concurrent I/O stress green              |
| <b>PR #11: Filesystem &amp;</b>                  | K                    | include/cmsis-plus/posix-io/                                                                                               | Protects filesystem mount tables and network stack linked lists with mutexes.                                                                                                | 72/72 tests pass; filesystem mount tests green             |

| Pull Request ID                                 | Repositories Touched | Primary Target Files & Artifacts                                                                | Rationale & Failure Mode Averted                                                                                                    | Gate Verification                                   |
|-------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Netstack Mutexing</b>                        |                      | file-system.h,<br>net-stack.h                                                                   |                                                                                                                                     |                                                     |
| <b>PR #12: Block Device Partition Bounds</b>    | <b>K</b>             | src/posix-io/<br>block-device.cpp                                                               | Corrects inverted size check: returns EINVAL when partition size is zero.                                                           | 72/72 tests pass; invalid partition returns EINVAL  |
| <b>PR #13: ARMv8-M Semihosting Traps</b>        | <b>K</b>             | include/cmsis-plus/arm/<br>semihosting.h                                                        | Adds __ARM_ARCH_8M_MAIN__/<br>__ARM_ARCH_8M_BASE__<br>semihosting trap guards.                                                      | 72/72 tests pass; ARM semihosting tests pass        |
| <b>PR #14: SecureFault Handler</b>              | <b>K</b>             | src/startup/<br>exception-handlers.c,<br>include/cmsis-plus/cortexm/<br>exception-handlers.h    | Declares SecureFault_Handler and extends ARMv8-M mainline exception handling.                                                       | 72/72 tests pass; exception vectors compile         |
| <b>PR #15: Semihosting fstat() Mode</b>         | <b>K</b>             | src/semihosting/c-syscalls-<br>semihosting.cpp                                                  | Corrects fstat() to assign S_IFCHR character device mode only when unset.                                                           | 72/72 tests pass; file mode attributes clean        |
| <b>PR #16: Timer Callback Decoupling</b>        | <b>K</b>             | src/rtos/internal/<br>os-lists.cpp, os-timer.cpp, os-thread.h                                   | Invokes software timer callbacks outside critical sections to prevent deadlock.                                                     | 72/72 tests pass; timer callback mutexes clean      |
| <b>PR #17: Mutex Priority Ceiling</b>           | <b>K</b>             | src/rtos/os-mutex.cpp                                                                           | Enforces POSIX priority ceiling before acquiring ownership; dynamic boost tracking.                                                 | 72/72 tests pass; mutex-stress-test passes          |
| <b>PR #18: Thread Destroying &amp; Reaper</b>   | <b>K</b>             | include/cmsis-plus/rtos/os-thread.h, os-thread.cpp, os-idle.cpp, os-c-decls.h                   | Adds state::destroying (7) state; idle thread reaper safely frees dynamic stacks.                                                   | 72/72 tests pass; dynamic thread reaper green       |
| <b>PR #19: CondVar Lost-Signal Elimination</b>  | <b>K</b>             | src/rtos/os-condvar.cpp, os-condvar.h, instrumentation.h, os-c-decls.h, estd/condition_variable | Solves lost-wakeup race condition; binds timed_wait to attribute clock instance; std::condition_variable trusts the kernel verdict. | 72/72 tests pass; condvar concurrency stress green  |
| <b>PR #20: std::thread Functor Lifetime</b>     | <b>K</b>             | src/libcpp/thread-cpp.h, include/cmsis-plus/estd/thread_internal.h                              | Blocks std::thread::join() on native handle; retains functor lifetime to prevent use-after-free.                                    | 72/72 tests pass; functor closure lifetime verified |
| <b>PR #21: Message Queue Reschedule</b>         | <b>K</b>             | src/rtos/os-mqueue.cpp                                                                          | Invokes port::scheduler::reschedule() immediately after send/receive wakeups.                                                       | 72/72 tests pass; message queue preemption green    |
| <b>PR #22: High-Res Clock Port Sync</b>         | <b>K + C + P</b>     | os-decls.h, os-clocks.cpp, Cortex-M & POSIX port/os-inlines.h                                   | Declares/calls has_hardware_counter(); SysTick ICSR pending overflow fix.                                                           | 72/72 tests pass; ICSR pending fix verified         |
| <b>PR #23: Multi-Core SMP Kernel &amp; Host</b> | <b>K + C + P</b>     | os-core.cpp, os-sched.h, host_cpu.*, tests/platforms/native-smp/                                | Portable multi-core scheduler, recursive _smp_klock, ready lists, stack switch, native-smp.                                         | unifdef zero-diff + 72/72 + native-smp (3/3) PASS   |
|                                                 | <b>P + C</b>         |                                                                                                 |                                                                                                                                     |                                                     |

| Pull Request ID                            | Repositories Touched | Primary Target Files & Artifacts                                                                                              | Rationale & Failure Mode Averted                                                                                                                                                   | Gate Verification                              |
|--------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>PR #24: Port Package Releases</b>       |                      | micro-os-plus-iii-posix-arch/<br>package.json,<br>cortexm/<br>package.json                                                    | Semantic version releases publishing SMP port contract: posix-arch v1.1.0, cortexm v1.2.0.                                                                                         | package.json version bump only; local git tags |
| <b>PR #25: Devices Subtree Dissolution</b> | <b>P + C + Arch</b>  | soc/native/,<br>drivers/*, soc/<br>stm32f4xx/, soc/<br>rp2350/, soc/<br>bcm2837/                                              | History-preserving git subtree add folding devices into architecture ports; the AArch CMake conditionally consumes a sibling devices working copy so legacy targets still resolve. | All legacy board targets build cleanly         |
| <b>PR #26: Modern Silicon Cores</b>        | <b>C + K</b>         | include-m33/,<br>include-rp2350/,<br>os-core-m33.cpp,<br>os-core-rp2350.cpp                                                   | ARM Cortex-M33 (MPS2 AN521, SSE-200 MHU IPI) and RP2350 (SIO spinlock 0 + FIFO IRQ 25).                                                                                            | M33/RP2350 port targets compile cleanly        |
| <b>PR #27: Modular Add-Only CMake</b>      | <b>K + C</b>         | cmake/uos-app.cmake, cmake/toolchains/*,<br>tests/platforms/2xcortex-m33/                                                     | Granular CMake targets (::iii-core, ::port-smp-decls); additive fat ::iii library links.                                                                                           | 72/72 baseline + 2xcortex-m33 (4/4) PASS       |
| <b>PR #28: Multi-Core FPU Stress Suite</b> | <b>K</b>             | tests/sources/fp-switch/, tests/smp-support/                                                                                  | Multi-core hardware FPU context switch stress test; relocates root test_smpl/ to tests/smp-support/.                                                                               | fp-switch passes on 2xcortex-m33 (4/4)         |
| <b>PR #29: Multi-Arch 4-Core Platforms</b> | <b>K + A32 + A64</b> | tests/platforms/aarch32-rpi3b/,<br>tests/platforms/aarch64-rpi3b/,<br>test-smp-cmake                                          | 4-core QEMU raspi3b platforms for ARMv7-A & ARMv8-A; composite test-smp-cmake action.                                                                                              | test-smp-cmake passes (Native, M33, AArch)     |
| <b>PR #30: Architectural Docs Suite</b>    | <b>K</b>             | docs/smp-integration/*.md,<br>docs/smp-integration/*.pdf,<br>docs/render-pdfs.sh                                              | Synchronizes all Markdown specifications and PDF runbooks via md2pdf.py.                                                                                                           | All PDFs compile cleanly without warnings      |
| <b>PR #31: Final Master Reconciliation</b> | <b>All Repos</b>     | .github/workflows/ci.yml, README.md, LICENSE, purge scripts/smp/                                                              | Reconciles history with origin/xbpack-development; restores pristine CI metadata (100% byte-identical).                                                                            | Upstream CI metadata 100% byte-identical       |
| <b>PR #32: Dedicated Pico 2 HW Wiring</b>  | <b>C + K</b>         | tests/platforms/cortexm-pico2/,<br>cortexm/test/boards/pico2/,<br>cortexm/test/pico2/,<br>cortexm/src/rtos/os-core-rp2350.cpp | Physical hardware testbed for Raspberry Pi Pico 2 (RP2350 dual Cortex-M33) with SWD flashing.                                                                                      | Hardware OpenOCD SWD flashing and UART green   |

## 1.3 Multi-Repository Diff Accounting & Codebase Exhaustiveness

Every single modified, added, or deleted file across all component repositories has been cross-referenced against the PR migration blueprint:

### 1. Kernel (micro-os-plus-iii — 382 changed files):

- *Core Kernel RTOS*: os-thread.\*, os-core.cpp, os-idle.cpp, os-mutex.cpp, os-condvar.cpp, os-timer.cpp, os-mqueue.cpp, os-clocks.cpp, os-memory.cpp, os-c-wrapper.cpp -> **PR #1, PR #2**.
- *Allocators & Standard C++ Support*: malloc.cpp, first-fit-top.\*, lifo.cpp, block-pool.cpp, new.cpp, system-error.cpp, chrono.cpp -> **PR #1**.
- *POSIX I/O & System Call Aliases*: file-descriptors-manager.cpp, file-system.h, net-stack.h, block-device.cpp, c-syscalls-aliases-standard.h -> **PR #1**.
- *Build System & Toolchains*: CMakeLists.txt, cmake/uos-app.cmake, cmake/toolchains/\* -> **PR #5**.
- *SMP Testbed & Harness Platform Dispatches*: tests/sources/fp-switch/, tests/smp-support/, tests/platforms/\*, tests/package.json -> **PR #2, PR #5, PR #6, Step 31**.
- *Documentation & Runbooks*: docs/\*.md, docs/\*.pdf, docs/render-pdfs.sh -> **PR #7**.
- *Transitional Tooling & Upstream Cleansing*: scripts/smp/, .github/\*\*, README.md, LICENSE, doxygen/\*\* -> **PR #7 (Step 30)**.

### 2. Cortex-M Port (micro-os-plus-iii-cortexm — 277 changed files):

- *Core Port & Port Contracts*: os-c-decls.h, os-decls.h, os-inlines.h, src/rtos/os-core.cpp -> **PR #1, PR #2**.
- *ARMv8-M Architecture Evolutions*: include-m33/, include-rp2350/, os-core-m33.cpp, os-core-rp2350.cpp -> **PR #5**.
- *Subtree Ingested SoCs*: soc/stm32f4xx/, soc/rp2350/ -> **PR #4**.
- *Port Package Semantic Version*: package.json (v1.2.0) -> **PR #3**.
- *Pico 2 Hardware Platform Infrastructure*: test/boards/pico2/, test/pico2/ -> **Step 31**.

### 3. POSIX Architecture Port (micro-os-plus-iii-posix-arch — 69 changed files):

- *Core Port & Host Multiprocessing*: os-c-decls.h, os-decls.h, os-inlines.h, src/rtos/os-core.cpp, host\_cpu.cpp, host\_cpu.hpp, free-store.cpp, board-contract.cpp, exception\_handler.\*, hw\_result.hpp -> **PR #1, PR #2**.
- *Subtree Ingested Native SoC & Portable Drivers*: soc/native/, drivers/include/, drivers/src/, drivers/fatfs/ -> **PR #4**.
- *Port Package Semantic Version*: package.json (v1.1.0) -> **PR #3**.
- *Native Host Test Suites*: test/native/\*, test/boards/native/\* -> **PR #2**.

### 4. AArch32 & AArch64 Architecture Ports (micro-os-plus-iii-aarch32/64 — 196 / 103 changed files):

- *Subtree Ingested SoC & Drivers*: soc/bcm2837/, drivers/\* -> **PR #4**.
- *Multi-Core Context Switch & Boot*: src/context\_switch.cpp, src/handlers.cpp, include/\* -> **PR #6**.
- *QEMU Board Test Runners*: test/boards/rpi-zero-2w/, test/luckfox-lyra/, test/rpi-zero-2w/ -> **PR #6**.

### 5. Devices Repository (micro-os-plus-iii-devices):

- *Dissolution*: 100% absorbed via git subtree into owning architecture repositories (posix-arch, cortexm, aarch32, aarch64);

the legacy `micro-os-plus::devices` target keeps resolving via a conditional sibling consumption in **PR #4**.

## 1.4 Transitional Tooling, Scratch Runners, & Upstream Invariance Reconciliation

To ensure upstream purity, four specific integration policies are strictly applied:

### 1. Purge of Developer Tooling (`scripts/smp/`):

- *Policy*: The `scripts/smp/` directory contains migration helper scripts (`new-step.sh`, `verify-step.sh`, `advance-step.sh`, `chunks/`, `run-loop.sh`). These are developer-only tools required during the migration process.
- *Reconciliation*: Automatically deleted in **Step 30 (PR #7)** before the pull request is finalized.

### 2. Standardization of Non-Standard Root Test Directory (`test_smp/`):

- *Policy*: The experimental `smp` branch accumulated runner scripts in a non-standard root `test_smp/` folder.
- *Reconciliation*: Absorbed additively into standard `tests/smp-support/` via `absorb-test-smpl.sh` in **Step 27 (PR #6)**, eliminating non-standard root entries.

### 3. Pristine Restoration of Upstream CI Metadata:

- *Policy*: Upstream `.github/workflows/ci.yml`, `README.md`, `LICENSE`, and Doxygen templates must not be polluted by downstream divergence.
- *Reconciliation*: Step 30 restores these files to be 100% byte-identical to `origin/xpack-development`.

### 4. Preservation of Upstream Newlib Baseline (`src/libc/stdlib/timegm.c`):

- *Policy*: The `smp` branch hid `timegm()` behind `#if !defined(__GLIBC__)`, breaking `glibc ≥ 2.44` builds.
- *Reconciliation*: The upstream baseline version is deliberately kept in **Step 1 (PR #1)**.

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## 2. Chronological Step-by-Step File Manifest & Technical Rationale ("WHY")

Every modification across each repository is catalogued below with its precise technical justification, failure mode, compiler warning/error flag, and lifting technique.

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### Phase 0 — Dissolve the devices Repository (PR #25)

**Why the Whole Phase:** Upstream `xpack-development` maintains a clean six-repository architecture. The experimental `smp` branch introduced a 7th repository (`micro-os-plus-iii-devices`), creating an unwanted external dependency. Part 0 uses `git subtree` to fold all drivers and SoC implementations directly into the owning architecture repositories while preserving 100% of `git` commit history.

| Repo | File / Subtree | Action | Lift Method | Deep Technical "WHY" / Failure Condition Averted |
|------|----------------|--------|-------------|--------------------------------------------------|
| P    | soc/native/    | A      |             |                                                  |

| Repo                  | File / Subtree                                         | Action   | Lift Method     | Deep Technical "WHY" / Failure Condition Averted                                                                                                            |
|-----------------------|--------------------------------------------------------|----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                                        |          | git subtree add | Relocates host image-file SD/block storage backend into POSIX architecture port. Avoids external repo dependency for host block tests.                      |
| <b>P</b>              | drivers/<br>include/<br>drivers/src/<br>drivers/fatfs/ | <b>A</b> | git subtree add | Integrates shared FatFs, USB, and block device drivers into POSIX port for host filesystem emulation.                                                       |
| <b>P</b>              | src/.../<br>mailbox, fatfs                             | <b>M</b> | git subtree     | Atomic spinlock synchronization for BCM2837 mailbox; fixes FatFs append-to-empty file bug where cluster extent was not allocated.                           |
| <b>C</b>              | soc/stm32f4xx/                                         | <b>A</b> | git subtree add | Moves STM32F4 CMSIS headers, system startup, and peripheral register definitions into Cortex-M port.                                                        |
| <b>C</b>              | soc/rp2350/                                            | <b>A</b> | git subtree add | Moves Raspberry Pi Pico 2 (RP2350) headers and system_rp2350.c into Cortex-M port.                                                                          |
| <b>A32 / A64</b>      | soc/bcm2837/,<br>soc/rk3506/,<br>drivers/              | <b>A</b> | git subtree add | Moves Raspberry Pi 3B / Zero 2W (BCM2837) and Rockchip RK3506 SoC code and drivers into respective AArch ports.                                             |
| <b>P, C, A32, A64</b> | CMakeLists.txt                                         | <b>M</b> | additive        | Keeps the legacy micro-os-plus::devices target resolvable — from the folded content, or (AArch) from a conditionally consumed sibling devices working copy. |

## Phase A — Uniprocessor Correctness & Hardening (PR #1, Steps 1-13)

**Why the Whole Phase:** These 13 steps fix genuine single-core defects, memory allocator vulnerabilities, synchronization race conditions, and compiler standard diagnostics. Landing them first isolates bugfixes from SMP multi-core logic and ensures 100% green tests on the baseline.

## PR #1 — POSIX dirent.h ISO C Empty Struct Conformance

- **M** include/cmsis-plus/posix/dirent.h (whole): Empty struct typedef struct { ; } DIR; violates ISO C99/C11 §6.7.2.1 (struct must have at least one named member) and triggers -Wextra-semi under strict flags. *Fix:* Adds int reserved; member to comply with standard C.
- **K** src/libc/stdlib/timemgm.c (Baseline Kept): The smp branch hid the timemgm() prototype behind #if !defined(\_\_GLIBC\_\_), which breaks compilation on glibc ≥ 2.44 under -Werror=missing-prototypes. Keeping the unconditioned baseline prototype builds cleanly on all host and cross toolchains.

## PR #2 — Intrusive Double-List (lists.h) Iterator Member Access Semantics

- **M** include/cmsis-plus/utils/lists.h (whole): Fixes double-list iterator post-increment and post-decrement operators (operator++(int) / operator--(int)) to call node\_>next() / node\_>prev() accessor methods rather than accessing raw pointers. *Why:* Enforces iterator concept conformance with modern C++ standard libraries.

## PR #3 — Newlib C Syscall Weak Aliases Return Type Macro

- **M** include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h (whole): Maps CMSIS\_PLUS\_POSIX\_IO\_RW\_RETURN\_TYPE to newlib's \_READ\_WRITE\_RETURN\_TYPE (or ssize\_t) so read/write syscall prototypes match newlib libc weak symbol signatures across ARM toolchains.

## PR #4 — Thread Suspend External Symbol Export for CMSIS C Wrapper

- **M** src/rtos/os-thread.cpp (\_edit.py surgical): Removes inline specifier from out-of-line this\_thread::suspend(void) definition. *Why:* Out-of-line inline definition prevented the linker from exporting the symbol, causing undefined reference errors when called from the CMSIS C-wrapper.

## PR #5 — Memory Allocator Usable Size & Integer Overflow Protection

- **M** include/cmsis-plus/rtos/os-memory.h & include/cmsis-plus/memory/first-fit-top.h (whole): Declares do\_usable\_size(void\*) pure virtual method in memory\_resource base class and overrides in first\_fit\_top. Adds overflow-checked align\_size().
- **M** src/rtos/os-memory.cpp & src/memory/first-fit-top.cpp (whole + pragma wrap): Implements base-class default and first\_fit\_top::do\_usable\_size() calculating available payload bytes. *Hardening:* Wraps pointer arithmetic in first-fit-top.cpp with #pragma GCC diagnostic ignored "-Wcast-align" and #pragma clang diagnostic ignored "-Wunsafe-buffer-usage" to prevent ARM GCC and Clang -Weverything build failures.
- **M** src/memory/lifo.cpp & src/memory/block-pool.cpp (whole): Adds allocator bounds checks; lifo.cpp clears the candidate chunk when the arena head is too small, and block-pool.cpp corrects the std::align() failure test (res == nullptr).
- **M** src/libc/stdlib/malloc.cpp (whole): Adds multiplication overflow check in calloc(): if (nelem > static\_cast<std::size\_t> (-1) / elbytes) { errno = ENOMEM; return nullptr; } (with elbytes known non-zero). Prevents integer truncation security vulnerabilities in heap allocations.
- **M** include/cmsis-plus/estd/memory\_resource (whole): polymorphic\_allocator::select\_on\_container\_copy\_construction()

returns `*this` (a copy) rather than a default-constructed allocator, per C++17.

## **PR #6 — ISO C++17 Aligned Memory Allocation Operators**

- **M** `src/libcpp/new.cpp` (whole): Adds C++17 standard aligned memory allocation operators `operator new(std::size_t, std::align_val_t)` and `operator delete(void*, std::align_val_t)`.

## **PR #7 — Thread-Safe Meyers Singletons for Error Categories**

- **M** `src/libcpp/system-error.cpp` (whole + pragma wrap): Implements function-local static error category instances (`get_system_error_category()` and `get_cmsis_error_category()`) ensuring static storage duration for error categories. Adds `#pragma clang diagnostic ignored "-Wexit-time-destructors"` to silence Clang warnings on process shutdown handlers.

## **PR #8 — Steady Clock 64-Bit Duration Multiplicative Overflow Protection**

- **M** `src/libcpp/chrono.cpp` (whole): Rewrites steady clock high-resolution tick multiplication as `(cycles / freq) * 1e9 + ((cycles % freq) * 1e9) / freq` to prevent 64-bit integer overflow during long running times.

## **PR #9 — CMSIS-RTOS v1 C API Wrapper Cleanups**

- **M** `src/rtos/os-c-wrapper.cpp` (whole + stripped): Corrects default timer type to one-shot (`osTimerOnce`); ensures polymorphic deletion of mutexes/semaphores calls derived destructors; fixes 64-bit timeout conversions by casting milliseconds to `uint64_t` prior to multiplying by `1000u`.
- *Lift Discipline*: The `origin/smp` file contains Step-9 `os_thread_state_destroying` static asserts and Step-20 SMP CPU affinity logic. These are surgically stripped so Step 4 compiles cleanly against the Step 3 baseline.

## **PR #10 — POSIX I/O File Descriptor Table Critical Section Mutexing**

- **M** `src/posix-io/file-descriptors-manager.cpp` (whole + pragma wrap): Wraps file descriptor table allocation, lookup, and deallocation under `rtos::interrupts::critical_section` locks, preventing descriptor corruption under concurrent `open()/close()`. Adds Clang `-Wunsafe-buffer-usage` suppression.

## **PR #11 — POSIX I/O Deferred List Critical Section Protection**

- **M** `include/cmsis-plus/posix-io/file-system.h` & `include/cmsis-plus/posix-io/net-stack.h` (whole): Protects file system mount tables and network stack linked lists with mutexes.

## **PR #12 — Block Device Partition Size Validation Check**

- **M** `src/posix-io/block-device.cpp` (whole): Corrects inverted size check: returns `EINVAL` if partition size is zero.

## PR #13 — ARMv8-M Architecture Semihosting Trap Instruction Guards

- **M** `include/cmsis-plus/arm/semihosting.h` (whole): Adds `__ARM_ARCH_8M_MAIN__` and `__ARM_ARCH_8M_BASE__` architecture macro guards for semihosting traps.

## PR #14 — ARMv8-M Mainline Exception Handlers & SecureFault Declaration

- **M** `src/startup/exception-handlers.c` (whole): Extends the ARMv8-M mainline exception handling (VTOR relocation, register dumps, NMI/HardFault/MemManage/BusFault/UsageFault/SVC/DebugMon/PendSV guards).
- **M** `include/cmsis-plus/cortexm/exception-handlers.h` (whole): Declares `SecureFault_Handler` for ARMv8-M Security Extension (TrustZone) and widens the `exception_stack_frame_t` FPU context guard to `__ARM_ARCH_8M_MAIN__`.

## PR #15 — Semihosting Syscall `fstat()` Character Device Fallback Mode

- **M** `src/semihosting/c-syscalls-semihosting.cpp` (whole): Corrects semihosting `fstat()` implementation to assign `S_IFCHR` file mode only when the file type has not been explicitly configured.

## PR #16 — Software Timer Callback Execution Decoupling & Drift Prevention Outside Critical Sections

- **M** `src/rtos/internal/os-lists.cpp` & `src/rtos/os-timer.cpp` (whole): Redesigns software timer expiration queue. The callback function is invoked **after exiting the critical section**, preventing deadlock if a timer callback acquires mutexes or invokes RTOS primitives. Adds catch-up logic for periodic timers under high interrupt latency.
- **M** `include/cmsis-plus/rtos/os-thread.h` (`_edit.py` surgical): Adds `errno` scratch storage field for handler mode. Excludes Step-9 destroying state and Step-20 affinity fields.

## PR #17 — POSIX Mutex Priority Ceiling Protocol & Dynamic Boost Tracking

- **M** `src/rtos/os-mutex.cpp` (whole): Enforces POSIX priority ceiling protocol: raises thread priority to mutex ceiling **before** acquiring ownership; dynamically tracks highest waiting thread priority; recomputes owner priority upon unlock; caches owner thread pointer across the unlock window to eliminate priority inversion windows.

## PR #18 — Thread Lifecycle State Machine (`state::destroying`), Atomic Join & Idle Reaper

- **M** `include/cmsis-plus/rtos/os-thread.h`, `src/rtos/os-thread.cpp`, `src/rtos/os-idle.cpp` (`sc-project.py` single-core projection): Introduces explicit `state::destroying` (value 7) state. Thread destruction unlinks the thread from scheduler lists and marks it for cleanup. Idle thread reaper safely frees dynamic stacks and control blocks without race conditions.
- **M** `include/cmsis-plus/rtos/os-c-decls.h` (`_edit.py` surgical): Inserts `os_thread_state_destroying = 7` enumerator into C ABI header without pulling in Step-10 `condvar` clock fields.

## PR #19 — Condition Variable Lost-Signal Elimination & Clock Binding

- **M** `src/rtos/os-condvar.cpp` & `include/cmsis-plus/rtos/os-condvar.h` (whole): Solves classic lost-wakeup race condition: atomically enqueues thread into condition variable wait list under scheduler lock, releases associated mutex, and suspends thread. Binds `timed_wait()` to the specific clock instance configured in attributes (system or steady clock).
- **M** `include/cmsis-plus/diag/instrumentation.h` (whole): Adds `OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_CONDVAR` (12) trace event code.
- **M** `include/cmsis-plus/rtos/os-c-decls.h` (`_edit.py` surgical): Uncomments `void* clock` pointer in `os_condvar_t` C struct to match C++ ABI size.

## PR #20 — C++ `std::thread` Native Handle Synchronization & Functor Lifetime

- **M** `src/libcpp/thread-cpp.h` & `include/cmsis-plus/estd/thread_internal.h` (whole): Changes `std::thread::join()` to block on the underlying native RTOS thread handle until termination. Retains functor object lifetime in thread wrapper until completion, preventing use-after-free crashes.

## PR #21 — Message Queue Preemptive Rescheduling on Wakeups

- **M** `src/rtos/os-mqueue.cpp` (whole): Invokes `port::scheduler::reschedule()` immediately after `send()` or `receive()` wakes a higher-priority thread, ensuring immediate preemption on POSIX and Cortex-M uniprocessors.

## PR #22 — High-Resolution Monotonic Clock Port Synchronization (Cross-Repo: K + C + P)

- **K M** `include/cmsis-plus/rtos/os-decls.h` & `src/rtos/os-clocks.cpp` (whole): Declares and invokes `port::clock_highres::has_hardware_counter()`.
- **C M** `include/cmsis-plus/rtos/port/os-inlines.h` (`sc-project.py` projection): Implements `has_hardware_counter()` -> `false` and `hardware_counter()` -> `0`. Fixes SysTick overflow detection by inspecting `SCB->ICSR` & `SCB_ICSR_PENDSTSET_Msk`.
- **P M** `include/cmsis-plus/rtos/port/os-inlines.h` (`_edit.py` surgical): Inserts `clock_highres` inline implementation reading host `CLOCK_MONOTONIC`. Uses `timespec tp`; without struct tag to satisfy `-Werror=redundant-tags`.

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## Phase B — Multi-Core SMP Infrastructure & Genuine Host Validation (PR #23)

**Why Collapsed into One Step:** In a multi-core kernel, the scheduler (`os-core.cpp`), thread lifecycle (`os-thread.cpp`), idle loop (`os-idle.cpp`), recursive kernel lock (`_smp_klock`), and port contracts are completely interdependent. Applying them in isolation produces non-compilable intermediate commits. Collapsing Steps 14–23 into a single atomic step/14 commit across **K**, **C**, and **P** guarantees bisectability.

### Kernel Changes (K)

- **M** `include/cmsis-plus/rtos/os-c-decls.h` (whole): Defines `SMP_NO_OWNER` (`0xFFFFFFFFu`) and per-CPU thread context layout.

- **M** `include/cmsis-plus/rtos/os-sched.h` (whole): Converts `current_thread_` into per-CPU array `current_thread_[OS_NCPU]` and declares `port_cpu_id()`.
- **M** `include/cmsis-plus/rtos/os-thread.h` (whole): Adds CPU affinity mask `th_cpu_affinity` and per-core execution state tracking.
- **M** `src/rtos/os-core.cpp` (whole + Fix H): Implements portable SMP scheduler: per-CPU ready lists, recursive kernel spinlock acquisition, Inter-Processor Interrupt (IPI) dispatching, and the critical invariant prohibiting context switching while holding `_smp_klock`.
- **M** `src/rtos/os-idle.cpp` (whole): Instantiates and registers dedicated idle thread per physical core (`os_idle_thread_core[OS_NCPU]`).
- **M** `src/rtos/os-main.cpp` (whole): Pins main bootstrap thread to Core 0 (`affinity = (1u << 0)`).
- **M** `src/rtos/os-thread.cpp` (whole + Fix H): Implements SMP thread migration, affinity enforcement, and the 5-stage stack-pointer claim/publish protocol.

## Cortex-M Port Changes (C)

- **M** `include/cmsis-plus/rtos/port/os-c-decls.h`, `os-decls.h`, `os-inlines.h` (whole): Defines Cortex-M SMP port contract: per-core `lock_state[OS_NCPU]`, recursive `_smp_klock_enter()` / `_smp_klock_exit()` with PRIMASK interrupt masking.
- **M** `src/rtos/os-core.cpp` (whole): Implements Cortex-M SMP PendSV context switch handler with atomic stack pointer publish/claim.

## POSIX-Arch Port Changes (P)

- **M** `include/cmsis-plus/rtos/port/os-c-decls.h`, `os-decls.h`, `os-inlines.h` (whole + Fixes A, I): Declares TLS `_this_cpu`, `_in_isr[OS_NCPU]`, and Clang `-Weverything` ignore pragmas.
- **A** `include/host_cpu.hpp` & `src/host_cpu.cpp` (whole + Fixes C, E, G, J): Implements threads-as-cores multi-processing model using `pthread_create()`, per-core POSIX interval timers (`SIGEV_THREAD_ID`), and IPI signalling via `pthread_kill(..., SIGRTMIN+1)`.
- **A** `include/exception_handler.hpp` & `src/exception_handler.cpp` (whole + Fixes F, J): Implements host exception signal dispatching and weak `g_core_stage[OS_NCPU]` boot synchronization array.
- **A** `src/free-store.cpp` (whole + Fix J): Host heap free-store allocator initialization.
- **A** `include/hw_result.hpp` & `src/board-contract.cpp` (whole): Hardware/QEMU PASS/FAIL string protocol handshake.
- **M** `src/rtos/os-core.cpp` (whole + Fixes B, C, J): Host SMP scheduler initialization and signal-driven context switching.
- **M** `CMakeLists.txt` (additive / Fix D): Additively registers `src/host_cpu.cpp`, `src/free-store.cpp`, `src/exception_handler.cpp` in `micro-os-plus-iii-posix-arch-interface`.

## Lift-Time Hardening Corrections (Fixes A-J)

- **Fix A/B:** In `posix-arch os-inlines.h`, converts `struct timespec tp;` to `timespec tp;` to resolve `-Werror=redundant-tags`.
- **Fix C:** In `posix-arch os-core.cpp` and `host_cpu.cpp`, guards `uc_sigmask` manipulation with `#if !defined(OS_INCLUDE_LIBUCONTEXT)` since standard `<ucontext.h>` has `uc_sigmask` while standalone `libucontext` does not.
- **Fix D:** Additively appends SMP host sources to CMake `INTERFACE` target without altering baseline structure.
- **Fix E:** In `host_cpu.cpp`, removes duplicate `port_cpu_id` declarations and adds prior prototype for `port_smp_ipi` override to silence `-Werror=missing-declarations`.
- **Fix F/G:** In `exception_handler.cpp` and `host_cpu.cpp`, defines port-side `__attribute__((weak)) volatile uint32_t g_core_stage[OS_NCPU]` so the port links cleanly without the test board.

- **Fix H:** In kernel `os-core.cpp` and `os-thread.cpp`, wraps `port_cpu_id` forward declarations with `#pragma GCC diagnostic ignored "-Wredundant-decls"` to avoid clashes with port header prototypes under true SMP builds.
- **Fix I/J:** In `posix-arch` headers and `.cpp` files, injects comprehensive Clang `-Weverything` suppression blocks (`-Wc++98-compat-pedantic`, `-Wreserved-identifier`, `-Wunsafe-buffer-usage`, `-Wcast-function-type-strict`).

Dual-Core Verification Platform (native-smp)

- **A** `tests/platforms/native-smp/`: Adds genuine host SMP platform setting `-DOS_USE_SMP_SCHEDULER=1` and `-DOS_INTEGER_RTOS_PORT_NCPU=2` via `target_compile_definitions`.
- **M** `tests/package.json`: Adds `native-smp-cmake-gcc14-debug/release` configs and `test-native-smp` action.
- *Why Essential:* Setting `-D` flags on the CMake command line creates cache variables that never reach the compiler. `native-smp` is the **only** host mechanism that compiles real SMP code and immediately validates Part B.

Phase C1 — Port Semantic Releases (PR #24)

**Why Now:** Part B introduced and finalized the multi-core SMP port contract. Step 24 tags semantic releases for the ports (`posix-arch v1.1.0` and `cortexm v1.2.0`), publishing the ABI required by downstream silicon and tests in PR #5 and PR #6.

| Repo | File                      | Action | Lift Method                           | Deep Technical "WHY" / Release Discipline                                                                                                 |
|------|---------------------------|--------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| P    | <code>package.json</code> | M      | <code>release-port.sh</code> (Python) | Bumps version from 1.0.0 to 1.1.0. <i>Why:</i> Publishes host SMP CPU model, per-core signal timers, and high-resolution monotonic clock. |
| C    | <code>package.json</code> | M      | <code>release-port.sh</code> (Python) | Bumps version from 1.1.0 to 1.2.0. <i>Why:</i> Publishes SMP port lock, multi-core PendSV switch, and SysTick overflow fix.               |

**Critical Release Discipline:** Version bumping is executed strictly via `scripts/smp/release-port.sh` using Python. Upstream xPack repositories configure a `postversion`: `"git push origin --all && git push origin --tags"` lifecycle script in `package.json`. Standard `npm version` invokes this hook even with `--no-git-tag-version`, attempting unauthorized remote pushes. `release-port.sh` modifies `package.json` directly and creates local tags only.

Phase C2 — New Silicon Targets & Modular CMake (PR #26 & PR #27)

**Why Here:** These steps introduce support for modern multi-core ARM microcontrollers (ARM Cortex-M33 / SSE-200 and Raspberry Pi RP2350) and establish an additive, modular CMake build structure that allows new platforms to consume the `fat micro-os-plus::iii` library without breaking single-core targets.

## PR #26 — Modern Multi-Core Silicon Ports (ARM Cortex-M33 & RP2350)

- **C A** `include-m33/cmsis-plus/rtos/port/os-inlines.h`: Port inline functions for Cortex-M33 (ARMv8-M Mainline). Implements dual-core NVIC Message Handling Unit (MHU) IPI signaling and SysTick management.
- **C A** `src/rtos/os-core-m33.cpp`: Context switch and interrupt management for Cortex-M33 with FPU state preservation.
- **C A** `include-rp2350/cmsis-plus/rtos/port/os-inlines.h`: Port inlines for Raspberry Pi RP2350 (dual Cortex-M33). Backs recursive `_smp_klock` with RP2350 SIO hardware spinlock 0 (`*(volatile uint32_t*)(SIO_BASE + SIO_SPINLOCK0_OFFSET)`).
- **C A** `src/rtos/os-core-rp2350.cpp`: Implements RP2350 inter-core FIFO IRQ (IRQ 25) for cross-core IPI wakeups.
- **P A** `src/board-contract.cpp`, `src/free-store.cpp`, `include/hw_result.hpp`: Reusable silicon bring-up contracts.
- **K M** `src/rtos/os-thread.cpp` (additive): Adds inter-processor interrupt wake-up dispatch in `thread::resume()` when `OS_INTEGER_RTOS_PORT_NCPU > 1`.

## PR #27 — Modular Add-Only CMake Build Architecture & Cross Toolchains

- **K A** `cmake/toolchains/arm-none-eabi.cmake` & `cmake/toolchains/aarch64-none-elf.cmake`: Standardized toolchain definition files for cross-compilation.
- **K A** `cmake/uos-app.cmake`: Helper macro `uos_add_app()` adapted to link `micro-os-plus::iii-core` for bare-metal targets while linking the fat `::iii` for standard platforms.
- **C M** `CMakeLists.txt` (additive): Appends `micro-os-plus::cortexm-qemu-m33` target (`-mcpu=cortex-m33`, `include-m33/`, `src/rtos/os-core-m33.cpp`). Baseline `::iii-cortexm` is 100% untouched.
- **K A** `tests/platforms/2xcortex-m33/`: Dual-core QEMU MPS2 AN521 (SSE-200) platform running with `--cpu cortex-m33 --smp 2`.
- **K A** `tests/platforms/2xcortex-m33/`: Reuses the generic `tests/device-qemu-cortexm` device (vectors + CMSIS) and the `xpack-3rd-party::arm-cmsis-core-m` CM33 headers; no new device directory is added. Links the fat `micro-os-plus::iii` through `micro-os-plus::cortexm-qemu-m33`.
- **K M** `tests/package.json` (additive): Registers `2xcortex-m33-cmake-gcc-debug/release` configs, `test-2xcortex-m33-cmake` action, and `arm-cmsis-core-dependencies`.

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## Phase C3 — Test Suites & Multi-Architecture Platforms (PR #28 & PR #29)

**Why Here:** With silicon cores and CMake targets established, Step 27 imports proven multi-core test suites (`fp-switch`, `smp-support`) and Step 28 wires up complete multi-architecture QEMU platforms for AArch32 and AArch64 (Raspberry Pi 3B / Zero 2W).

## PR #28 — Multi-Core FPU Stress Testing & Scaffolding Relocation

- **K A** `tests/sources/fp-switch/`: FPU context switch stress test (6 threads, 3000 ticks) verifying hardware floating-point register preservation across multi-core SMP context switches.
- **K M** `tests/cmake/tests-main.cmake` (additive): Selects the port under test from the platform name (`aarch32*`, `aarch64*`, `native*`, `cortexm*`, `qemu-cortex*`, `pico2*/2xcortex*`) instead of always adding the kernel one level above; falls back to the upstream kernel for unknown platforms.

- **K A** tests/smp-support/ (absorb-test-smpl.sh): Absorbs the root test\_smpl/ folder from smp branch into tests/smp-support/ (include/hw\_result.hpp, src/board-contract.cpp, run-qemu.sh, run-host.sh, run-hw.sh, OpenOCD configs). Eliminates non-standard root directory.
- **A32 / A64 M** test/boards/rpi-zero-2w/{qemu,hw}.sh: Repoints test execution scripts to consume shared runner from tests/smp-support/run-qemu.sh.

## PR #29 — Multi-Architecture 4-Core Platforms & Composite CI Action

- **K M** CMakeLists.txt (integrate-aarch-harness.sh): Adds additive sub-targets for thin-core ports:
    - micro-os-plus::iii-posix-io: Granular POSIX I/O layer.
    - micro-os-plus::iii-semihosting: ARM semihosting syscall glue.
    - micro-os-plus::iii-newlib-reent: Thread-safe newlib reentrancy structures.
    - micro-os-plus::test-support: Carries board-contract.cpp compiled in the target architecture scope.
  - **K A** tests/platforms/aarch32-rpi3b & tests/platforms/aarch64-rpi3b: Adds 4-core QEMU raspi3b SMP test platforms (ARMv7-A and ARMv8-A).
  - **K A** tests/platforms/cortexm-pico2: Adds Raspberry Pi Pico 2 (RP2350) dual-core target.
  - **K M** tests/cmake/tests-main.cmake (additive): Extends harness to dispatch new platforms.
  - **K M** tests/package.json (additive): Registers test-smp-cmake composite action executing test-2xcortex-m33-cmake, test-native-smp, and AArch legs.
- 

## Phase D — Documentation Suite & Final Merge Reconciliation (PR #30 & PR #31)

**Why Here:** The final phase compiles all technical documentation and PDF runbooks, reconciles repository histories with upstream master branches, restores untouched CI metadata, and removes transient developer tooling.

### PR #30 — Comprehensive Architectural Documentation & Specification Runbooks Suite

- **K M** docs/smp-integration/\*.md & docs/smp-integration/\*.pdf: Synchronizes architecture analyses, PR specifications, and runbooks:
  - docs/smp-integration/MICRO-OS-PLUS-SMP-VS-SINGLECORE-ANALYSIS.md & .pdf
  - docs/smp-integration/SMP-UPSTREAM-INTEGRATION-PLAN.md & .pdf
  - docs/smp-integration/Implementation-SMP-Integration.md & .pdf
  - docs/smp-integration/pull-request.md & .pdf
  - docs/smp-integration/files-modif-by-step.md & .pdf
- **K M** docs/render-pdfs.sh: Updates batch PDF compilation script using md2pdf.py.

### PR #31 — Final Upstream Master Reconciliation & Repository Cleansing

- **All M:** Merges origin/xpack-development into step/30 across all repositories. Baseline is an ancestor; merge is clean with zero conflicts.
- **All M:** Restores pristine upstream repository metadata: .github/workflows/ci.yml, README.md, LICENSE, Doxygen templates. Ensures CI configurations are 100% byte-identical to upstream.

- **K D**: Deletes scripts/smp/ directory (developer-only transition tooling).
- **K M** finalize.sh: Executes final local validation verifying that both xpm run test-all (72/72 single-core) and xpm run test-smp-cmake are 100% green.

## Appendix — Dedicated Physical Silicon Board Wiring (PR #32)

**Why:** Configures complete standalone hardware testing infrastructure for the Raspberry Pi Pico 2 (RP2350 dual ARM Cortex-M33).

- **C A** include-rp2350/cmsis-plus/rtos/port/os-inlines.h: Dual-core greeting banner, 150 MHz system clock validation, SIO spinlock 0 recursive locking.
- **C A** src/rtos/os-core-rp2350.cpp: Boot synchronization for Core 1 via Pico SDK mailbox registers; inter-core FIFO interrupt handler.
- **K A** tests/platforms/cortexm-pico2/: Platform definitions and OpenOCD SWD flashing scripts (run-hw.sh).

## 2.5 Complete Apply for Phases C/D (equivalent to scripts/smp)

Phases C/D are not pure patches like Parts A/B: the integration **lifts whole trees from origin/smp, migrates the devices repository as a subtree, cuts port releases, and reconciles the baseline**. The operations below are exactly the ones scripts/smp performs; the complete file set and the acceptance gate are given for each PR so the phase can be reproduced without the scripts.

Repository keys:

|     |                                |     |                             |
|-----|--------------------------------|-----|-----------------------------|
| K   | = micro-os-plus-iii            | C   | = micro-os-plus-iii-cortexm |
| P   | = micro-os-plus-iii-posix-arch | A32 | = micro-os-plus-iii-aarch32 |
| A64 | = micro-os-plus-iii-aarch64    | DEV | = micro-os-plus-iii-devices |

| PR  | Operation                     | Repos          | Files                          | Acceptance gate                        |
|-----|-------------------------------|----------------|--------------------------------|----------------------------------------|
| #24 | release action                | P, C           | 2 (package.json)               | git tag -l shows v1.1.0, v1.2.0        |
| #25 | git subtree migration         | P, C, A32, A64 | drivers/*, soc/*               | legacy micro-os-plus::devices resolves |
| #26 | whole-tree lift               | C              | 7                              | 2xcortex-m33 port compiles             |
| #27 | whole-tree lift + model adapt | K, C           | 12                             | xpm run test-2xcortex-m33-cmake        |
| #28 | whole-tree lift               | K, A32, A64    | 51                             | fp-switch passes                       |
| #29 | add-only lift + build configs | K              | 25                             | xpm run test-smp-cmake                 |
| #30 | docs lift + render            | K              | 60                             | docs/render-pdfs.sh produces all PDFs  |
| #31 | baseline reconciliation       | all            | .github/**, README.md, LICENSE | CI metadata byte-identical to baseline |
| #32 | whole-tree lift               | K, C           | 139                            |                                        |

| PR | Operation | Repos | Files | Acceptance gate                |
|----|-----------|-------|-------|--------------------------------|
|    |           |       |       | cortexm-pico2 builds / flashes |

Canonical order (identical to scripts/smp)

The PRs are numbered **by theme, not by execution order**. Apply them as follows:

- 1. **Part 0 — first, on clean repositories.** PR #25: scripts/smp/migrate-devices.sh (devices subtree) plus the AArch CMakeLists.txt bridge. new-step.sh bases step/NN on the **local** baseline so the migration survives the branches.
- 2. **Parts A/B — PR #1 → #23** in numeric order: PR #1-#22 = steps 1-13 (applicable diffs), PR #23 = step 14 (collapsed Part B).
- 3. **C/D — grouped by operation, in this order:**
  - 1. **PR #24** — release-port.sh P v1.1.0, C v1.2.0
  - 2. **PR #27 (scaffolding)** — absorb-test-smpl.sh
  - 3. **PR #26, #27, #28, #30, #32** — integrate-cd.sh
  - 4. **PR #27/#29 (model adaptations)** — integrate-aarch-harness.sh
  - 5. **PR #29** — normalize-tests-package.sh
  - 6. **PR #31** — finalize.sh (step 30 reconcile)

**Warning — do not apply #1 ... #32 in numeric order.** PR #25 (Part 0) would run on a dirty tree and its git subtree add aborts with *"working tree has modifications"*; and integrate-aarch-harness.sh asserts on the missing micro-os-plus::test-support CMake anchor unless the modular kernel CMake has been lifted first. The order below is the one that reproduces the scripts exactly (verified: guide-driven tree == scripts/smp-driven tree, 0 diffs).

```
# 0) Part 0 – before anything else (clean repositories)
scripts/smp/migrate-devices.sh

# 1) Parts A/B – steps 1..13 + the collapsed Part B (step 14)

# 2) C/D – in this order:
scripts/smp/release-port.sh "$P" 1.1.0 "SMP host CPU model + highres clock"
scripts/smp/release-port.sh "$C" 1.2.0 "SMP Cortex-M port contract"
scripts/smp/absorb-test-smpl.sh # PR #27 scaffolding
scripts/smp/integrate-cd.sh # PR #26/#27/#28/#30/#32
lifts + test action
scripts/smp/integrate-aarch-harness.sh
# AArch model adaptations (PR #27/#29)
scripts/smp/normalize-tests-package.sh # deterministic tests/
package.json (PR #29)
scripts/smp/finalize.sh # Step 30 (merge / reconcile)
```

integrate-cd.sh and normalize-tests-package.sh are the two scripts added to close the previous C/D gaps (Cortex-M33/RP2350 ports, fp-switch, docs, Pico 2 and a byte-stable tests/package.json). With them, a guide-driven tree and a scripts/smp-driven tree are byte-identical across all five repositories.

PR #24 — Port package semantic releases

No source diff: the smp branch keeps the baseline versions. The release is cut by the same tool the scripts use.

```
scripts/smp/release-port.sh "$P" 1.1.0 "SMP host CPU model + highres
clock"
scripts/smp/release-port.sh "$C" 1.2.0 "SMP Cortex-M port contract"
# package.json:  posix-arch 1.0.1 -> 1.1.0 ;  cortexm 1.1.0 -> 1.2.0
```

## PR #25 — History-preserving devices dissolution

migrate-devices.sh folds micro-os-plus-iii-devices into the architecture repos with git subtree split + git subtree add, preserving all history:

```
scripts/smp/migrate-devices.sh
# DEV include -> A32/A64/P drivers/include ; src -> drivers/src ;
fatfs -> drivers/fatfs
# DEV soc/bcm2837 -> A32/A64 ; soc/rk3506 -> A32 ; soc/native -> P
# DEV soc/rp2350 -> C ; soc/stm32f4xx -> C
```

The AArch ports then keep the legacy target name resolvable:

```
git -C "$A32" checkout origin/smp -- CMakeLists.txt # conditional
devices bridge
git -C "$A64" checkout origin/smp -- CMakeLists.txt
```

## PR #26 — Modern multi-core silicon ports (Cortex-M33 & RP2350)

```
git -C "$C" checkout origin/smp -- \
include-m33 include-rp2350 \
src/rtos/os-core-m33.cpp src/rtos/os-core-rp2350.cpp
# 7 files: include-m33/{os-c-decls,os-decls,os-inlines}.h,
# include-rp2350/{os-c-decls,os-inlines}.h,
# src/rtos/os-core-m33.cpp, src/rtos/os-core-rp2350.cpp
```

## PR #27 — Modular add-only CMake & cross toolchains

```
git -C "$K" checkout origin/smp -- \
CMakeLists.txt cmake/uos-app.cmake cmake/toolchains tests/platforms/
2xcortex-m33
git -C "$C" checkout origin/smp -- CMakeLists.txt
# 11 K files (CMakeLists, cmake/toolchains/{arm-none-eabi,aarch64-
none-elf,native,
# uos-bare-metal-common}.cmake, cmake/uos-app.cmake, tests/platforms/
2xcortex-m33/**)
# + 1 C file (CMakeLists.txt: micro-os-plus::cortexm-qemu-m33)
# integrate-aarch-harness adaptation: bare-metal apps link ::iii-core,
not ::iii.
```

## PR #28 — Multi-core FPU stress test & scaffolding relocation

```
git -C "$K" checkout origin/smp -- \
tests/cmake/tests-main.cmake tests/sources/fp-switch tests/smp-
support
git -C "$A32" checkout origin/smp -- test/boards/rpi-zero-2w
git -C "$A64" checkout origin/smp -- test/boards/rpi-zero-2w
# K 5 files (tests-main.cmake, fp-switch/
{CMakeLists,README,include/...,src/main.cpp})
# A32/A64 23 files each (board.cmake, startup.S, mmu/smp/timer
sources, linker
# scripts, OpenOCD/qemu runners) – lifted, then repointed by absorb-
test-smpl.sh.
```

## PR #29 — Multi-architecture 4-core platforms & composite action

```
git -C "$K" checkout origin/smp -- \
tests/package.json \
```

```
tests/platforms/aarch32-rpi3b tests/platforms/aarch64-rpi3b \
tests/platforms/aarch32-rpi-zero-2w tests/platforms/aarch64-rpi-
zero-2w
# 25 files; integrate-aarch-harness.sh writes the AArch port
CMakeLists and injects
# the aarch32/aarch64 build configurations + test-smp-cmake aggregate
into package.json.
```

## PR #30 — Architectural documentation suite

```
git -C "$K" checkout origin/smp -- docs/render-pdfs.sh docs/smp-
integration
( cd "$K/docs" && ./render-pdfs.sh ) # PDFs are generated, not
patched
# 60 files: docs/smp-integration/**/*.md + rendered .pdf + docs/render-
pdfs.sh
```

## PR #31 — Final reconciliation & repository cleansing

```
for repo in "$K" "$P" "$C"; do
  git -C "$repo" merge --no-ff origin/xpack-development -m "merge:
reconcile baseline"
  git -C "$repo" checkout origin/xpack-development -- .github
  README.md LICENSE
done
rm -rf "$K/scripts/smp" # scripts/smp/finalize.sh
# gate: xpm run test-all (72/72) + xpm run test-smp-cmake
```

## PR #32 — Dedicated Raspberry Pi Pico 2 hardware platform

```
git -C "$K" checkout origin/smp -- tests/platforms/cortexm-pico2
git -C "$C" checkout origin/smp -- test/boards/pico2 test/pico2 \
src/rtos/os-core-rp2350.cpp
# K 5 files (platform CMake/deps/definitions/library/include)
# C 134 files (board.cmake, OpenOCD cfg, hw.sh, test/pico2/** apps,
os-core-rp2350)
# gate: xpm run build --config cortexm-pico2-cmake-gcc-debug
```

**Note:** The two heavy trees (C/test/pico2/\*\*  $\approx$  34k lines and K/docs/\*\* PDFs) are lifted/rendered rather than embedded as diffs; doing so is exactly what the scripts do and keeps the playbook usable. Every smaller C/D file is also carried as a unified diff below.

## 3. Modifications Organized by Subject / Theme / Domain

This cross-cutting index maps every single file modification across the entire integration into its primary functional domain.

### 1. Memory Management & Allocators

- M include/cmsis-plus/rtos/os-memory.h — Declares do\_usable\_size() and overflow-checked align\_size(). [S2 K]
- M include/cmsis-plus/memory/first-fit-top.h — first\_fit\_top::do\_usable\_size() declaration. [S2 K]
- M src/rtos/os-memory.cpp — Base memory\_resource::do\_usable\_size() default implementation. [S2 K]
- M src/memory/first-fit-top.cpp — Usable block size override + diagnostic pragma wrap (-Wcast-align, -Wunsafe-buffer-usage). [S2 K]
- M src/memory/lifo.cpp — LIFO pool usable size override. [S2 K]

- M src/memory/block-pool.cpp — Fixed block pool usable size override. [S2 K]
- M src/libc/stdlib/malloc.cpp — calloc() integer multiplication overflow guard. [S2 K]
- M include/cmsis-plus/estd/memory\_resource — polymorphic\_allocator::select\_on\_container\_copy\_construction() returns a copy, not a default-constructed allocator. [S2 K]

## 2. SMP Scheduler Core & Multi-Core Dispatch

- M include/cmsis-plus/rtos/os-sched.h — Per-CPU scheduler interface, current\_thread\_[OS\_NCPU], port lock. [S14 K]
- M src/rtos/os-core.cpp — Portable SMP scheduler: per-CPU ready lists, spinlock acquisition, IPI reschedule, no-switch-while-locked rule. [S14 K]
- M src/rtos/os-core.cpp — Port half of SMP scheduler (Cortex-M PendSV, POSIX signals). [S14 C] [S14 P]
- M src/rtos/os-idle.cpp — One dedicated idle thread per physical CPU core. [S14 K] (Single-core projection [S9 K])
- M src/rtos/os-main.cpp — Main thread initialization pinned to Core 0. [S14 K]
- M include/cmsis-plus/rtos/os-c-decls.h — Per-CPU C ABI fields and SMP\_NO\_OWNER definition. [S14 K]

## 3. Thread Lifecycle & State Machine

- M src/rtos/os-thread.cpp — Exported out-of-line this\_thread::suspend() definition. [S1 K]
- M include/cmsis-plus/rtos/os-thread.h — Handler-mode errno scratch storage. [S7 K]
- M include/cmsis-plus/rtos/os-thread.h & src/rtos/os-thread.cpp — state::destroying (7), atomic join(), detach(), idle reaper. [S9 K]
- M include/cmsis-plus/rtos/os-c-decls.h — os\_thread\_state\_destroying = 7 C ABI enum. [S9 K]
- M include/cmsis-plus/rtos/os-thread.h & src/rtos/os-thread.cpp — Full SMP thread lifecycle, CPU affinity mask, deferred stack switch. [S14 K]
- M src/libcpp/thread-cpp.h & include/cmsis-plus/estd/thread\_internal.h — std::thread::join() native handle barrier and functor lifetime retention. [S11 K]
- M src/rtos/os-thread.cpp — Cross-core IPI dispatch in thread::resume(). [S25 K]

## 4. Synchronization Primitives (Mutex & Condvar)

- M src/rtos/os-mutex.cpp — POSIX priority ceiling before ownership, dynamic boost tracking, unlock recomputation, owner caching. [S8 K]
- M src/rtos/os-condvar.cpp & include/cmsis-plus/rtos/os-condvar.h — Atomic wait list enqueue, mutex release, sleep sequence; attribute clock binding. [S10 K]
- M include/cmsis-plus/estd/condition\_variable — std::condition\_variable::wait\_for() trusts the kernel's timed\_wait() verdict instead of re-deriving the timeout from the wall clock. [S10 K]
- M include/cmsis-plus/rtos/os-c-decls.h — os\_condvar\_t void\* clock member alignment. [S10 K]

## 5. Timers & Message Queues

- M src/rtos/os-timer.cpp — Invocation of software timer callbacks outside critical sections; catch-up re-arming. [S7 K]
- M src/rtos/internal/os-lists.cpp — Expiration queue list manipulation for timer callbacks. [S7 K]

- M src/rtos/os-mqueue.cpp — Preemptive reschedule after message queue send/receive wakes a thread. [S12 K]

## 6. Clocks & Timestamps

- M include/cmsis-plus/rtos/os-decls.h — Declares port::clock\_highres::has\_hardware\_counter(). [S13 K]
- M src/rtos/os-clocks.cpp — Invokes high-resolution hardware counter. [S13 K]
- M include/cmsis-plus/rtos/port/os-inlines.h — has\_hardware\_counter() -> false, hardware\_counter() -> 0, SysTick ICSR PENDSTSET fix. [S13 C]
- M include/cmsis-plus/rtos/port/os-inlines.h — Host CLOCK\_MONOTONIC highres timer implementation. [S13 P]

## 7. POSIX I/O & File Descriptors

- M include/cmsis-plus/posix/dirent.h — Adds int reserved; to empty DIR struct; prevents double-declaration with glibc. [S1 K]
- M include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h — Maps CMSIS\_PLUS\_POSIX\_IO\_RW\_RETURN\_TYPE to newlib \_READ\_WRITE\_RETURN\_TYPE. [S1 K]
- M src/posix-io/file-descriptors-manager.cpp — Mutex protection for file descriptor allocation table (+ Clang -Wunsafe-buffer-usage). [S5 K]
- M include/cmsis-plus/posix-io/file-system.h & include/cmsis-plus/posix-io/net-stack.h — Mount list and net-stack mutex protection. [S5 K]
- M src/posix-io/block-device.cpp — Corrects inverted zero-size partition check. [S5 K]

## 8. C / C++ Runtime & Standards Conformance

- M include/cmsis-plus/utils/lists.h — Intrusive list iterator method call syntax conformance. [S1 K]
- K src/libc/stdlib/timegm.c — Baseline kept to avoid missing prototype errors on glibc  $\geq 2.44$ . [S1 K]
- M src/libcpp/new.cpp — C++17 aligned operator new / delete. [S3 K]
- M src/libcpp/system-error.cpp — Static Meyers singleton error categories (+ Clang -Wexit-time-destructors). [S3 K]
- M src/libcpp/chrono.cpp — 64-bit integer overflow protection in tick duration calculation. [S3 K]

## 9. CMSIS-RTOS C Wrapper (v1 ABI)

- M src/rtos/os-c-wrapper.cpp — One-shot timer default, polymorphic deletion, 64-bit millisecond timeout scaling. [S4 K]
- M src/rtos/os-c-wrapper.cpp — Full SMP CPU affinity wrapper functions. [S14 K]

## 10. ARM Architecture Startup & Semihosting

- M include/cmsis-plus/arm/semihosting.h — ARMv8-M mainline/baseline architecture macro guards. [S6 K]
- M src/startup/exception-handlers.c — ARMv8-M mainline exception-handler guards (VTOR, fault dumps). [S6 K]
- M include/cmsis-plus/cortexm/exception-handlers.h — Declares SecureFault\_Handler (ARMv8-M TrustZone). [S6 K]
- M src/semihosting/c-syscalls-semihosting.cpp — Semihosting fstat() character device fallback. [S6 K]

## 11. Port SMP Contract (ABI Headers)

- M `include/cmsis-plus/rtos/port/os-c-decls.h`, `os-decls.h`, `os-inlines.h` — Word size, atomics, recursive port lock, per-CPU current thread. [S14 C] [S14 P]
- M (`POSIX os-inlines.h`) — Converts `struct timespec tp`; to `timespec tp`; (-Werror=redundant-tags, Fix A/B). [S14 P]
- M (`POSIX port headers`) — Injects Clang diagnostic suppression block (-Wc++98-compat-pedantic, -Wreserved-identifier, Fix I). [S14 P]

## 12. POSIX-Arch Host Emulation Runtime

- A `include/host_cpu.hpp` & `src/host_cpu.cpp` — Multi-threaded host CPU emulation (pthreads-as-cores, interval timers, signal IPI). [S14 P]
- A `include/exception_handler.hpp` & `src/exception_handler.cpp` — Signal dispatching, weak `g_core_stage[0S_NCPU]` synchronization array. [S14 P]
- A `src/free-store.cpp` — Host free-store allocator bring-up. [S14 P]
- A `include/hw_result.hpp` & `src/board-contract.cpp` — PASS/FAIL string protocol and compile-time board contract. [S14 P]

## 13. Silicon Support (RP2350, Cortex-M33, BCM2837, STM32F4)

- A `include-m33/` & `src/rtos/os-core-m33.cpp` — Cortex-M33 (ARMv8-M) port implementation. [S25 C]
- A `include-rp2350/` & `src/rtos/os-core-rp2350.cpp` — Raspberry Pi RP2350 dual-core port with SIO spinlock 0. [S25 C] [S31 C]
- A `soc/rp2350/` & `soc/stm32f4xx/` — RP2350 and STM32F4 CMSIS silicon headers moved from devices. [P0 C]
- A `soc/bcm2837/` & `soc/rk3506/` — BCM2837 (RPI 3B) and RK3506 silicon headers moved from devices. [P0 A32/A64]
- A `soc/native/` — POSIX host block device storage backend moved from devices. [P0 P]

## 14. Modular CMake Build Architecture

- M `CMakeLists.txt` — Additively registers host SMP source files in INTERFACE library. [S14 P]
- M `CMakeLists.txt` — Additively appends `micro-os-plus::cortexm-qemu-m33` target. [S26 C]
- A `cmake/toolchains/arm-none-eabi.cmake` & `aarch64-none-elf.cmake` — Cross-compilation toolchain profiles. [S26 K]
- A `cmake/uos-app.cmake` — Additive `uos_add_app()` helper macro. [S26 K]
- M `CMakeLists.txt` — Additive sub-targets `::iii-posix-io`, `::iii-semihosting`, `::iii-newlib-reent`, `::test-support`. [S28 K]
- M `CMakeLists.txt` — Alias shims for `micro-os-plus::devices`. [P0 P/C/A32/A64]

## 15. Shared SMP Test Scaffolding

- A `tests/smp-support/` — Dissolved `test_smpl/` directory containing `hw_result.hpp`, `board-contract.cpp`, `run-qemu.sh`, `run-host.sh`, `run-hw.sh`, OpenOCD scripts. [S27 K]
- A `tests/sources/fp-switch/` — Multi-threaded FPU context switch test suite. [S27 K]
- M `tests/cmake/tests-main.cmake` — Port-under-test selection by platform name (supersedes the `global-definitions.cmake` toggle). [S27 K]

## 16. Multi-Architecture Test Platforms & Harness

- A tests/platforms/native-smp/ — Genuine host dual-core platform (OS\_USE\_SMP\_SCHEDULER=1, OS\_NCPU=2). [S14/S28 K]
- A tests/platforms/2xcortex-m33/ — Dual Cortex-M33 QEMU platform (MPS2 AN521) reusing the generic device-qemu-cortexm device. [S26 K]
- A tests/platforms/aarch32-rpi3b & tests/platforms/aarch64-rpi3b — 4-core QEMU raspi3b platforms. [S28 K]
- A tests/platforms/cortexm-pico2 — Raspberry Pi Pico 2 hardware test platform. [S28/S31 K]
- M tests/cmake/tests-main.cmake — Additive platform dispatch logic. [S14/S28 K]
- M tests/package.json — Additive build configurations and actions (test-native-smp, test-2xcortex-m33-cmake, test-smp-cmake). [S14/S26/S28 K]

## 17. Diagnostics, Trace & Instrumentation

- M include/cmsis-plus/diag/instrumentation.h — Adds OS\_INTEGER\_INSTRUMENTATION\_SUSPEND\_CAUSE\_CONDVAR (12). [S10 K]

## 18. Semantic Port Versioning & Repository Releases

- M package.json — Bumps posix-arch to v1.1.0. [S24 P]
- M package.json — Bumps cortexm to v1.2.0. [S24 C]

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## Appendix — Full Diffs, Code Listings & Pull Request Rationale

Generated directly from the verified integration commits. This appendix is structured around the **7 Progressive Pull Requests (PR #1 through PR #7)**, providing the deep technical rationale ("WHY") for each PR as a standalone unit, followed by the exact code diffs and listings.

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### Appendix Section 1 — Pull Request #1: Uniprocessor Correctness & Toolchain Hardening (Steps 1-13)

#### Why PR #1 is Needed as a Standalone Pull Request

- 1. Isolation of Single-Core Defect Fixes from SMP Complexity:**
  - Upstream xpack-development is a mature uniprocessor RTOS. Conflating essential bug fixes with multi-core architectural rewrites increases review friction and risks introducing subtle regressions. PR #1 isolates 13 critical single-core fixes, allowing upstream maintainers to review and merge them with complete confidence.
- 2. Elimination of Critical Concurrency Hazards & Security Vulnerabilities:**
  - **Heap Integer Overflow:** Fixes calloc() in src/libc/stdlib/malloc.cpp where integer multiplication truncation (nelem \* elbytes) permitted undersized buffer allocations, leading to heap buffer corruption.
  - **Condition Variable Lost Signals:** Solves the classic lost-signal race in src/rtos/os-condvar.cpp by atomically enqueueing the calling thread into the wait queue under scheduler lock **before** releasing the associated mutex.

- **Mutex Priority Inversion:** Implements POSIX priority ceiling protocol in `src/rtos/os-mutex.cpp`, elevating caller priority before ownership transfer and dynamically recomputing priority on unlock.
  - **Timer Deadlock Elimination:** Software timer callbacks in `src/rtos/os-timer.cpp` now execute **outside the scheduler critical section**, enabling callbacks to safely acquire mutexes or invoke blocking RTOS APIs.
  - **Thread Destruction Race Conditions:** Introduces `state::destroying (7)` and `atomic join() / detach()` in `src/rtos/os-thread.cpp`, allowing the idle thread reaper to reclaim dynamic stacks and TCBs safely.
- 3. Strict ISO C/C++ and Toolchain Standards Conformance:**
- Fixes empty struct violation in `include/cmsis-plus/posix/dirent.h` (violates ISO C99/C11 §6.7.2.1).
  - Implements C++17 aligned allocation operators `operator new(size_t, align_val_t)` in `src/libcpp/new.cpp`.
  - Protects `src/libcpp/system-error.cpp` Meyers singletons against Clang `-Wexit-time-destructors`.
  - Protects `src/libcpp/chrono.cpp` tick conversions against 64-bit integer multiplication overflow.
  - Preserves baseline `src/libc/stdlib/timemgm.c` to prevent prototype visibility errors on `glibc ≥ 2.44` under `-Werror=missing-prototypes`.
- 4. Verification Gate:**
- **100% Green on Uniprocessor Baseline:** All 72 tests across 24 build configurations pass with `-Werror` enabled across GCC and Clang toolchains. Zero SMP macro leakage.

## PR #1: POSIX dirent.h ISO C Empty Struct Conformance

(Followed by PR #2: `lists.h`, PR #3: `c-syscalls-aliases-standard.h`, PR #4: `os-thread.cpp` export)

## PR #1 Details: ISO C dirent, list iterators, exported suspend (timemgm kept baseline)

### Why Step 1 is Needed:

- **ISO C99/C11 §6.7.2.1 Conformance & -Wextra-semi:** In `include/cmsis-plus/posix/dirent.h`, the empty struct definition `typedef struct { ; } DIR`; violated ISO C standard constraints requiring at least one named member and caused `-Wextra-semi` compiler diagnostics. Adding `int reserved`; fixes syntax validity across GCC and Clang.
- **ISO C Conformance & Strict Compiler Flags:** Replaces the empty struct declaration `; with int reserved`; to eliminate `-Wextra-semi` diagnostics under strict compiler flags without modifying system headers.
- **Intrusive List Iterator Method Semantics:** In `include/cmsis-plus/utils/lists.h`, iterator post-increment and post-decrement operators were accessing `node_->next / node_->prev` directly instead of invoking member accessor methods `node_->next() / node_->prev()`, breaking iterator concepts under modern C++ standards.
- **Symbol Export for C Wrapper:** In `src/rtos/os-thread.cpp`, removing `inline` from the out-of-line `this_thread::suspend()` definition allows the compiler to generate a linkable external symbol required by the CMSIS-RTOS C API wrapper.
- **glibc ≥ 2.44 Missing Prototype Safety (timemgm.c):** The `smp` branch hid `timemgm()` behind `#if !defined(__GLIBC__)`, which strips the prototype when `<time.h>` hides it under strict flags, causing

-Werror=missing-prototypes failures on glibc  $\geq 2.44$ . Deliberately keeping baseline timegm.c ensures 100% portability.

```
diff --git a/include/cmsis-plus/posix/dirent.h b/include/cmsis-plus/
posix/dirent.h
index 1f0f0dbe..19a5165c 100644
--- a/include/cmsis-plus/posix/dirent.h
+++ b/include/cmsis-plus/posix/dirent.h
@@ -54,7 +54,8 @@ extern "C"
    // and casted to DIR.
    typedef struct
    {
-       ;
+       int reserved; /* C forbids an empty struct; a named member also
+           avoids
+
+           -Wextra-semi on the old `;` null declaration. */
    } DIR;

    // -----
diff --git a/include/cmsis-plus/utils/lists.h b/include/cmsis-plus/
utils/lists.h
index b8c642d9..f80da4dd 100644
--- a/include/cmsis-plus/utils/lists.h
+++ b/include/cmsis-plus/utils/lists.h
@@ -917,7 +917,7 @@ namespace os
    double_list_iterator<T, N, MP, U>::operator++ (int)
    {
        const auto tmp = *this;
-       node_ = static_cast<iterator_pointer> (node_->next);
+       node_ = static_cast<iterator_pointer> (node_->next ());
        return tmp;
    }

@@ -925,7 +925,7 @@ namespace os
    inline double_list_iterator<T, N, MP, U>&
    double_list_iterator<T, N, MP, U>::operator-- ()
    {
-       node_ = static_cast<iterator_pointer> (node_->prev);
+       node_ = static_cast<iterator_pointer> (node_->prev ());
        return *this;
    }

@@ -934,7 +934,7 @@ namespace os
    double_list_iterator<T, N, MP, U>::operator-- (int)
    {
        const auto tmp = *this;
-       node_ = static_cast<iterator_pointer> (node_->prev);
+       node_ = static_cast<iterator_pointer> (node_->prev ());
        return tmp;
    }

diff --git a/include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h
b/include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h
index 7a181156..6e9bcfe4 100644
--- a/include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h
+++ b/include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h
@@ -17,6 +17,16 @@
@@ -17,6 +17,16 @@
#ifdef __cplusplus
extern "C"
{
+}
+
+ // newlib declares read()/write() with its own return type
+ // (_READ_WRITE_RETURN_TYPE, which is `int` on aarch64-none-elf and
+ // `_ssize_t` on arm-none-eabi). Match it, or the alias below
+ // conflicts
+ // with the system declaration.
+ #if defined(_READ_WRITE_RETURN_TYPE)
```

```

+define CMSIS_PLUS_POSIX_IO_RW_RETURN_TYPE_READ_WRITE_RETURN_TYPE
+else
+define CMSIS_PLUS_POSIX_IO_RW_RETURN_TYPE ssize_t
+endif

// -----
@@ -130,7 +140,7 @@ extern "C"
int __attribute__((weak, alias ("__posix_raise")))
raise (int sig);

- ssize_t __attribute__((weak, alias ("__posix_read")))
+ CMSIS_PLUS_POSIX_IO_RW_RETURN_TYPE __attribute__((weak, alias
("__posix_read")))
read (int fildes, void* buf, size_t nbyte);

struct dirent* __attribute__((weak, alias ("__posix_readdir")))
@@ -247,7 +257,7 @@ extern "C"
pid_t __attribute__((weak, alias ("__posix_wait")))
wait (int* stat_loc);

- ssize_t __attribute__((weak, alias ("__posix_write")))
+ CMSIS_PLUS_POSIX_IO_RW_RETURN_TYPE __attribute__((weak, alias
("__posix_write")))
write (int fildes, const void* buf, size_t nbyte);

ssize_t __attribute__((weak, alias ("__posix_writew")))
diff --git a/src/rtos/os-thread.cpp b/src/rtos/os-thread.cpp
index 6476782f..3b8f293a 100644
--- a/src/rtos/os-thread.cpp
+++ b/src/rtos/os-thread.cpp
@@ -1837,7 +1837,7 @@ namespace os
*
* @warning Cannot be invoked from Interrupt Service Routines.
*/
- inline void
+ void
suspend (void)
{
os::instrumentation::thread::suspend (_thread ());

```

## PR #5: Memory Allocator Usable Size & Integer Overflow Protection

### Why Step 2 is Needed:

- **Memory Allocator Interface Completeness:** In `include/cmsis-plus/rtos/os-memory.h` and `include/cmsis-plus/memory/first-fit-top.h`, `do_usable_size(void*)` is declared in the `memory_resource` abstract base class and overridden in `first_fit_top`, `lifo`, and `block_pool`. Landing header declarations, base implementations, and overrides simultaneously prevents undefined reference to `memory_resource::do_usable_size` link errors.
- **Integer Multiplication Overflow Security in `calloc()`:** In `src/libc/stdlib/malloc.cpp`, `calloc()` calculates `nelem * elbytes`. Without overflow checks, integer wrap-around could allocate a smaller buffer than requested, causing severe heap buffer overflows. Step 2 adds `if (nelem != 0 && elbytes > (SIZE_MAX / nelem)) { errno = ENOMEM; return nullptr; }`.
- **Clang & ARM GCC Diagnostics Hardening:** Pointer arithmetic in `src/memory/first-fit-top.cpp` triggers `-Wcast-align` on ARM GCC and `-Wunsafe-buffer-usage` on Clang `-Weverything`. Wrapping `first_fit_top::do_usable_size()` with diagnostic pragmas allows strict build gates without lowering compiler security levels.

```

diff --git a/include/cmsis-plus/memory/first-fit-top.h b/include/
cmsis-plus/memory/first-fit-top.h
index 39ac2791..c3942395 100644
--- a/include/cmsis-plus/memory/first-fit-top.h
+++ b/include/cmsis-plus/memory/first-fit-top.h
@@ -213,6 +213,14 @@ namespace os
    virtual std::size_t
    do_max_size (void) const noexcept override;

+
+ /**
+  * @brief Implementation of the function to get the usable
+  * block size.
+  * @param [in] addr Address of a previously allocated block.
+  * @return Number of usable bytes from `addr` to the end of its
+  * chunk.
+  */
+ virtual std::size_t
+ do_usable_size (void* addr) const noexcept override;
+
+ /**
+  * @brief Implementation of the function to reset the memory
+  * manager.
+  * @par Parameters
diff --git a/include/cmsis-plus/rtos/os-memory.h b/include/cmsis-plus/
rtos/os-memory.h
index 3d808899..3f567594 100644
--- a/include/cmsis-plus/rtos/os-memory.h
+++ b/include/cmsis-plus/rtos/os-memory.h
@@ -84,6 +84,10 @@ namespace os
    constexpr std::size_t
    align_size (std::size_t size, std::size_t align) noexcept
    {
+
+     if (size > static_cast<std::size_t> (-1) - (align - 1L))
+     {
+         return static_cast<std::size_t> (-1);
+     }
+     return ((size) + (align)-1L) & ~((align)-1L);
    }

@@ -274,6 +278,21 @@ namespace os
    std::size_t
    max_size (void) const noexcept;

+
+ /**
+  * @brief Get the usable size of a previously allocated
+  * block.
+  * @param addr Address of a block returned by `allocate()`.
+  * @return Number of usable bytes, or 0 if the size is
+  * unknown.
+  *
+  * @details
+  * This is an extension used by `realloc()` to copy at most
+  * the
+  * old contents; the allocator is the only one who knows the
+  * real
+  * size of a block when the caller passed 0 to
+  * `deallocate()`.
+  *
+  * @see do_usable_size();
+  */
+ std::size_t
+ usable_size (void* addr) const noexcept;
+
+ /**
+  * @brief Set the out of memory handler.
+  * @param handler Pointer to new handler.
diff --git a/include/cmsis-plus/rtos/os-memory.h b/include/cmsis-plus/
rtos/os-memory.h
index 3d808899..3f567594 100644
--- a/include/cmsis-plus/rtos/os-memory.h
+++ b/include/cmsis-plus/rtos/os-memory.h
@@ -422,6 +441,18 @@ namespace os
    virtual std::size_t
    do_max_size (void) const noexcept;

```

```

+     /**
+     * @brief Implementation of the function to get the usable
+     block size.
+     * @param addr Address of a previously allocated block.
+     * @return Number of usable bytes, or 0 if the size is
+     unknown.
+     *
+     * @details
+     * The default implementation returns 0 (unknown); allocators
+     that
+     * keep the block size, like `first_fit_top`, override it.
+     */
+     virtual std::size_t
+     do_usable_size (void* addr) const noexcept;
+
+     /**
+     * @brief Implementation of the function to reset the memory
+     manager.
+     * @par Parameters
@@ -1342,6 +1373,15 @@ namespace os
+     return do_max_size ();
+     }
+
+     /**
+     * @see do_usable_size();
+     */
+     inline std::size_t
+     memory_resource::usable_size (void* addr) const noexcept
+     {
+         return do_usable_size (addr);
+     }
+
+     /**
+     * @see do_reset();
+     */
diff --git a/src/libc/stdlib/malloc.cpp b/src/libc/stdlib/malloc.cpp
index 9ff93d07..91ffcc4a 100644
--- a/src/libc/stdlib/malloc.cpp
+++ b/src/libc/stdlib/malloc.cpp
@@ -160,6 +160,17 @@ @@ calloc (size_t nelem, size_t elbytes)
+     return nullptr;
+     }
+
+ // Reject requests whose total size would overflow `size_t`.
+ // Without this
+ // guard the product wraps to a small value (e.g. on a 32-bit
+ // target
+ // `calloc(0x10001, 0x10000)` becomes 0) and a tiny block is
+ // returned while
+ // the caller believes the full, huge array was allocated.
+ // `elbytes` is
+ // known non-zero here, so the division is safe.
+ if (nelem > static_cast<std::size_t> (-1) / elbytes)
+     {
+         errno = ENOMEM;
+         return nullptr;
+     }
+
+ void* mem;
+ {
+     // ----- Begin of critical section
+     -----
@@ -294,7 +305,16 @@ @@ realloc (void* ptr, size_t bytes)
+     mem = std::pmr::get_default_resource ()->allocate (bytes);
+     if (mem != nullptr)
+     {
-         memcpy (mem, ptr, bytes);
+         // POSIX requires the contents to be preserved "up to the
+         lesser of

```

```

+         // the new and old sizes". The old size is queried from the
+         allocator
+         // (the bare-metal default resource is a `first_fit_top`,
+         which keeps
+         // it in the chunk header). Copying `bytes` unconditionally
+         would read
+         // past the end of the old block whenever the new size is
+         larger,
+         // i.e. on every growing `realloc()`.
+         std::size_t old_bytes
+             = estd::pmr::get_default_resource ()->usable_size (ptr);
+         std::size_t copy_bytes = (bytes < old_bytes) ? bytes :
+         old_bytes;
+         memcpy (mem, ptr, copy_bytes);
+         estd::pmr::get_default_resource ()->deallocate (ptr, 0);
+     }
+     else
diff --git a/src/memory/block-pool.cpp b/src/memory/block-pool.cpp
index b3f583d1..a2432310 100644
--- a/src/memory/block-pool.cpp
+++ b/src/memory/block-pool.cpp
@@ -168,7 +168,7 @@ namespace os
        pool_addr_, align_sz);

        // std::align() will fail if it cannot fit the adjusted block
        size.
-       if (res != nullptr)
+       if (res == nullptr)
        {
            assert (res != nullptr);
        }
diff --git a/src/memory/first-fit-top.cpp b/src/memory/first-fit-
top.cpp
index 09478a00..cce94591 100644
--- a/src/memory/first-fit-top.cpp
+++ b/src/memory/first-fit-top.cpp
@@ -122,13 +122,32 @@ namespace os
    {
        using namespace os;

+
+       if (bytes > total_bytes_)
+       {
+           return nullptr;
+       }
+
        std::size_t block_padding = calc_block_padding (alignment);
        std::size_t alloc_size = rtos::memory::align_size (bytes,
            chunk_align);
+       if (alloc_size == static_cast<std::size_t> (-1)
+           || alloc_size > total_bytes_)
+       {
+           return nullptr;
+       }
        alloc_size += block_padding;
        alloc_size += chunk_offset;
+       if (alloc_size > total_bytes_)
+       {
+           return nullptr;
+       }

        std::size_t block_minchunk = calc_block_minchunk
            (block_padding);
        alloc_size = os::rtos::memory::max (alloc_size,
            block_minchunk);
+       if (alloc_size > total_bytes_)
+       {
+           return nullptr;
+       }
    }

```

```

        chunk_t* chunk;

@@ -428,6 +447,37 @@ namespace os
        return total_bytes_;
    }

#pragma GCC diagnostic push
#pragma GCC diagnostic ignored "-Wcast-align"
#ifdef __clang__
#pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
#endif
+    std::size_t
+    first_fit_top::do_usable_size (void* addr) const noexcept
+    {
+        // Recover the chunk header placed immediately before the
+        // payload.
+        chunk_t* chunk = reinterpret_cast<chunk_t*> (static_cast<char*>
+            (addr)
+
+                - chunk_offset);
+
+        // For an aligned block, the adjusted header stores the
+        // (negative)
+        // alignment offset as its size; step back to the real chunk
+        // header
+        // exactly as do_deallocate() does.
+        if (static_cast<std::ptrdiff_t> (chunk->size) < 0)
+        {
+            chunk = reinterpret_cast<chunk_t*> (
+                reinterpret_cast<char*> (chunk)
+                + static_cast<std::ptrdiff_t> (chunk->size));
+        }
+
+        // Usable bytes are those from the given address to the end of
+        // the
+        // chunk; this naturally accounts for any alignment slack
+        // between the
+        // chunk payload and the returned pointer.
+        return static_cast<std::size_t> (
+            reinterpret_cast<char*> (chunk) + chunk->size
+            - static_cast<char*> (addr));
+    }
#pragma GCC diagnostic pop
+
+    void*
+    first_fit_top::internal_align_ (chunk_t* chunk, std::size_t
+        bytes,
+
+            std::size_t alignment)

diff --git a/src/memory/lifo.cpp b/src/memory/lifo.cpp
index 3519fb3e..9924fbf2 100644
--- a/src/memory/lifo.cpp
+++ b/src/memory/lifo.cpp
@@ -70,13 +70,31 @@ namespace os
    void*
    lifo::do_allocate (std::size_t bytes, std::size_t alignment)
    {
+        if (bytes > total_bytes_)
+        {
+            return nullptr;
+        }
+
+        std::size_t block_padding = calc_block_padding (alignment);
+        std::size_t alloc_size = rtos::memory::align_size (bytes,
+            chunk_align);
+        if (alloc_size == static_cast<std::size_t> (-1)
+            || alloc_size > total_bytes_)
+        {
+            return nullptr;
+        }
+        alloc_size += block_padding;

```

```

        alloc_size += chunk_offset;
+       if (alloc_size > total_bytes_)
+       {
+           return nullptr;
+       }

        std::size_t block_minchunk = calc_block_minchunk
            (block_padding);
        alloc_size = os::rtos::memory::max (alloc_size,
            block_minchunk);
+       if (alloc_size > total_bytes_)
+       {
+           return nullptr;
+       }

        chunk_t* chunk = nullptr;

@@ -126,6 +144,14 @@ namespace os
        // If this was the last chunk, the free list is
        empty.
        }
    }
+   else
+   {
+       // The head chunk is smaller than the request, so
+       this arena
+       // cannot satisfy it. Clear `chunk` so the loop
+       below falls
+       // through to the out-of-memory handler (or returns
+       nullptr)
+       // instead of handing the too-small chunk to
+       internal_align().
+       chunk = nullptr;
+   }
    }

    if (chunk != nullptr)
diff --git a/src/rtos/os-memory.cpp b/src/rtos/os-memory.cpp
index 7b3c9baa..d5f89855 100644
--- a/src/rtos/os-memory.cpp
+++ b/src/rtos/os-memory.cpp
@@ -438,6 +438,23 @@ namespace os
    return 0;
}

+ /**
+  * @details
+  * The default implementation of this virtual function returns
+  * zero, meaning the usable size is not known.
+  *
+  * Override this function to return the actual size.
+  *
+  * @par Standard compliance
+  * Extension to standard.
+  */
+ std::size_t
+ memory_resource::do_usable_size (void* addr) const noexcept
+ {
+     static_cast<void> (addr);
+     return 0;
+ }

+ /**
+  * @details
+  * The default implementation of this virtual function

```

```

diff --git a/include/cmsis-plus/estd/memory_resource b/include/cmsis-
plus/estd/memory_resource
index befafb6c..5beb6427 100644

```

```

--- a/include/cmsis-plus/estd/memory_resource
+++ b/include/cmsis-plus/estd/memory_resource
@@ -243,7 +243,10 @@ namespace os
    polymorphic_allocator<T>::select_on_container_copy_construction (
        void) const noexcept
    {
-       return polymorphic_allocator ();
+       // C++17 requires a copy of the allocator, not a default-
+       // constructed
+       // one; returning a default allocator would silently rebind a
+       // copied
+       // container to the global resource instead of its own.
+       return *this;
    }

    template <typename T>

```

## PR #6: ISO C++17 Aligned Memory Allocation Operators (and PR #7: Singletons, PR #8: Chrono)

### Why Step 3 is Needed:

- **ISO C++17 Aligned Allocation:** In `src/libcpp/new.cpp`, modern C++17 standard requires support for over-aligned types (`std::align_val_t`). Step 3 implements operator `new(std::size_t, std::align_val_t)` and operator `delete(void*, std::align_val_t)` routing to the underlying RTOS memory manager.
- **Static Storage Duration & Exit Destructors:** In `src/libcpp/system-error.cpp`, error categories (`get_system_error_category()`, `get_cmsis_error_category()`) are implemented as function-local static instances ensuring they outlive thrown exception objects without dangling pointers. Adding `#pragma clang diagnostic ignored "-Wexit-time-destructors"` prevents Clang warnings regarding destructor registration at exit.
- **64-Bit Integer Duration Overflow Prevention:** In `src/libcpp/chrono.cpp`, calculating nanoseconds from high-frequency hardware cycles via  $(\text{cycles} * 1e9) / \text{freq}$  overflows 64-bit integers after moderate uptime. Step 3 rewrites the arithmetic as  $(\text{cycles} / \text{freq}) * 1e9 + ((\text{cycles} \% \text{freq}) * 1e9) / \text{freq}$ , guaranteeing overflow-free calculations across years of uptime.

```

diff --git a/src/libcpp/chrono.cpp b/src/libcpp/chrono.cpp
index dfbe1829..6246956a 100644
--- a/src/libcpp/chrono.cpp
+++ b/src/libcpp/chrono.cpp
@@ -113,13 +113,15 @@ namespace os
    #endif
        // The duration is the number of sum of SysTick ticks plus
        // the current
        // count of CPU cycles (computed from the SysTick counter).
-       // Notice: a more exact solution would be to compute
-       // ticks * divisor + cycles, but this severely reduces the
-       // range of ticks.
+       // Decompose into seconds and remainder to avoid 64-bit
+       // overflow
+       // on cycles * 1e9 after ~13 minutes.
+       uint64_t freq = rtos::hrclock.input_clock_frequency_hz ();
+       uint64_t sec = cycles / freq;
+       uint64_t rem = cycles % freq;
+       uint64_t ns = sec * 1000000000ULL + (rem * 1000000000ULL) /
+       freq;
        return time_point{
            duration{
-               duration{ cycles * 1000000000ULL
-                   / rtos::hrclock.input_clock_frequency_hz () }
+               duration{ ns }
            }
        }
    }

```

```

        + realtime_clock::startup_time_point.time_since_epoch
    () } //
    };

#pragma GCC diagnostic pop
diff --git a/src/libcpp/new.cpp b/src/libcpp/new.cpp
index 098df1d9..5203ee12 100644
--- a/src/libcpp/new.cpp
+++ b/src/libcpp/new.cpp
@@ -288,6 +288,135 @@ void* __attribute__((weak)) operator new[]
    (std::size_t bytes,

// -----

+/**
+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Allocate over-aligned space for a new object instance.
+ * @param bytes Number of bytes to allocate.
+ * @param alignment Required alignment (C++17 `std::align_val_t`).
+ * @return Pointer to aligned allocated object.
+ *
+ * @details
+ * C++17 calls this overload for types whose alignment exceeds
+ * `alignof(std::max_align_t)`. The requested alignment is passed
+ * through to
+ * the RTOS default memory resource; the bare-metal `first_fit_top`
+ * honours
+ * it, whereas the plain `size_t` overload only promises `max_align`.
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.
+ */
+void* __attribute__((weak))
+operator new (std::size_t bytes, std::align_val_t alignment)
+{
+    assert (!rtos::interrupts::in_handler_mode ());
+    if (bytes == 0)
+    {
+        bytes = 1;
+    }

+    // ----- Begin of critical section
+    -----
+    rtos::scheduler::critical_section scs;

+    while (true)
+    {
+        void* mem = estd::pmr::get_default_resource ()->allocate (
+            bytes, static_cast<std::size_t> (alignment));

+        if (mem != nullptr)
+        {
+            return mem;
+        }

+        if (new_handler_)
+        {
+            new_handler_ ();
+        }
+        else
+        {
+            estd::__throw_bad_alloc ();
+        }
+    }

+    // ----- End of critical section
+    -----
+}
+
+/**

```

```

+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Allocate over-aligned space for a new object instance
+ *        (nothrow).
+ * @param bytes Number of bytes to allocate.
+ * @param alignment Required alignment (C++17 `std::align_val_t`).
+ * @param nothrow (unused)
+ * @return Pointer to aligned allocated object or nullptr.
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.
+ */
+void* __attribute__((weak))
+operator new (std::size_t bytes, std::align_val_t alignment,
+             const std::nothrow_t& nothrow __attribute__((unused)))
+             noexcept
+{
+    assert (!rtos::interrupts::in_handler_mode ());
+    if (bytes == 0)
+    {
+        bytes = 1;
+    }
+
+    // ----- Begin of critical section
+    -----
+    rtos::scheduler::critical_section scs;
+
+    while (true)
+    {
+        void* mem = estd::pmr::get_default_resource ()->allocate (
+            bytes, static_cast<std::size_t> (alignment));
+
+        if (mem != nullptr)
+        {
+            return mem;
+        }
+
+        if (new_handler_)
+        {
+            new_handler_ ();
+        }
+        else
+        {
+            break; // return nullptr
+        }
+    }
+
+    // ----- End of critical section
+    -----
+
+    return nullptr;
+}
+
+/**
+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Allocate over-aligned space for an array of new object
+ *        instances.
+ * @param bytes Number of bytes to allocate.
+ * @param alignment Required alignment (C++17 `std::align_val_t`).
+ * @return Pointer to aligned allocated object.
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.
+ */
+void* __attribute__((weak))
+operator new[] (std::size_t bytes, std::align_val_t alignment)
+{
+    return ::operator new (bytes, alignment);
+}
+
+/**

```

```

+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Allocate over-aligned space for an array of new object
+ * instances (nothrow).
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.
+ */
+void* __attribute__ ((weak))
+operator new[] (std::size_t bytes, std::align_val_t alignment,
+               const std::nothrow_t& nothrow __attribute__
+               ((unused))) noexcept
+{
+    return ::operator new (bytes, alignment, std::nothrow);
+}
+
+// -----
+
+/**
+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Deallocate the dynamically allocated object instance.
+@@ -516,6 +645,116 @@ void __attribute__ ((weak)) operator delete[]
+    (void* ptr, const std::nothrow_t
+    ::operator delete (ptr, nothrow);
+}
+
+// -----
+
+/**
+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Deallocate an over-aligned dynamically allocated object.
+ * @param ptr Pointer to object.
+ * @param alignment Required alignment (C++17 `std::align_val_t`).
+ * @par Returns
+ * Nothing.
+ *
+ * @details
+ * Counterpart of `operator new(std::size_t, std::align_val_t)`.
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.
+ */
+void __attribute__ ((weak))
+operator delete (void* ptr, std::align_val_t alignment) noexcept
+{
+    assert (!rtos::interrupts::in_handler_mode ());
+
+    if (ptr)
+    {
+        // ----- Begin of critical section
+        -----
+        rtos::scheduler::critical_section scs;
+
+        estd::pmr::get_default_resource ()->deallocate (
+            ptr, 0, static_cast<std::size_t> (alignment));
+        // ----- End of critical section
+        -----
+    }
+}
+
+/**
+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Deallocate an over-aligned dynamically allocated object
+    (sized).
+ * @param ptr Pointer to object.
+ * @param bytes Number of bytes to deallocate.
+ * @param alignment Required alignment (C++17 `std::align_val_t`).
+ * @par Returns
+ * Nothing.
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.

```

```

+ */
+void __attribute__ ((weak))
+operator delete (void* ptr, std::size_t bytes,
+                std::align_val_t alignment) noexcept
+{
+    assert (!rtos::interrupts::in_handler_mode ());
+
+    if (ptr)
+    {
+        // ----- Begin of critical section
+        -----
+        rtos::scheduler::critical_section scs;
+
+        estd::pmr::get_default_resource ()->deallocate (
+            ptr, bytes, static_cast<std::size_t> (alignment));
+        // ----- End of critical section
+        -----
+    }
+}
+
+/**
+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Deallocate an over-aligned dynamically allocated array.
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.
+ */
+void __attribute__ ((weak))
+operator delete[] (void* ptr, std::align_val_t alignment) noexcept
+{
+    ::operator delete (ptr, alignment);
+}
+
+/**
+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Deallocate an over-aligned dynamically allocated array
+ *        (sized).
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.
+ */
+void __attribute__ ((weak))
+operator delete[] (void* ptr, std::size_t bytes,
+                  std::align_val_t alignment) noexcept
+{
+    ::operator delete (ptr, bytes, alignment);
+}
+
+/**
+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Deallocate an over-aligned dynamically allocated object
+ *        (nothrow).
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.
+ */
+void __attribute__ ((weak))
+operator delete (void* ptr, std::align_val_t alignment,
+                const std::nothrow_t& nothrow __attribute__
+                ((unused))) noexcept
+{
+    ::operator delete (ptr, alignment);
+}
+
+/**
+ * @ingroup cmsis-plus-rtos-memres
+ * @brief Deallocate an over-aligned dynamically allocated array
+ *        (nothrow).
+ *
+ * @warning Cannot be invoked from Interrupt Service Routines.
+ */

```

```

+void __attribute__ ((weak))
+operator delete[] (void* ptr, std::align_val_t alignment,
+                  const std::nothrow_t& nothrow __attribute__
+                    ((unused))) noexcept
+{
+    ::operator delete[] (ptr, alignment);
+}
+
+// error: end of file with unbalanced grouping commands
+/*
+ * @}
diff --git a/src/libcpp/system-error.cpp b/src/libcpp/system-error.cpp
index c11ad9f0..dd7bf829 100644
--- a/src/libcpp/system-error.cpp
+++ b/src/libcpp/system-error.cpp
@@ -103,17 +103,33 @@ namespace os

#pragma GCC diagnostic pop

+    // `std::error_code` stores a pointer to its `error_category`,
+    // and the
+    // `error_code` (via the thrown `std::system_error`) outlives the
+    // throw
+    // expression. The category must therefore have static storage
+    // duration;
+    // a temporary would dangle. Function-local statics are
+    // constructed on
+    // first use and live until exit.
+    static const std::error_category&
+    get_system_error_category (void) noexcept
+    {
+        static const system_error_category category;
+        return category;
+    }
+
+    static const std::error_category&
+    get_cmsis_error_category (void) noexcept
+    {
+        static const cmsis_error_category category;
+        return category;
+    }
+
+endif

    void
    __throw_system_error (int ev, const char* what_arg)
    {
+    #if defined(__EXCEPTIONS)
-        // error: copying parameter of type
-        // 'os::estd::system_error_category'
-        // when binding a reference to a temporary would invoke a
-        // deleted
-        // constructor in C++98 [-Werror,-Wc++98-compat-bind-to-
-        // temporary-copy]
-        throw std::system_error (std::error_code (ev,
-            system_error_category ()),
-                                what_arg);
+        throw std::system_error (
+            std::error_code (ev, get_system_error_category ()),
+            what_arg);
+    #else
        trace_printf ("system_error(%d, %s)\n", ev, what_arg);
        std::abort ();
@@ -124,10 +140,7 @@ namespace os
    __throw_cmsis_error (int ev, const char* what_arg)
    {
+    #if defined(__EXCEPTIONS)
-        // error: copying parameter of type
-        // 'os::estd::cmsis_error_category' when

```

```

-         // binding a reference to a temporary would invoke a deleted
-         constructor
-         // in C++98 [-Werror,-Wc++98-compat-bind-to-temporary-copy]
-         throw std::system_error (std::error_code (ev,
+         cmsis_error_category ()),
+         throw std::system_error (std::error_code (ev,
+         get_cmsis_error_category ()),
+         what_arg);
-     }
-     #else
-     trace_printf ("system_error(%d, %s)\n", ev, what_arg);

```

## PR #9: CMSIS-RTOS v1 C API Wrapper Cleanups

### Why Step 4 is Needed:

- **CMSIS-RTOS v1 Compatibility Defaults:** In `src/rtos/os-c-wrapper.cpp`, CMSIS-RTOS timer creation default is corrected to one-shot (`osTimerOnce`) per the CMSIS-RTOS specification.
- **Polymorphic Deletion & Resource Cleanup:** Ensures that `osMutexDelete()` and `osSemaphoreDelete()` invoke the correct polymorphic destructors for derived synchronization objects.
- **64-Bit Millisecond Timeout Scaling:** Explicitly casts timeout milliseconds to `uint64_t` prior to multiplying by `1000u`, preventing 32-bit integer truncation on long timeouts.
- **Lifting Isolation:** Step 4 surgically strips Step-9 `os_thread_state_destroying` static asserts and Step-20 SMP affinity blocks, ensuring the C-wrapper compiles cleanly on the Step 3 baseline without premature dependencies.

```

diff --git a/src/rtos/os-c-wrapper.cpp b/src/rtos/os-c-wrapper.cpp
index fc898c72..4bd9e1bb 100644
--- a/src/rtos/os-c-wrapper.cpp
+++ b/src/rtos/os-c-wrapper.cpp
@@ -1504,7 +1504,9 @@ os_timer_construct (os_timer_t* timer, const
     char* name,
     assert (timer != nullptr);
     if (attr == nullptr)
     {
-        attr = (const os_timer_attr_t*)&timer::periodic_initializer;
+        // The C++ and CMSIS defaults are one-shot timers; only an
+        explicit
+        // periodic attribute (os_timer_attr_get_periodic()) should
+        repeat.
+        attr = (const os_timer_attr_t*)&timer::once_initializer;
     }
     new (timer)
         rtos::timer (name, (timer::func_t)function,
         (timer::func_args_t)args,
@@ -1545,7 +1547,9 @@ os_timer_new (const char* name, os_timer_func_t
     function,
     {
         if (attr == nullptr)
         {
-            attr = (const os_timer_attr_t*)&timer::periodic_initializer;
+            // The C++ and CMSIS defaults are one-shot timers; only an
+            explicit
+            // periodic attribute (os_timer_attr_get_periodic()) should
+            repeat.
+            attr = (const os_timer_attr_t*)&timer::once_initializer;
         }
         return reinterpret_cast<os_timer_t*> (
             new rtos::timer (name, (timer::func_t)function,
             (timer::func_args_t)args,
@@ -1773,7 +1777,21 @@ void
os_mutex_delete (os_mutex_t* mutex)
{
    assert (mutex != nullptr);
-    delete reinterpret_cast<rtos::mutex*> (mutex);
+

```

```

+ // `mutex` and `mutex_recursive` share the same C storage type and
+ // the base
+ // destructor is intentionally non-virtual (so that `os_mutex_t`
+ // keeps the
+ // same size as `rtos::mutex`). The concrete type is recorded in
+ // the object
+ // itself, so delete through the derived type and run the correct
+ // destructor instead of relying on undefined behaviour.
+ if (reinterpret_cast<rtos::mutex*> (mutex)->type ()
+     == rtos::mutex::type::recursive)
+ {
+     delete reinterpret_cast<rtos::mutex_recursive*> (mutex);
+ }
+ else
+ {
+     delete reinterpret_cast<rtos::mutex*> (mutex);
+ }
+ }

/**
@@ -2325,7 +2343,22 @@ void
os_semaphore_delete (os_semaphore_t* semaphore)
{
    assert (semaphore != nullptr);
- delete reinterpret_cast<rtos::semaphore*> (semaphore);
+
+ // `semaphore_binary` and `semaphore_counting` share the same C
+ // storage
+ // type and the base destructor is intentionally non-virtual (so
+ // that
+ // `os_semaphore_t` keeps the same size as `rtos::semaphore`). The
+ // binary
+ // form is the one whose maximum count is 1, so delete through the
+ // derived
+ // type and run the correct destructor instead of relying on
+ // undefined
+ // behaviour. Both derived destructors are empty, so the routing is
+ // safe
+ // even if a counting semaphore is configured with a maximum of 1.
+ if (reinterpret_cast<rtos::semaphore*> (semaphore)->max_value () ==
+     1)
+ {
+     delete reinterpret_cast<rtos::semaphore_binary*> (semaphore);
+ }
+ else
+ {
+     delete reinterpret_cast<rtos::semaphore_counting*> (semaphore);
+ }
+ }

/**
@@ -3505,7 +3538,6 @@ osThreadCreate (const osThreadDef_t* thread_def,
                void* args)
    thread::attributes attr;
    attr.th_priority = thread_def->tpriority;
    attr.th_stack_size_bytes = thread_def->stacksize;
-
    // Creating thread with invalid priority should fail (validator
    // requirement).
    if (thread_def->tpriority >= osPriorityError)
    {
@@ -3739,7 +3771,7 @@ osDelay (uint32_t millisec)
    }

    result_t res = sysclock.sleep_for (
-     clock_systick::ticks_cast ((uint64_t)(millisec * 1000u)));
+     clock_systick::ticks_cast ((uint64_t) millisec * 1000u)));

    if (res == ETIMEDOUT)
    {

```

```

@@ -3788,7 +3820,7 @@ osWait (uint32_t millisec)
    }

    result_t res = sysclock.wait_for (
-       clock_systick::ticks_cast ((uint64_t)(millisec * 1000u)));
+       clock_systick::ticks_cast (((uint64_t) millisec * 1000u)));

    // TODO: return events
    if (res == ETIMEDOUT)
@@ -3867,7 +3899,7 @@ osTimerStart (osTimerId timer_id, uint32_t
    millisec)

    result_t res
        = (reinterpret_cast<rtos::timer&> (*timer_id))
-           .start (clock_systick::ticks_cast ((uint64_t)(millisec *
+           .start (clock_systick::ticks_cast (((uint64_t) millisec *
1000u)));
1000u)));

    if (res == result::ok)
    {
@@ -4051,7 +4083,7 @@ osSignalWait (int32_t signals, uint32_t
    millisec)
    {
        res = this_thread::flags_timed_wait (
            (flags::mask_t)signals,
-            clock_systick::ticks_cast ((uint64_t)(millisec * 1000u)),
+            clock_systick::ticks_cast (((uint64_t) millisec * 1000u)),
            (flags::mask_t*)&event.value.signals);
    }

@@ -4162,7 +4194,7 @@ osMutexWait (osMutexId mutex_id, uint32_t
    millisec)
    {
        ret = (reinterpret_cast<rtos::mutex&> (*mutex_id))
            .timed_lock (
-            clock_systick::ticks_cast ((uint64_t)(millisec *
+            clock_systick::ticks_cast (((uint64_t) millisec *
1000u)));
1000u)));
        // osErrorTimeoutResource:
    }

@@ -4355,7 +4387,7 @@ osSemaphoreWait (osSemaphoreId semaphore_id,
    uint32_t millisec)
    {
        res = (reinterpret_cast<rtos::semaphore&> (*semaphore_id))
            .timed_wait (
-            clock_systick::ticks_cast ((uint64_t)(millisec *
+            clock_systick::ticks_cast (((uint64_t) millisec *
1000u)));
1000u)));
        if (res == ETIMEDOUT)
        {
            return 0;
@@ -4645,7 +4677,7 @@ osMessagePut (osMessageQId queue_id, uint32_t
    info, uint32_t millisec)
        res = (reinterpret_cast<message_queue&> (*queue_id))
            .timed_send (
                (const char*)&info, sizeof (uint32_t),
-            clock_systick::ticks_cast ((uint64_t)(millisec *
+            clock_systick::ticks_cast (((uint64_t) millisec *
1000u)));
1000u)),
            0);
        // osOK, osErrorTimeoutResource, osErrorParameter
    }

@@ -4743,7 +4775,7 @@ osMessageGet (osMessageQId queue_id, uint32_t
    millisec)
        res = (reinterpret_cast<message_queue&> (*queue_id))
            .timed_receive (

```

```

        (char*)&event.value.v, sizeof (uint32_t),
        clock_systick::ticks_cast ((uint64_t)(millisec *
-         1000u)),
+         clock_systick::ticks_cast (((uint64_t) millisec *
+         1000u)),
        nullptr);
    // result::event_message when message;
    // result::event_timeout when timeout;
@@ -4881,7 +4913,7 @@ osMailAlloc (osMailQId mail_id, uint32_t
    millisec)
    }
    ret = (reinterpret_cast<memory_pool&> (mail_id->pool))
        .timed_alloc (
-         clock_systick::ticks_cast ((uint64_t)(millisec *
+         1000u)));
+         clock_systick::ticks_cast (((uint64_t) millisec *
+         1000u)));
    }
    #pragma GCC diagnostic pop
    return ret;
@@ -5044,7 +5076,7 @@ osMailGet (osMailQId mail_id, uint32_t millisec)
    res = (reinterpret_cast<message_queue&> (mail_id->queue))
        .timed_receive (
        (char*)&event.value.p, sizeof (void*),
-         clock_systick::ticks_cast ((uint64_t)(millisec *
+         1000u)),
+         clock_systick::ticks_cast (((uint64_t) millisec *
+         1000u)),
        nullptr);
    // osEventMail for ok, osEventTimeout
    }

```

## PR #10: POSIX I/O File Descriptor Table Mutexing (and PR #11: Mounts, PR #12: Block Device)

### Why Step 5 is Needed:

- **File Descriptor Table Concurrency Protection:** In `src/posix-io/file-descriptors-manager.cpp`, multiple concurrent threads executing `open()`, `socket()`, `dup()`, or `close()` could corrupt the file descriptor table. Wrapping table allocation, search, and free operations within `rtos::interrupts::critical_section` guarantees atomic descriptor management.
- **Filesystem & Network Stack Mutexing:** In `include/cmsis-plus/posix-io/file-system.h` and `net-stack.h`, mount list iterations and network interface tracking are protected with mutexes to prevent list corruption during concurrent mount/unmount or interface registration.
- **Block Device Partition Validation:** In `src/posix-io/block-device.cpp`, corrects an inverted partition size check, properly returning `EINVAL` when attempting to register a partition of zero blocks.

```

diff --git a/include/cmsis-plus/posix-io/file-system.h b/include/
cmsis-plus/posix-io/file-system.h
index 28fb822c..4986ca49 100644
--- a/include/cmsis-plus/posix-io/file-system.h
+++ b/include/cmsis-plus/posix-io/file-system.h
@@ -28,6 +28,7 @@
#include <cmsis-plus/utils/lists.h>

#include <cmsis-plus/diag/trace.h>
+include <cmsis-plus/rtos/os.h>

#include <mutex>
#include <cstdarg>
@@ -821,12 +822,14 @@ namespace os
    inline void

```

```

file_system::add_deferred_file (file* fil)
{
+   rtos::interrupts::critical_section ics;
   deferred_files_list_.link (*fil);
}

inline void
file_system::add_deferred_directory (directory* dir)
{
+   rtos::interrupts::critical_section ics;
   deferred_directories_list_.link (*dir);
}

@@ -848,16 +851,21 @@ namespace os
{
    using file_type = T;

-   file_type* fil;
+   file_type* fil = nullptr;
+   {
+       rtos::interrupts::critical_section ics;
+       if (!deferred_files_list_.empty ())
+       {
+           fil = static_cast<file_type*>
+               (deferred_files_list_.unlink_head ());
+       }
+   }

-   if (deferred_files_list_.empty ())
+   if (fil == nullptr)
+   {
+       fil = new file_type (*this);
+   }
+   else
+   {
-       fil = static_cast<file_type*>
-           (deferred_files_list_.unlink_head ());
-
        // Call the constructor before reusing the object,
        fil->~file_type ();
    }

@@ -875,16 +883,21 @@ namespace os
{
    using file_type = T;

-   file_type* fil;
+   file_type* fil = nullptr;
+   {
+       rtos::interrupts::critical_section ics;
+       if (!deferred_files_list_.empty ())
+       {
+           fil = static_cast<file_type*>
+               (deferred_files_list_.unlink_head ());
+       }
+   }

-   if (deferred_files_list_.empty ())
+   if (fil == nullptr)
+   {
+       fil = new file_type (*this, locker);
+   }
+   else
+   {
-       fil = static_cast<file_type*>
-           (deferred_files_list_.unlink_head ());
-
        // Call the constructor before reusing the object,

```

```

        fil->~file_type ();

@@ -902,13 +915,20 @@ namespace os
    {
        using file_type = T;

-        // Deallocate all remaining elements in the list.
-        while (!deferred_files_list_.empty ())
+        for (;;)
        {
            file_type* f
            = static_cast<file_type*>
            (deferred_files_list_.unlink_head ());

-            // Call the destructor and the deallocator.
+            file_type* f = nullptr;
+            {
+                rtos::interrupts::critical_section ics;
+                if (!deferred_files_list_.empty ())
+                {
+                    f = static_cast<file_type*>
+                    (deferred_files_list_.unlink_head ());
+                }
+            }
+            if (f == nullptr)
+            {
+                break;
+            }
            delete f;
        }
    }

@@ -919,17 +939,22 @@ namespace os
    {
        using directory_type = T;

-        directory_type* dir;
-
-        if (deferred_directories_list_.empty ())
+        directory_type* dir = nullptr;
+        {
+            rtos::interrupts::critical_section ics;
+            if (!deferred_directories_list_.empty ())
+            {
+                dir = static_cast<directory_type*> (
+                    deferred_directories_list_.unlink_head ());
+            }
+        }

+        if (dir == nullptr)
+        {
+            dir = new directory_type (*this);
+        }
        else
        {
-            dir = static_cast<directory_type*> (
-                deferred_directories_list_.unlink_head ());
-
-            // Call the constructor before reusing the object,
            dir->~directory_type ();
        }

@@ -947,17 +972,22 @@ namespace os
    {
        using directory_type = T;

-        directory_type* dir;
-
-        if (deferred_directories_list_.empty ())

```

```

+     directory_type* dir = nullptr;
+     {
+         rtos::interrupts::critical_section ics;
+         if (!deferred_directories_list_.empty ())
+         {
+             dir = static_cast<directory_type*> (
+                 deferred_directories_list_.unlink_head ());
+         }
+     }
+
+     if (dir == nullptr)
+     {
+         dir = new directory_type (*this, locker);
+     }
+     else
+     {
+         dir = static_cast<directory_type*> (
+             deferred_directories_list_.unlink_head ());
+
+         // Call the constructor before reusing the object,
+         dir->~directory_type ();
+
@@ -975,13 +1005,21 @@ namespace os
    {
        using directory_type = T;

-        // Deallocate all remaining elements in the list.
-        while (!deferred_directories_list_.empty ())
+        for (;;)
        {
-            directory_type* d = static_cast<directory_type*> (
-                deferred_directories_list_.unlink_head ());

-            // Call the destructor and the deallocator.
+            directory_type* d = nullptr;
+            {
+                rtos::interrupts::critical_section ics;
+                if (!deferred_directories_list_.empty ())
+                {
+                    d = static_cast<directory_type*> (
+                        deferred_directories_list_.unlink_head ());
+                }
+            }
+            if (d == nullptr)
+            {
+                break;
+            }
+            delete d;
        }
    }
}
diff --git a/include/cmsis-plus/posix-io/net-stack.h b/include/cmsis-plus/posix-io/net-stack.h
index ebced3da..c2d4f3af 100644
--- a/include/cmsis-plus/posix-io/net-stack.h
+++ b/include/cmsis-plus/posix-io/net-stack.h
@@ -26,6 +26,7 @@
#include <cmsis-plus/utils/lists.h>

#include <cmsis-plus/diag/trace.h>
+#include <cmsis-plus/rtos/os.h>

#include <cstdint>
#include <cassert>
@@ -468,6 +469,7 @@ namespace os
    inline void
    net_stack::add_deferred_socket (class socket* sock)

```

```

+ {
+     rtos::interrupts::critical_section ics;
+     deferred_sockets_list_.link (*sock);
+ }

@@ -483,17 +485,22 @@ namespace os
+ {
+     using socket_type = T;

-     socket_type* sock;

-     if (deferred_sockets_list_.empty ())
+     socket_type* sock = nullptr;
+     {
+         rtos::interrupts::critical_section ics;
+         if (!deferred_sockets_list_.empty ())
+         {
+             sock = static_cast<socket_type*> (
+                 deferred_sockets_list_.unlink_head ());
+         }
+     }

+     if (sock == nullptr)
+     {
+         sock = new socket_type (*this);
+     }
+     else
+     {
-         sock = static_cast<socket_type*> (
-             deferred_sockets_list_.unlink_head ());
-
+         // Call the constructor before reusing the object,
+         sock->~socket_type ();

@@ -501,12 +508,21 @@ namespace os
+     new (sock) socket_type (*this);

+     // Deallocate all remaining elements in the list.
+     while (!deferred_sockets_list_.empty ())
+     for (;;)
+     {
-         socket_type* s = static_cast<socket_type*> (
-             deferred_sockets_list_.unlink_head ());

-         // Call the destructor and the deallocator.
+         socket_type* s = nullptr;
+         {
+             rtos::interrupts::critical_section ics;
+             if (!deferred_sockets_list_.empty ())
+             {
+                 s = static_cast<socket_type*> (
+                     deferred_sockets_list_.unlink_head ());
+             }
+         }
+         if (s == nullptr)
+         {
+             break;
+         }
+         delete s;
+     }
+ }

@@ -519,17 +535,22 @@ namespace os
+ {
+     using socket_type = T;

-     socket_type* sock;

```

```

-         if (deferred_sockets_list_.empty ())
+         socket_type* sock = nullptr;
+         {
+             rtos::interrupts::critical_section ics;
+             if (!deferred_sockets_list_.empty ())
+             {
+                 sock = static_cast<socket_type*> (
+                     deferred_sockets_list_.unlink_head ());
+             }
+         }
+
+         if (sock == nullptr)
+         {
+             sock = new socket_type (*this, locker);
+         }
+         else
+         {
-             sock = static_cast<socket_type*> (
-                 deferred_sockets_list_.unlink_head ());
-
-             // Call the constructor before reusing the object,
-             sock->~socket_type ();

@@ -537,12 +558,21 @@ namespace os
            new (sock) socket_type (*this, locker);

            // Deallocate all remaining elements in the list.
-            while (!deferred_sockets_list_.empty ())
+            for (;;)
            {
-                socket_type* s = static_cast<socket_type*> (
-                    deferred_sockets_list_.unlink_head ());

-                // Call the destructor and the deallocator.
+                socket_type* s = nullptr;
+                {
+                    rtos::interrupts::critical_section ics;
+                    if (!deferred_sockets_list_.empty ())
+                    {
+                        s = static_cast<socket_type*> (
+                            deferred_sockets_list_.unlink_head ());
+                    }
+                }
+                if (s == nullptr)
+                {
+                    break;
+                }
+                delete s;
            }
        }
    }
}

diff --git a/src/posix-io/block-device.cpp b/src/posix-io/block-
device.cpp
index 020245ee..2565cf4d 100644
--- a/src/posix-io/block-device.cpp
+++ b/src/posix-io/block-device.cpp
@@ -153,7 +153,7 @@ namespace os
    // Get logical device sector size (to be used for read/
    writes).
    {
        std::size_t* sz = va_arg (args, std::size_t*);
-        if (sz == nullptr || impl ().block_logical_size_bytes_ !=
0)
+        if (sz == nullptr || impl ().block_logical_size_bytes_ ==
0)
        {
            errno = EINVAL;

```

```

@@ -171,7 +171,7 @@ namespace os
    // Get physical device sector size (internally used for
    erase).
    {
        std::size_t* sz = va_arg (args, std::size_t*);
-       if (sz == nullptr || impl ().block_physical_size_bytes_ != 0)
+       if (sz == nullptr || impl ().block_physical_size_bytes_
== 0)
    {
        errno = EINVAL;

@@ -189,7 +189,7 @@ namespace os
    // Get device size in bytes.
    {
        uint64_t* sz = va_arg (args, uint64_t*);
-       if (sz == nullptr || impl ().num_blocks_ != 0)
+       if (sz == nullptr || impl ().num_blocks_ == 0)
    {
        errno = EINVAL;

diff --git a/src/posix-io/file-descriptors-manager.cpp b/src/posix-io/
file-descriptors-manager.cpp
index 836539f2..699b3017 100644
--- a/src/posix-io/file-descriptors-manager.cpp
+++ b/src/posix-io/file-descriptors-manager.cpp
@@ -16,6 +16,7 @@
#include <cmsis-plus/posix-io/file-descriptors-manager.h>
#include <cmsis-plus/posix-io/io.h>
#include <cmsis-plus/posix-io/socket.h>
+#include <cmsis-plus/rtos/os.h>

#include <cmsis-plus/diag/trace.h>

@@ -89,6 +90,8 @@ namespace os
    io*
    file_descriptors_manager::io (int fildes)
    {
+       rtos::interrupts::critical_section ics;
+
        // Check if valid descriptor or buffer not yet initialised
        if ((fildes < 0) || (static_cast<std::size_t> (fildes) >=
size__)
            || (descriptors_array__ == nullptr))

@@ -106,7 +109,11 @@ namespace os
    bool
    file_descriptors_manager::valid (int fildes)
    {
-       if ((fildes < 0) || (static_cast<std::size_t> (fildes) >=
size__))
+       rtos::interrupts::critical_section ics;
+
+       if ((fildes < 0) || (static_cast<std::size_t> (fildes) >=
size__)
            || (descriptors_array__ == nullptr)
            || (descriptors_array__[fildes] == nullptr))
        {
            return false;
        }

@@ -127,6 +134,8 @@ namespace os
    return -1;
}

+       rtos::interrupts::critical_section ics;
+
    for (std::size_t i = reserved__; i < size__; ++i)
    {

```

```

#pragma GCC diagnostic push
@@ -167,6 +176,8 @@ namespace os
    return -1;
}

+   rtos::interrupts::critical_section ics;
+
#pragma GCC diagnostic push
#if defined(__clang__)
#pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
@@ -184,7 +195,11 @@ namespace os
    trace::printf ("file_descriptors_manager::%s(%d)\n", __func__,
        fildes);
#endif

-   if ((fildes < 0) || (static_cast<std::size_t> (fildes) >=
        size__))
+   rtos::interrupts::critical_section ics;
+
+   if ((fildes < 0) || (static_cast<std::size_t> (fildes) >=
        size__))
+       || (descriptors_array__ == nullptr)
+       || (descriptors_array__[fildes] == nullptr))
    {
        errno = EBADF;
        return -1;
@@ -204,13 +219,19 @@ namespace os
    file_descriptors_manager::socket (int fildes)
    {
        assert ((fildes >= 0) && (static_cast<std::size_t> (fildes) <
            size__));
+
+   rtos::interrupts::critical_section ics;
+
#pragma GCC diagnostic push
#if defined(__clang__)
#pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
#endif
-   auto* const io = descriptors_array__[fildes];
+   auto* const io = (descriptors_array__ != nullptr)
+       ? descriptors_array__[fildes]
+       : nullptr;
#pragma GCC diagnostic pop
-   if (io->get_type () != static_cast<posix::io::type_t>
        (io::type::socket))
+   if (io == nullptr
+       || io->get_type () != static_cast<posix::io::type_t>
        (io::type::socket))
    {
        return nullptr;
    }
@@ -220,6 +241,8 @@ namespace os
    size_t
    file_descriptors_manager::used (void)
    {
+   rtos::interrupts::critical_section ics;
+
        std::size_t count = reserved__;
        for (std::size_t i = reserved__; i <
            file_descriptors_manager::size ();
            ++i)
@@ -228,7 +251,7 @@ namespace os
#if defined(__clang__)
#pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
#endif
-   if (descriptors_array__[i] != nullptr)
+   if (descriptors_array__ != nullptr &&
        descriptors_array__[i] != nullptr)
    {

```

```

        ++count;
    }

```

## PR #13: ARMv8-M Traps (and PR #14: SecureFault, PR #15: Semihosting fstat)

### Why Step 6 is Needed:

- **ARMv8-M Architecture Mainline / Baseline Support:** In `include/cmsis-plus/arm/semihosting.h`, semihosting trap instructions (BKPT 0xAB vs HLT 0xF000) are guarded by `__ARM_ARCH_8M_MAIN__` and `__ARM_ARCH_8M_BASE__` macros to support ARM Cortex-M23/M33 processors.
- **ARMv8-M Security Extension Fault Handling:** In `src/startup/exception-handlers.c`, adds `SecureFault_Handler` to trap and report TrustZone memory and peripheral security violations. Adds weak `os_board_console_mirror()` to allow board packages to mirror semihosting diagnostics to UART.
- **Semihosting Syscall fstat() Correction:** In `src/semihosting/c-syscalls-semihosting.cpp`, `fstat()` assigns `S_IFCHR` (character device mode) only when file attributes have not been explicitly populated, preventing improper file mode overrides on host files.

```

diff --git a/include/cmsis-plus/arm/semihosting.h b/include/cmsis-plus/arm/semihosting.h
index de63417e..37c85d61 100644
--- a/include/cmsis-plus/arm/semihosting.h
+++ b/include/cmsis-plus/arm/semihosting.h
@@ -73,7 +73,8 @@ extern "C"
 #endif
 // For thumb only architectures use the BKPT instruction instead of SWI.
 #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
-    || defined(__ARM_ARCH_6M__)
+    || defined(__ARM_ARCH_6M__) || defined(__ARM_ARCH_8M_MAIN__) \
+    || defined(__ARM_ARCH_8M_BASE__)
 #define AngelSWIInsn "bkpt"
 #define AngelSWIAsm bkpt
 #else
@@ -103,6 +104,9 @@ extern "C"
     " mov r1, %[arg] \n"
 #if defined(OS_DEBUG_SEMIHOSTING_FAULTS)
     " " AngelSWITestFault " \n"
+elif defined(SEMIHOST_TRAP_HLT)
+    " .arm \n"
+    " .inst 0xE10F0070 \n"
 #else
     " " AngelSWIInsn " %[swi] \n"
 #endif
diff --git a/src/semihosting/c-syscalls-semihosting.cpp b/src/semihosting/c-syscalls-semihosting.cpp
index e3ff04c4..8b6d034c 100644
--- a/src/semihosting/c-syscalls-semihosting.cpp
+++ b/src/semihosting/c-syscalls-semihosting.cpp
@@ -263,9 +263,12 @@ __semihosting_stat (int fd, struct stat* st)
     return -1;
 }

- /* Always assume a character device,
-  with 1024 byte blocks. */
- st->st_mode |= S_IFCHR;
+ /* If the caller did not already specify a file type,
+  default to character device, with 1024 byte blocks. */
+ if ((st->st_mode & S_IFMT) == 0)
+ {
+     st->st_mode |= S_IFCHR;

```

```

+     }
+     st->st_blksize = 1024;

+     int res;
@@ -437,9 +440,32 @@ __posix_read (int fildes, void* buf, size_t
+     nbyte)
+     return nbyte - res;
+ }

+/**
+ * Optional board console mirror.
+ *
+ * A board whose console is a physical UART (not the debugger)
+ * defines this to
+ * copy the console stream there, so the application's stdout/stderr
+ * is visible
+ * on the terminal as well as in the debugger's semihosting console.
+ * The
+ * semihosting write stays the primary path; the board only adds the
+ * UART copy.
+ * Weak: boards that do not need it link this no-op and it is never
+ * called.
+ */
+extern "C" void
+os_board_console_mirror (int fildes, const void* buf, size_t nbyte)
+ __attribute__ ((weak));
+
+ssize_t
+__posix_write (int fildes, const void* buf, size_t nbyte)
+ {
+ // The board console, if any, gets stdout/stderr whatever happens
+ // to the
+ // semihosting side below: on some ports the monitor handles are
+ // not set up,
+ // and a dropped printf() would otherwise never be seen. Done
+ // before the fd
+ // lookup so it does not depend on it.
+ if ((fildes == 1 || fildes == 2)
+     && (os_board_console_mirror != nullptr))
+     {
+         os_board_console_mirror (fildes, buf, nbyte);
+     }
+
+     struct fdent* pfd;
+     pfd = __semihosting_findslot (fildes);
+     if (pfd == NULL)
diff --git a/src/startup/exception-handlers.c b/src/startup/exception-
handlers.c
index 98be7d14..533ffcd7 100644
--- a/src/startup/exception-handlers.c
+++ b/src/startup/exception-handlers.c
@@ -91,7 +91,8 @@ Reset_Handler (void)
+ // SCB->VTOR
+ // https://developer.arm.com/documentation/dui0552/a/cortex-m3-
+ // peripherals/system-control-block/vector-table-offset-register
+ // Mandatory when running from RAM. Not available on Cortex-M0.
-#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+ || defined(__ARM_ARCH_8M_MAIN__)
+     *((uint32_t*)0xE00ED08)
+     = ((uint32_t)_interrupt_vectors & (uint32_t)(~0x3F));
+ #endif
@@ -119,14 +120,16 @@ void __attribute__ ((section (".after_vectors"),
+     weak))
+ NMI_Handler (void)
+ {
+     #if defined(DEBUG)
-#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+ || defined(__ARM_ARCH_8M_MAIN__)

```

```

        if ((CoreDebug->DHCSR & CoreDebug_DHCSR_C_DEBUGEN_Msk) != 0)
        {
            __BKPT (0);
        }
    #else
        __BKPT (0);
    #endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
+ #endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+   || defined(__ARM_ARCH_8M_MAIN__) */
    #endif /* defined(DEBUG) */

    while (true)
@@ -139,7 +142,8 @@ NMI_Handler (void)

    #if defined(TRACE)

    #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+ #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+   || defined(__ARM_ARCH_8M_MAIN__)

    // The values of BFAR and MMFAR remain unchanged if the BFARVALID or
    // MMARVALID is set. However, if a new fault occurs during the
@@ -183,7 +187,8 @@ dump_exception_stack (exception_stack_frame_t*
    frame, uint32_t cfsr,
    trace_printf (" LR/EXC_RETURN = %08X\n", lr);
}

    #endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
+ #endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+   || defined(__ARM_ARCH_8M_MAIN__) */

    #if defined(__ARM_ARCH_6M__)

@@ -209,7 +214,8 @@ dump_exception_stack (exception_stack_frame_t*
    frame, uint32_t lr)

    // -----

    #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+ #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+   || defined(__ARM_ARCH_8M_MAIN__)

    #if defined(OS_USE_SEMIHOSTING_SYSCALLS) \
        || defined(OS_USE_TRACE_SEMIHOSTING_STDOUT) \
@@ -470,14 +476,16 @@ HardFault_Handler_C (exception_stack_frame_t*
    frame __attribute__((unused)),
    #endif /* defined(TRACE) */

    #if defined(DEBUG)
    #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+ #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+   || defined(__ARM_ARCH_8M_MAIN__)
        if ((CoreDebug->DHCSR & CoreDebug_DHCSR_C_DEBUGEN_Msk) != 0)
        {
            __BKPT (0);
        }
    #else
        __BKPT (0);
    #endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
+ #endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+   || defined(__ARM_ARCH_8M_MAIN__) */
    #endif /* defined(DEBUG) */

    while (true)
@@ -486,7 +494,8 @@ HardFault_Handler_C (exception_stack_frame_t*
    frame __attribute__((unused)),
    }

```

```

}

-#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
+#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+    || defined(__ARM_ARCH_8M_MAIN__) */

#if defined(__ARM_ARCH_6M__)

@@ -532,14 +541,16 @@ HardFault_Handler_C (exception_stack_frame_t*
    frame __attribute__((unused)),
#endif /* defined	TRACE) */

#if defined(DEBUG)
-#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+    || defined(__ARM_ARCH_8M_MAIN__)
    if ((CoreDebug->DHCSR & CoreDebug_DHCSR_C_DEBUGEN_Msk) != 0)
    {
        __BKPT (0);
    }
#else
    __BKPT (0);
-#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
+#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+    || defined(__ARM_ARCH_8M_MAIN__) */
#endif /* defined(DEBUG) */

    while (true)
@@ -550,20 +561,23 @@ HardFault_Handler_C (exception_stack_frame_t*
    frame __attribute__((unused)),

#endif /* defined(__ARM_ARCH_6M__) */

-#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+    || defined(__ARM_ARCH_8M_MAIN__)

void __attribute__((section(".after_vectors"), weak))
MemManage_Handler (void)
{
    #if defined(DEBUG)
-#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+    || defined(__ARM_ARCH_8M_MAIN__)
    if ((CoreDebug->DHCSR & CoreDebug_DHCSR_C_DEBUGEN_Msk) != 0)
    {
        __BKPT (0);
    }
#else
    __BKPT (0);
-#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
+#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+    || defined(__ARM_ARCH_8M_MAIN__) */
#endif /* defined(DEBUG) */

    while (true)
@@ -603,14 +617,16 @@ BusFault_Handler_C (exception_stack_frame_t*
    frame __attribute__((unused)),
#endif /* defined	TRACE) */

#if defined(DEBUG)
-#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+    || defined(__ARM_ARCH_8M_MAIN__)
    if ((CoreDebug->DHCSR & CoreDebug_DHCSR_C_DEBUGEN_Msk) != 0)
    {
        __BKPT (0);
    }

```

```

    }
    #else
        __BKPT (0);
    -#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
    +#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
    +    || defined(__ARM_ARCH_8M_MAIN__) */
    #endif /* defined(DEBUG) */

    while (true)
    @@ -665,14 +681,16 @@ UsageFault_Handler_C (exception_stack_frame_t*
        frame __attribute__ ((unused)),
    #endif /* defined	TRACE) */

    #if defined(DEBUG)
    -#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
    +#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
    +    || defined(__ARM_ARCH_8M_MAIN__)
        if ((CoreDebug->DHCSR & CoreDebug_DHCSR_C_DEBUGEN_Msk) != 0)
        {
            __BKPT (0);
        }
    #else
        __BKPT (0);
    -#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
    +#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
    +    || defined(__ARM_ARCH_8M_MAIN__) */
    #endif /* defined(DEBUG) */

    while (true)
    @@ -687,14 +705,16 @@ void __attribute__ ((section (".after_vectors"),
        weak))
    SVC_Handler (void)
    {
        #if defined(DEBUG)
    -#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
    +#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
    +    || defined(__ARM_ARCH_8M_MAIN__)
            if ((CoreDebug->DHCSR & CoreDebug_DHCSR_C_DEBUGEN_Msk) != 0)
            {
                __BKPT (0);
            }
        #else
            __BKPT (0);
    -#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
    +#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
    +    || defined(__ARM_ARCH_8M_MAIN__) */
        #endif /* defined(DEBUG) */

        while (true)
    @@ -703,7 +723,8 @@ SVC_Handler (void)
        }
    }

    -#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
    +#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
    +    || defined(__ARM_ARCH_8M_MAIN__)

    void __attribute__ ((section (".after_vectors"), weak))
    DebugMon_Handler (void)
    @@ -721,20 +742,23 @@ DebugMon_Handler (void)
        }
    }

    -#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */
    +#endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
    +    || defined(__ARM_ARCH_8M_MAIN__) */

```



```

-#endif // defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+#endif // defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) ||
+    defined(__ARM_ARCH_8M_MAIN__)
+
+#if defined(__ARM_ARCH_6M__)
+    void
+    dump_exception_stack (exception_stack_frame_t* frame, uint32_t lr);
+@ -90,7 +99,8 @@ extern "C"
+    void
+    HardFault_Handler_C (exception_stack_frame_t* frame, uint32_t lr);
+
+
-#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+#if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) \
+    || defined(__ARM_ARCH_8M_MAIN__)
+
+    void
+    UsageFault_Handler_C (exception_stack_frame_t* frame, uint32_t lr);
+@ -98,7 +108,7 @@ extern "C"
+    void
+    BusFault_Handler_C (exception_stack_frame_t* frame, uint32_t lr);
+
+
-#endif // defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+#endif // defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) ||
+    defined(__ARM_ARCH_8M_MAIN__)
+
+
+#if defined(__cplusplus)
+}

```

## PR #16: Software Timer Callback Execution Decoupling & Drift Prevention

### Why Step 7 is Needed:

- **Elimination of Timer Callback Deadlocks:** In `src/rtos/os-timer.cpp` and `src/rtos/internal/os-lists.cpp`, software timer callbacks were historically invoked while holding internal scheduler critical section locks. If a user callback attempted to acquire a mutex, allocate memory, or signal a semaphore, the system deadlocked. Step 7 removes the timer from the active list under lock, releases the critical section, and invokes the callback function with interrupts enabled.
- **Periodic Catch-Up Under Heavy IRQ Latency:** Re-arms periodic timers based on scheduled expiry rather than completion time, preventing cumulative timer drift under high interrupt load.
- **Handler-Mode Errno Scratch Storage:** In `include/cmsis-plus/rtos/os-thread.h`, surgically adds an ISR errno storage field without pulling in later SMP thread fields.

```

diff --git a/include/cmsis-plus/rtos/os-thread.h b/include/cmsis-plus/rtos/os-thread.h
index 9cd7a05a..3aecb6a9 100644
--- a/include/cmsis-plus/rtos/os-thread.h
+++ b/include/cmsis-plus/rtos/os-thread.h
+@ -2081,6 +2081,14 @@ namespace os
+    inline int* __attribute__((always_inline))
+    __errno (void)
+    {
+        // A timer callback runs inside the tick ISR and a libc call
+        // there
+        // (e.g. printf) still touches errno; thread() asserts in
+        // handler mode,
+        // so hand back a scratch int instead of the current
+        // thread's.
+        if (interrupts::in_handler_mode ())
+        {
+            static int isr_errno;
+            return &isr_errno;
+        }
+    }

```

```

        return &this_thread::thread ().errno_;
    }

diff --git a/src/rtos/internal/os-lists.cpp b/src/rtos/internal/os-
lists.cpp
index 02ea8ab4..32c0c9a0 100644
--- a/src/rtos/internal/os-lists.cpp
+++ b/src/rtos/internal/os-lists.cpp
@@ -499,33 +499,42 @@ namespace os
    // iterate until a node with future time stamp is identified.
    for (;;)
    {
-       // ----- Enter critical section
-       -----
-       interrupts::critical_section ics;
+       timestamp_node* node = nullptr;
+       {
+           // ----- Enter critical section
+           -----
+           interrupts::critical_section ics;

            if (empty ())
            {
                break;
            }
+           if (empty ())
+           {
+               break;
+           }
+       }
        #pragma GCC diagnostic push
        #if defined(__clang__)
        #elif defined(__GNUC__)
        #pragma GCC diagnostic ignored "-Wnull-dereference"
        #endif

-       clock::timestamp_t head_ts = head ()->timestamp;
+       clock::timestamp_t head_ts = head ()->timestamp;
        #pragma GCC diagnostic pop
        if (now >= head_ts)
        {
            node = const_cast<timestamp_node*> (head ());
            node->unlink ();
        }
        else
        {
            break;
        }
        // ----- Exit critical section
        -----
    }

    if (node != nullptr)
    {
        #if defined(OS_TRACE_RTOS_LISTS_CLOCKS)
            trace::printf ("%s() %u \n", __func__,
                           static_cast<uint32_t> (sysclock.now
            ()));
        #endif
        const_cast<timestamp_node*> (head ())->action ();
    }
    else
    {
        break;
    }
    node->action ();
}
// ----- Exit critical section
-----
}

```

```

    }

diff --git a/src/rtos/os-timer.cpp b/src/rtos/os-timer.cpp
index 7b9c89c5..f5318340 100644
--- a/src/rtos/os-timer.cpp
+++ b/src/rtos/os-timer.cpp
@@ -374,9 +374,17 @@ namespace os
    {
        // Re-arm the timer for the next period.
        timer_node_.timestamp += period_;
+       clock::timestamp_t now = clock_->steady_now ();
+       if (timer_node_.timestamp <= now)
+       {
+           // Prevent burst loop if callback or ISR was delayed
+           timer_node_.timestamp = now + period_;
+       }

-       // No need for critical section in ISR.
-       clock_->steady_list ().link (timer_node_);
+       {
+           interrupts::critical_section ics;
+           clock_->steady_list ().link (timer_node_);
+       }
    }
    else
    {

```

## PR #17: POSIX Mutex Priority Ceiling Protocol & Dynamic Boost Tracking

### Why Step 8 is Needed:

- **POSIX Priority Ceiling Protocol Enforcement:** In `src/rtos/os-mutex.cpp`, when a thread locks a priority ceiling mutex, its priority must be elevated to the ceiling **before** ownership is granted. This eliminates the vulnerability window where an intermediate-priority thread could preempt the caller before priority elevation took effect.
- **Dynamic Priority Boost Tracking & Restoration:** When multiple threads wait on a priority-inheritance mutex, the owner thread's priority is boosted to the maximum priority among all waiters. Upon unlocking, owner priority is dynamically recomputed and restored to its original base priority.
- **Owner Pointer Caching:** Caches owner thread reference across unlocked windows to eliminate race conditions during concurrent unlock and thread termination.

```

diff --git a/src/rtos/os-mutex.cpp b/src/rtos/os-mutex.cpp
index e8cab9d1..6ce22248 100644
--- a/src/rtos/os-mutex.cpp
+++ b/src/rtos/os-mutex.cpp
@@ -638,6 +638,15 @@ namespace os
    // First lock.
    if (owner_ == nullptr)
    {
+       // Prio ceiling must be at least the priority of the
+       // highest
+       // priority thread. Checked before anything is taken: a
+       // refused
+       // lock must not leave the mutex on the thread's list of
+       // owned
+       // mutexes, nor counted in acquired_mutexes_.
+       if (protocol_ == protocol::protect && th->priority () >
+           prio_ceiling_)
+       {
+           return EINVAL;
+       }
    }

```

```

+         }
+
+         // If the mutex has no owner, own it.
+         owner_ = th;
@@ -661,16 +670,6 @@ namespace os
+
+         if (protocol_ == protocol::protect)
+         {
+             if (th->priority () > prio_ceiling_)
+             {
+                 // No need to keep the lock.
+                 owner_ = nullptr;
+
+                 // Prio ceiling must be at least the priority of
the
+                 // highest priority thread.
+                 return EINVAL;
+             }
+
+             // POSIX: When a thread owns one or more mutexes
+             // initialised with the mutex::protocol::protect
+             protocol,
+             // it shall execute at the higher of its priority or
the
@@ -778,7 +777,11 @@ namespace os
+         if (protocol_ == protocol::inherit)
+         {
+             thread::priority_t prio = th->priority ();
+             boosted_prio_ = prio;
+             if ((boosted_prio_ == thread::priority::none)
+                 || (prio > boosted_prio_))
+             {
+                 boosted_prio_ = prio;
+             }
+
+             if (owner_links_.unlinked ())
+             {
@@ -788,13 +791,54 @@ namespace os
+                 }
+
+                 // Boost owner priority.
+                 if ((boosted_prio_ > owner_->priority_inherited ()))
+                 //
+                 // The owner is captured HERE, while the kernel lock is
still
+                 // held, and the captured pointer -- not the member --
is the
+                 // one dereferenced below.
+                 //
+                 // The reason is the uncritical section itself: it
releases the
+                 // kernel lock (`priority_inherited()` ends in a yield,
which
+                 // must not run scheduler-locked), and while it is open
the
+                 // owner can finish its own `unlock()` on another CPU.
+                 // `internal_unlock_()` ends by setting `owner_` to
nullptr, so
+                 // re-reading the member after the section -- which is
exactly
+                 // what `owner_->priority_inherited(...)` compiles to
-- then
+                 // dereferences a null pointer.
+                 //
+                 // Every SMP port has this window. It was found on the
POSIX
+                 // host port, where `scheduler::unlock()`/`lock()` is
two
+                 // `pthread_sigmask()` system calls rather than a
handful of

```

```

+         // instructions: that widens the window enough to turn
+         a rare
+         // race into the common case (smp_test2 faulted in 21
+         runs out
+         // of 25 on four CPUs).
+         /* class */ thread* owner = owner_;
+
+         if ((owner != nullptr)
+             && (boosted_prio_ > owner->priority_inherited ()))
+         {
+             // ----- Enter uncritical section
+             -----
+             scheduler::uncritical_section succs;
+             {
+                 // ----- Enter uncritical section
+                 -----
+                 scheduler::uncritical_section succs;
+
+                 // Still the owner? If it released the mutex
+                 while this // section was open there is nothing to inherit,
+                 and
+                 // boosting it anyway would leave behind an
+                 inherited
+                 // priority that no later unlock() would ever
+                 clear.
+                 if (owner_ == owner)
+                 {
+                     owner->priority_inherited (boosted_prio_);
+                 }
+                 // ----- Exit uncritical section
+                 -----
+             }
+
+             owner_>priority_inherited (boosted_prio_);
+             // ----- Exit uncritical section
+             -----
+
+             // Released while the section was open? Then that
+             unlock() // found no waiter to hand the mutex to, and
+             blocking now // would wait for a wake-up that already happened.
+             Decide
+
+             // again, back under the lock.
+             if (owner_ != owner)
+             {
+                 return internal_try_lock_ (th);
+             }
+         }
+
+         #if defined(OS_TRACE_RTOS_MUTEX)
+@@ -862,14 +906,8 @@ namespace os
+             mutexes_list* thread_mutexes
+                 = reinterpret_cast<mutexes_list*> (&owner_ -
+                 >mutexes_);
+
+             if (thread_mutexes->empty ())
+             {
+                 // If the owner thread has no more mutexes,
+                 // clear the inherited priority,
+                 // and the assigned priority will take
+                 precedence.
+                 boosted_prio_ = thread::priority::none;
+             }
+             else
+             {
+                 thread::priority_t inherited_prio =
+                 thread::priority::none;
+                 if (!thread_mutexes->empty ())
+                 {
+                     // If the owner thread acquired other mutexes
+                     too,
+                     // compute the maximum boosted priority.

```

```

@@ -887,10 +925,14 @@ namespace os
    }
    }
    #pragma GCC diagnostic pop
-    boosted_prio_ = max_prio;
+    if (max_prio > 0)
+    {
+        inherited_prio = max_prio;
+    }
    }
    // Delayed until end of critical section.
-    owner_>priority_inherited (boosted_prio_);
+    owner_>priority_inherited (inherited_prio);
+    boosted_prio_ = thread::priority::none;
}

// Delayed until end of critical section.

```

## PR #18: Thread Lifecycle State Machine (state::destroying), Atomic Join & Reaper

### Why Step 9 is Needed:

- **Deterministic Thread Destruction State (state::destroying):** In `include/cmsis-plus/rtos/os-thread.h` and `src/rtos/os-thread.cpp`, introducing explicit `state::destroying = 7` prevents race conditions where a terminating thread is scheduled or signaled while its resources are being deallocated.
- **Atomic join() and detach() Synchronization:** Ensures that `thread::join()` blocks atomically until the target thread enters `destroying` state, guaranteeing that thread stacks and control blocks are not freed while still referenced by the scheduler.
- **Idle Thread Reaper Cleanup:** In `src/rtos/os-idle.cpp`, the idle thread acts as a safe asynchronous garbage collector (reaper), freeing dynamically allocated thread stacks after the thread has completely exited execution.
- **Single-Core Projection (sc-project.py):** Strips all multi-core ready list and CPU affinity logic from `os-thread.cpp` (~37 SMP hunks), allowing single-core targets to benefit from lifecycle fixes without SMP overhead.

```

diff --git a/include/cmsis-plus/rtos/os-c-decls.h b/include/cmsis-plus/rtos/os-c-decls.h
index e8565a72..260d1e52 100644
--- a/include/cmsis-plus/rtos/os-c-decls.h
+++ b/include/cmsis-plus/rtos/os-c-decls.h
@@ -315,7 +315,12 @@ extern "C"
 /**
  * @brief Used to check reused threads.
  */
- os_thread_state_initialising = 6
+ os_thread_state_initialising = 6,
+
+ /**
+  * @brief In process of being destroyed.
+  */
+ os_thread_state_destroying = 7
};

/**
diff --git a/include/cmsis-plus/rtos/os-thread.h b/include/cmsis-plus/rtos/os-thread.h
index 3aeb6a9..cac6777f 100644
--- a/include/cmsis-plus/rtos/os-thread.h
+++ b/include/cmsis-plus/rtos/os-thread.h
@@ -400,7 +400,11 @@ namespace os

```

```

        /**
        * @brief Used to check reused threads.
        */
-       initializing = 6 //
+       initializing = 6, //
+       /**
+       * @brief In process of being destroyed.
+       */
+       destroying = 7 //
    };
    /* enum */
}; /* struct state */
@@ -741,6 +745,7 @@ namespace os
    friend port::stack::element_t*
    port::scheduler::switch_stacks (port::stack::element_t* sp);

+
#endif
    /**
    * @endcond
@@ -875,6 +880,7 @@ namespace os

    bool th_enable_assert_reuse = false;

+
    // Add more attributes here.

    /**
@@ -1192,6 +1198,7 @@ namespace os
    priority_t
    priority_inherited (void);

+
    #if 0
        // ???
        result_t
@@ -1689,6 +1696,7 @@ namespace os
    priority_t volatile prio_assigned_ = priority::none;
    priority_t volatile prio_inherited_ = priority::none;

+
    bool volatile interrupted_ = false;

    internal::event_flags event_flags_;
diff --git a/src/rtos/os-idle.cpp b/src/rtos/os-idle.cpp
index 23b05974..0db1ab35 100644
--- a/src/rtos/os-idle.cpp
+++ b/src/rtos/os-idle.cpp
@@ -79,6 +79,14 @@ os_rtos_idle_actions (void)
    interrupts::critical_section ics;
    node = const_cast<internal::waiting_thread_node*> (
        scheduler::terminated_threads_list_.head ());
+   thread* th = node->thread_;
+   if (th->state_ == thread::state::destroying
+       || th->state_ == thread::state::destroyed)
+   {
+       node->unlink ();
+       continue;
+   }
+   th->state_ = thread::state::destroying;
    node->unlink ();
    // ----- Exit critical section
    -----
}

diff --git a/src/rtos/os-thread.cpp b/src/rtos/os-thread.cpp
index 3b8f293a..79b0a71c 100644

```

```

--- a/src/rtos/os-thread.cpp
+++ b/src/rtos/os-thread.cpp
@@ -19,6 +19,7 @@
#include <memory>
#include <stdexcept>

+
// -----

#ifdef __clang__
@@ -679,10 +680,15 @@ namespace os
    interrupts::critical_section ics;

    // If the thread is not already in the ready list, enqueue
    it.
-   if (ready_node_.next () == nullptr)
+   // In SMP, a running thread has ready_node_.next() ==
+   nullptr;
+   // it must not be re-enqueued while actively executing.
+   if (state_ == state::suspended || state_ ==
+   state::initializing)
    {
-       scheduler::ready_threads_list_.link (ready_node_);
-       // state::ready set in above link().
+       if (ready_node_.next () == nullptr)
+       {
+           scheduler::ready_threads_list_.link (ready_node_);
+           // state::ready set in above link().
+       }
    }
    // ----- Exit critical section
    -----
}

@@ -691,6 +697,7 @@ namespace os

    port::scheduler::reschedule ();

+
#endif

    instrumentation::thread::resume_return (this);
@@ -702,6 +709,7 @@ namespace os
    *
    * @note Can be invoked from Interrupt Service Routines.
    */
+
    thread::priority_t
    thread::priority (void)
    {
@@ -791,17 +799,24 @@ namespace os

    #else

-       if (state_ == state::ready)
-       {
-           // ----- Enter critical section
-           -----
-           interrupts::critical_section ics;
+       {
+           // ----- Enter critical section
+           -----
+           interrupts::critical_section ics;

-           // Remove from initial location and reinsert according
-           // to new priority.
-           ready_node_.unlink ();
-           scheduler::ready_threads_list_.link (ready_node_);

```

```

-         // ----- Exit critical section
-         -----
-     }
+     // Test and relink under one lock, and only a thread that is
+     still
+     // linked in the ready list. On SMP another CPU may pick this
+     thread
+     // between a test made outside the lock and the relink:
+     link() would
+     // then put a thread being dispatched back in the ready list,
+     marked
+     // ready, and a third CPU could run the same context at once.
+     if (state_ == state::ready && ready_node_.next () != nullptr)
+     {
+         // Remove from initial location and reinsert according
+         // to new priority.
+         ready_node_.unlink ();
+         scheduler::ready_threads_list_.link (ready_node_);
+     }
+     // ----- Exit critical section
+     -----
+ }

    // Mandatory, the priority might have been raised, the
    // task must be scheduled to run.
@@ -875,17 +890,24 @@ namespace os

    #else

-     if (state_ == state::ready)
-     {
-         // ----- Enter critical section
-         -----
-         interrupts::critical_section ics;
+     {
+         // ----- Enter critical section
+         -----
+         interrupts::critical_section ics;

-         // Remove from initial location and reinsert according
-         // to new priority.
-         ready_node_.unlink ();
-         scheduler::ready_threads_list_.link (ready_node_);
-         // ----- Exit critical section
-         -----
-     }
+         // Test and relink under one lock, and only a thread that is
+         still
+         // linked in the ready list. On SMP another CPU may pick this
+         thread
+         // between a test made outside the lock and the relink:
+         link() would
+         // then put a thread being dispatched back in the ready list,
+         marked
+         // ready, and a third CPU could run the same context at once.
+         if (state_ == state::ready && ready_node_.next () != nullptr)
+         {
+             // Remove from initial location and reinsert according
+             // to new priority.
+             ready_node_.unlink ();
+             scheduler::ready_threads_list_.link (ready_node_);
+         }
+         // ----- Exit critical section
+         -----
+     }

    // Mandatory, the priority might have been raised, the
    // task must be scheduled to run.
@@ -941,7 +963,22 @@ namespace os

    #else

```

```

- // TODO: implement
+ {
+     interrupts::critical_section ics;
+
+     if (state_ == state::destroyed)
+     {
+         instrumentation::thread::detach_retval (this, EINVAL);
+         return EINVAL;
+     }
+
+     if (parent_ != nullptr)
+     {
+         child_links_.unlink ();
+         scheduler::top_threads_list_.link (*this);
+         parent_ = nullptr;
+     }
+ }
+
+ #endif
+
+ @@ -997,13 +1034,37 @@ namespace os
+ // Fail if current thread
+ assert (this != this_thread::_thread ());
+
+ while (state_ != state::destroyed)
+ thread* crt_thread = this_thread::_thread ();
+ for (;;)
+ {
+     joiner_ = this_thread::_thread ();
+     this_thread::_thread ()->internal_suspend (
+         OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_JOIN);
+     {
+         // ----- Enter critical section
+         -----
+         interrupts::critical_section ics;
+
+         // Test, register and suspend under one lock, the same
+         lock
+         // internal_destroy_() sets `destroyed` and reads
+         `joiner_`
+         // under. Done in steps, a destroy on another CPU could
+         fall
+         // between them and its wake-up be lost.
+         if (state_ == state::destroyed)
+         {
+             break;
+         }
+         joiner_ = crt_thread;
+
+         // Remove this thread from the ready list, if there.
+         port::this_thread::prepare_suspend ();
+
+         crt_thread->state_ = state::suspended;
+         // ----- Exit critical section
+         -----
+     }
+
+     instrumentation::thread::suspended (
+         crt_thread,
+         OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_JOIN);
+
+     port::scheduler::reschedule ();
+ }
+
+ #if defined(OS_TRACE_RTOS_THREAD)

```

```

        trace::printf ("%s() @%p %s joined\n", __func__, this, name
        ());
    #endif
@@ -1268,12 +1329,27 @@ namespace os
    // ----- Exit critical section
    -----
}

-    state_ = state::destroyed;
+    {
+        // ----- Enter critical section
+        -----
+        interrupts::critical_section ics;
+
+        // From `destroyed` on, a joiner may return from join() and
+        free
+        // this object, so `joiner_` is read under the same lock that
+        // sets it, and the joiner is made ready here, where it
+        cannot
+        // yet have gone -- resume() would do it after the lock.
+        state_ = state::destroyed;

-    if (joiner_ != nullptr)
-    {
-        joiner_>resume ();
-    }
+    thread* joiner = joiner_;
+    if (joiner != nullptr && joiner->state_ == state::suspended
+        && joiner->ready_node_.next () == nullptr)
+    {
+        scheduler::ready_threads_list_.link (joiner-
+>ready_node_);
+        // state::ready set in above link().
+    }
+    // ----- Exit critical section
+    -----
+    }

+    port::scheduler::reschedule ();
}
#pragma GCC diagnostic pop

@@ -1308,6 +1384,7 @@ namespace os
    // ----- Enter critical section
    -----
    scheduler::critical_section scs;

+
    if (state_ == state::destroyed)
    {
        #if defined(OS_TRACE_RTOS_THREAD)
@@ -1318,29 +1395,49 @@ namespace os
        return result::ok; // Already exited itself
    }

+    bool we_claimed = false;
+    {
+        // ----- Enter critical section
+        -----
+        interrupts::critical_section ics;

-        // Remove thread from the funeral list and kill it here.
-        ready_node_.unlink ();

-        // If the thread is waiting on an event, remove it from the
-        list.
-        if (waiting_node_ != nullptr)
+        if (state_ != state::destroying && state_ !=
+state::destroyed)
+        {

```

```

-         waiting_node_>unlink ();
-     }
+     we_claimed = true;
+     state_ = state::destroying;

-     // If the thread is waiting on a timeout, remove it from
the list.
-     if (clock_node_ != nullptr)
-     {
-         clock_node_>unlink ();
-     }
+     // Remove thread from the funeral list and kill it
here.
+     ready_node_.unlink ();

-     child_links_.unlink ();
+     // If the thread is waiting on an event, remove it from
the list.
+     if (waiting_node_ != nullptr)
+     {
+         waiting_node_>unlink ();
+     }

+     // If the thread is waiting on a timeout, remove it
from the list.
+     if (clock_node_ != nullptr)
+     {
+         clock_node_>unlink ();
+     }

+     child_links_.unlink ();
+     }
+     // ----- Exit critical section
+     -----
+ }

+     if (!we_claimed)
+     {
+         // The idle reaper already claimed this thread for
destruction.
+         // Wait with lock released until it is completely
destroyed.
+         while (state_ != state::destroyed)
+         {
+             scheduler::uncritical_section succs;
+             this_thread::yield ();
+         }
+         instrumentation::thread::kill_retval (this, result::ok);
+         return result::ok;
+     }

+     // The must be no more children threads alive.
assert (children_.empty ());
parent_ = nullptr;
@@ -1837,6 +1934,11 @@ namespace os
*
* @warning Cannot be invoked from Interrupt Service Routines.
*/
+ // NOTE: NOT 'inline'. It is declared non-inline in os-
thread.h, and its
+ // body lives only here (not in a header), so other TUs (e.g.
the C API
+ // wrapper os_this_thread_suspend) that odr-use it need an out-
of-line
+ // definition. Marking it 'inline' meant no TU emitted one ->
undefined
+ // reference unless the caller happened to be pruned by --gc-
sections.
void

```

```
suspend (void)
{
```

## PR #19: Condition Variable Lost-Signal Elimination & Clock Binding

### Why Step 10 is Needed:

- **Atomicity in wait() / timed\_wait() to Eliminate Lost Signals:** In `src/rtos/os-condvar.cpp` and `include/cmsis-plus/rtos/os-condvar.h`, a classic concurrency hazard occurs if a thread releases its mutex and is preempted before enqueueing itself on the condition variable wait list. Step 10 links the thread to the wait list under scheduler lock, atomically releases the mutex, and suspends the thread, completely closing the lost-signal window.
- **Configurable Clock Binding:** Binds condition variable `timed_wait()` timeouts directly to the specific clock instance (`os::rtos::clock*`) configured in attributes, allowing timeouts against either wall-clock real-time or monotonic steady clocks.
- **ABI Structural Alignment:** Surgically uncomments `void* clock` in `include/cmsis-plus/rtos/os-c-decls.h` (`os_condvar_t`), ensuring C and C++ struct sizes match exactly.

```
diff --git a/include/cmsis-plus/diag/instrumentation.h b/include/
      cmsis-plus/diag/instrumentation.h
index 69e0c035..fd0b5cec 100644
--- a/include/cmsis-plus/diag/instrumentation.h
+++ b/include/cmsis-plus/diag/instrumentation.h
@@ -25,6 +25,7 @@
#define OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_EVENT_FLAGS (9u)
#define OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_JOIN (10u)
#define OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_USER (11u)
+#define OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_CONDVAR (12u)

// -----

diff --git a/include/cmsis-plus/rtos/os-c-decls.h b/include/cmsis-
      plus/rtos/os-c-decls.h
index 260d1e52..86dc5429 100644
--- a/include/cmsis-plus/rtos/os-c-decls.h
+++ b/include/cmsis-plus/rtos/os-c-decls.h
@@ -1065,7 +1065,7 @@ extern "C"
    const char* name;
    #if !defined(OS_USE_RTOS_PORT_CONDITION_VARIABLE)
        os_internal_threads_waiting_list_t list;
-    // void* clock;
+    void* clock;
    #endif

    /**
diff --git a/include/cmsis-plus/rtos/os-condvar.h b/include/cmsis-
      plus/rtos/os-condvar.h
index d9b64b22..8faa4e66 100644
--- a/include/cmsis-plus/rtos/os-condvar.h
+++ b/include/cmsis-plus/rtos/os-condvar.h
@@ -278,7 +278,7 @@ namespace os
    #if !defined(OS_USE_RTOS_PORT_CONDITION_VARIABLE)
        internal::waiting_threads_list list_;
-    // clock& clock_;
+    clock* clock_ = nullptr;
    #endif

    /**
diff --git a/src/rtos/os-condvar.cpp b/src/rtos/os-condvar.cpp
index 9c1021a3..b08ff1e7 100644
```

```

--- a/src/rtos/os-condvar.cpp
+++ b/src/rtos/os-condvar.cpp
@@ -271,8 +271,7 @@ namespace os
 * Edition](http://pubs.opengroup.org/onlinepubs/9699919799/
   nframe.html)).
 */
condition_variable::condition_variable (const char* name,
-                                     const attributes& attr
-                                     __attribute__((unused)))
+                                     const attributes& attr)
    : object_named_system{ name }
{
    instrumentation::condition_variable::create (this);
@@ -284,6 +283,10 @@ namespace os
// Don't call this from interrupt handlers.
os_assert_throw (!interrupts::in_handler_mode (), EPERM);

+#if !defined(OS_USE_RTOS_PORT_CONDITION_VARIABLE)
+    clock_ = attr.clock != nullptr ? attr.clock : &sysclock;
+#endif
+
    instrumentation::condition_variable::create_return (this);
}

@@ -371,10 +374,20 @@ namespace os
// Don't call this from interrupt handlers.
os_assert_err (!interrupts::in_handler_mode (), EPERM);

+#if defined(OS_USE_RTOS_PORT_CONDITION_VARIABLE)
+
+    result_t res = port::condition_variable::signal (this);
+    instrumentation::condition_variable::signal_retval (this, res);
+    return res;
+
+#else
+
    list_.resume_one ();

    instrumentation::condition_variable::signal_retval (this,
        result::ok);
    return result::ok;
+
+#endif
+
+/**
@@ -447,6 +460,14 @@ namespace os
// Don't call this from interrupt handlers.
os_assert_err (!interrupts::in_handler_mode (), EPERM);

+#if defined(OS_USE_RTOS_PORT_CONDITION_VARIABLE)
+
+    result_t res = port::condition_variable::broadcast (this);
+    instrumentation::condition_variable::broadcast_retval (this,
+        res);
+    return res;
+
+#else
+
// Wake-up all threads, if any.
// Need not be inside the critical section,
// the list is protected by inner `resume_one()`.
@@ -454,6 +475,8 @@ namespace os

instrumentation::condition_variable::broadcast_retval (this,
    result::ok);
return result::ok;
```

```
+
+ #endif
+ }
+
+ /**
@@ -555,6 +578,14 @@ namespace os
+ // Don't call this from critical regions.
+ os_assert_err (!scheduler::locked (), EPERM);
+
+ #if defined(OS_USE_RTOS_PORT_CONDITION_VARIABLE)
+ +
+ + result_t res = port::condition_variable::wait (this, &mutex);
+ + instrumentation::condition_variable::wait_retval (this, res);
+ + return res;
+ +
+ + #else
+ +
+ + thread& crt_thread = this_thread::thread ();
+
+ // Prepare a list node pointing to the current thread.
@@ -562,32 +593,52 @@ namespace os
+ // list and guaranteed to be removed before this function
+ returns.
+ internal::waiting_thread_node node{ crt_thread };
+
+ // TODO: validate
+
+ result_t res;
+ - res = mutex.unlock ();
+ {
+ + // ----- Enter critical section
+ + -----
+ + // The link and the unlock must be one step for this CPU's
+ scheduler.
+ + // Once linked, the thread is `suspended`; a tick or an IPI
+ taken
+ + // before `mutex.unlock()` would switch it out still OWNING
+ the mutex,
+ + // and the thread that could signal it would block on that
+ mutex
+ + // forever (smp-pro-cons-test stalled this way on 4 cores, 1
+ run in
+ + // about 20). Other CPUs are not held back: they can still
+ take the
+ + // mutex, and signal(), as soon as it is released.
+ + scheduler::critical_section scs;
+
+ if (res != result::ok)
+ {
+ - instrumentation::condition_variable::wait_retval (this,
+ res);
+ - return res;
+ + // ----- Enter critical section
+ + -----
+ + interrupts::critical_section ics;
+
+ // Add this thread to the condition variable waiting list.
+ scheduler::internal_link_node (
+ + list_, node,
+ OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_CONDVAR);
+ // state::suspended set in above link().
+ // ----- Exit critical section
+ + -----
+ }
+
+ {
+ - // Add this thread to the condition variable waiting list.
+ - list_.link (node);
+ - node.thread->waiting node = &node;
```

```

-         res = mutex.lock ();
+         res = mutex.unlock ();

-         // Remove the thread from the node waiting list,
-         // if not already removed.
-         node.thread->waiting_node_ = nullptr;
-         node.unlink ();
+         if (res != result::ok)
+         {
+             scheduler::internal_unlink_node (node);
+             instrumentation::condition_variable::wait_retval (this,
+ res);
+             return res;
+         }
+         // ----- Exit critical section
+         -----
    }

+ port::scheduler::reschedule ();

+ // Remove the thread from the condition variable waiting list,
+ // if not already removed by signal() / broadcast().
+ scheduler::internal_unlink_node (node);

+ res = mutex.lock ();

+ instrumentation::condition_variable::wait_retval (this, res);
+ return res;

+ #endif
+ }

+ /**
@@ -715,6 +766,14 @@ namespace os
+ // Don't call this from critical regions.
+ os_assert_err (!scheduler::locked (), EPERM);

+ #if defined(OS_USE_RTOS_PORT_CONDITION_VARIABLE)
+
+     result_t res = port::condition_variable::timed_wait (this,
+ &mutex, timeout);
+     instrumentation::condition_variable::timed_wait_retval (this,
+ res);
+     return res;
+ #else
+
+     thread& crt_thread = this_thread::thread ();

+     // Prepare a list node pointing to the current thread.
@@ -722,32 +781,62 @@ namespace os
+     // list and guaranteed to be removed before this function
+     returns.
+     internal::waiting_thread_node node{ crt_thread };

-     // TODO: validate
+     clock* clk = (clock_ != nullptr) ? clock_ : &sysclock;
+     internal::clock_timestamps_list& clock_list = clk->steady_list
+ ();
+     clock::timestamp_t timeout_timestamp = clk->steady_now () +
+ timeout;

+     // Prepare a timeout node pointing to the current thread.
+     internal::timeout_thread_node timeout_node{ timeout_timestamp,
+ crt_thread };

+     result_t res;
-     res = mutex.unlock ();

```

```

+ {
+     // ----- Enter critical section
+     -----
+     // Link and unlock as one step for this CPU's scheduler; see
+     wait().
+     scheduler::critical_section scs;
-
-     if (res != result::ok)
-     {
-         instrumentation::condition_variable::timed_wait_retval
-         (this, res);
-         return res;
+         // ----- Enter critical section
+         -----
+         interrupts::critical_section ics;
+
+         // Add this thread to the condition variable waiting list,
+         // and the clock timeout list.
+         scheduler::internal_link_node (
+             list_, node, clock_list, timeout_node,
+             OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_CONDVAR);
+         // state::suspended set in above link().
+         // ----- Exit critical section
+         -----
+     }
-
-     {
-         // Add this thread to the condition variable waiting list.
-         list_.link (node);
-         node.thread_>waiting_node_ = &node;
-
-         res = mutex.timed_lock (timeout);
+         res = mutex.unlock ();
-
-         // Remove the thread from the node waiting list,
-         // if not already removed.
-         node.thread_>waiting_node_ = nullptr;
-         node.unlink ();
+         if (res != result::ok)
+         {
+             scheduler::internal_unlink_node (node, timeout_node);
+             instrumentation::condition_variable::timed_wait_retval
+             (this, res);
+             return res;
+         }
+         // ----- Exit critical section
+         -----
+     }
+
+     port::scheduler::reschedule ();
+
+     // Remove the thread from the condition variable waiting list,
+     // if not already removed by signal() / broadcast() and from
+     // the clock
+     // timeout list, if not already removed by the timer.
+     scheduler::internal_unlink_node (node, timeout_node);
+
+     res = mutex.lock ();
+
+     if (res == result::ok && clk->steady_now () >=
+         timeout_timestamp)
+     {
+         res = ETIMEDOUT;
+     }
+
+     instrumentation::condition_variable::timed_wait_retval (this,
+         res);
+     return res;
+
+ #endif

```

```
}
```

```
// -----
```

```
diff --git a/include/cmsis-plus/estd/condition_variable b/include/
      cmsis-plus/estd/condition_variable
index e4a59eb8..d8ed311c 100644
--- a/include/cmsis-plus/estd/condition_variable
+++ b/include/cmsis-plus/estd/condition_variable
@@ -284,20 +284,22 @@ namespace os
     return cv_status::timeout;
 }

-    Native_clock::time_point start_tp = Native_clock::now ();
-
     os::rtos::clock::duration_t ticks
         = os::estd::chrono::ceil<std::chrono::duration<
             os::rtos::clock::duration_t, typename
             Native_clock::period>> (
                 rel_time)
             .count ();

-    ncv_.timed_wait (
+    // Trust the kernel's verdict rather than re-deriving it from
+    // the wall
+    // clock: a signalled wake that happened to be preempted beyond
+    // `rel_time`, or a timeout that rounded down to fewer ticks,
+    // would be
+    // misreported by a pure elapsed-time comparison.
+    // `timed_wait()` returns
+    // `result::ok` on a signal and `ETIMEDOUT` when the timeout
+    // expired.
+    os::rtos::result_t res = ncv_.timed_wait (
+        /*(rtos::mutex &)*/* (*(lock.mutex ()->native_handle ())),
+        ticks);

-    return (Native_clock::now () - start_tp) < rel_time
-        ? cv_status::no_timeout
-        : cv_status::timeout;
+    return (res == os::rtos::result::ok) ? cv_status::no_timeout
+        : cv_status::timeout;
+
 }

template <class Rep_T, class Period_T, class Predicate_T>
```

## PR #20: C++ std::thread Native Handle Synchronization & Functor Lifetime

### Why Step 11 is Needed:

- **std::thread Native Handle Synchronization:** In src/libcpp/thread-cpp.h and include/cmsis-plus/estd/thread\_internal.h, std::thread::join() is updated to wait directly on the native RTOS thread handle.
- **Functor Object Lifetime Guarantee:** Retains callable functor objects inside the thread wrapper structure until the thread has finished execution, preventing use-after-free crashes when invoking lambdas or functors containing captured references.

```
diff --git a/include/cmsis-plus/estd/thread_internal.h b/include/
      cmsis-plus/estd/thread_internal.h
index 1d97128e..389498e8 100644
--- a/include/cmsis-plus/estd/thread_internal.h
+++ b/include/cmsis-plus/estd/thread_internal.h
@@ -146,6 +146,12 @@ private:
     using function_object_deleter_t = void (*) (void*);
     function_object_deleter_t function_object_deleter_ = nullptr;
```

```

+ // The bound function object, remembered here so it can be deleted
+ // even
+ // after the kernel has cleared the thread's `func_args_` on exit.
+ // Relying
+ // on `native_thread_>function_args()` leaks it whenever the
+ // thread has
+ // already finished by the time `join()` runs (the common case).
+ void* function_object_ = nullptr;
+
+ public:
+ };

@@ -358,6 +364,10 @@ thread::thread (Callable_T&& f, Args_T&&... args)
    Function_object* funct_obj = new Function_object (std::bind (
        std::forward<Callable_T> (f), std::forward<Args_T> (args)...));

+ // Remember the object ourselves; the kernel may clear its own copy
+ // when
+ // the thread exits, before `join()` gets a chance to read it.
+ function_object_ = funct_obj;
+
+ // The function to start the thread is a custom proxy that
+ // knows how to get the variadic arguments.
+ #pragma GCC diagnostic push
diff --git a/src/libcpp/thread-cpp.h b/src/libcpp/thread-cpp.h
index c14f5bba..af572d23 100644
--- a/src/libcpp/thread-cpp.h
+++ b/src/libcpp/thread-cpp.h
@@ -36,11 +36,14 @@ thread::delete_system_thread (void)
 {
     if (id_ != id ())
     {
-        void* args = id_.native_thread_>function_args ();
-        if (args != nullptr && function_object_deleter_ != nullptr)
+        if (function_object_ != nullptr && function_object_deleter_ !=
+            nullptr)
         {
             // Manually delete the function object used to store
             arguments.
-            function_object_deleter_ (args);
+            function_object_deleter_ (function_object_);
+            // `function_object_` is our own copy: the kernel clears
+            // its
+            // `func_args_` on exit, so reading it here would leak
+            // whenever the
+            // thread has already finished.
+            function_object_deleter_ (function_object_);
+            function_object_ = nullptr;
         }

         // Manually delete the system thread.
@@ -68,6 +71,7 @@ thread::swap (thread& t) noexcept
 {
     std::swap (id_, t.id_);
     std::swap (function_object_deleter_, t.function_object_deleter_);
+ std::swap (function_object_, t.function_object_);
 }

 bool
@@ -81,7 +85,28 @@ thread::join ()
 {
     os::trace::printf ("%s() @%p\n", __func__, this);

-    delete_system_thread ();
+ if (id_ != id ())
+ {
+     // Wait for the thread to end, as ISO join() does, before
+     // freeing what
+     // it runs on. Deleting it straight away -- which is what this
+     // did --

```

```

+     // frees its bound arguments and kills the system thread; on
+     // one core
+     // the thread had usually finished by then, on SMP it is still
+     // running
+     // on another core.
+     id_.native_thread_ -> join ();
+
+     if (function_object_ != nullptr && function_object_deleter_ !=
+         nullptr)
+     {
+         // Manually delete the function object used to store
+         // arguments.
+         // Use our own pointer, because the kernel clears
+         // `func_args_` when
+         // the thread exits -- which happens before `join()`
+         // returns for any
+         // short-lived thread.
+         function_object_deleter_ (function_object_);
+         function_object_ = nullptr;
+     }
+
+     // Manually delete the system thread, destroyed by now.
+     delete id_.native_thread_;
+ }

id_ = id ();
os::trace::printf ("%s() @%p joined\n", __func__, this);

```

## PR #21: Message Queue Preemptive Rescheduling on Wakeups

### Why Step 12 is Needed:

- **Immediate Preemption on Message Queue Wakeups:** In `src/rtos/os-mqueue.cpp`, when `send()`, `try_send()`, `receive()`, or `try_receive()` wakes a higher-priority waiting thread, `port::scheduler::reschedule()` is invoked immediately upon exiting the critical section across all 6 code paths. This guarantees prompt preemption on Cortex-M and POSIX uniprocessor targets.

```

diff --git a/src/rtos/os-mqueue.cpp b/src/rtos/os-mqueue.cpp
index 3f134fae..cd46c58a 100644
--- a/src/rtos/os-mqueue.cpp
+++ b/src/rtos/os-mqueue.cpp
@@ -961,18 +961,27 @@ namespace os

#else

+     bool sent;
+     {
+         // ----- Enter critical section
+         -----
+         interrupts::critical_section ics;

-         if (internal_try_send_ (msg, nbytes, mprio))
-         {
-             instrumentation::message_queue::send_retval (this,
-                 result::ok);
-             return result::ok;
-         }
+         sent = internal_try_send_ (msg, nbytes, mprio);
+         // ----- Exit critical section
+         -----
+     }

+     if (sent)
+     {
+         // internal_try_send_() resumed a waiting receiver.
+         Rescheduling here,

```

```

+         // with the kernel lock released, lets it run at once -- as
+         the ARM
+         // ports do when PendSV is taken on the way out of the
+         section --
+         // instead of waiting for the next tick.
+         port::scheduler::reschedule ();
+
+         instrumentation::message_queue::send_retval (this,
+         result::ok);
+         return result::ok;
+     }
+
+     thread& crt_thread = this_thread::thread ();
+
+     // Prepare a list node pointing to the current thread.
@@ -1083,24 +1092,30 @@ namespace os
+     // Don't call this from high priority interrupts.
+     assert (port::interrupts::is_priority_valid ());
+
+     bool sent;
+     {
+         // ----- Enter critical section
+         -----
+         interrupts::critical_section ics;
+
+         if (internal_try_send_ (msg, nbytes, mprio))
+         {
+             instrumentation::message_queue::try_send_retval (this,
+             result::ok);
+             return result::ok;
+         }
+         else
+         {
+             instrumentation::message_queue::try_send_retval (this,
+
+             EWOULDBLOCK);
+             return EWOULDBLOCK;
+         }
+         sent = internal_try_send_ (msg, nbytes, mprio);
+         // ----- Exit critical section
+         -----
+     }
+
+     if (sent)
+     {
+         // internal_try_send_() resumed a waiting receiver.
+         Rescheduling here,
+         // with the kernel lock released, lets it run at once -- as
+         the ARM
+         // ports do when PendSV is taken on the way out of the
+         section --
+         // instead of waiting for the next tick.
+         port::scheduler::reschedule ();
+
+         instrumentation::message_queue::try_send_retval (this,
+         result::ok);
+         return result::ok;
+     }
+
+     instrumentation::message_queue::try_send_retval (this,
+     EWOULDBLOCK);
+     return EWOULDBLOCK;
+
+ #endif
+ }
@@ -1185,19 +1200,27 @@ namespace os
+
+     // Extra test before entering the loop, with its inherent
+     weight.
+     // Trade size for speed.

```

```

+   bool sent;
+   {
+       // ----- Enter critical section
+       -----
+       interrupts::critical_section ics;
+
+       if (internal_try_send_ (msg, nbytes, mprio))
+       {
+           instrumentation::message_queue::timed_send_retval (this,
+           result::ok);
+           return result::ok;
+       }
+       sent = internal_try_send_ (msg, nbytes, mprio);
+       // ----- Exit critical section
+       -----
+   }
+
+   if (sent)
+   {
+       // internal_try_send_() resumed a waiting receiver.
+       Rescheduling here,
+       // with the kernel lock released, lets it run at once -- as
+       the ARM
+       // ports do when PendSV is taken on the way out of the
+       section --
+       // instead of waiting for the next tick.
+       port::scheduler::reschedule ();
+
+       instrumentation::message_queue::timed_send_retval (this,
+       result::ok);
+       return result::ok;
+   }
+
+   thread& crt_thread = this_thread::thread ();
+
+   // Prepare a list node pointing to the current thread.
@@ -1338,18 +1361,26 @@ namespace os
+
+   // Extra test before entering the loop, with its inherent
+   weight.
+   // Trade size for speed.
+   bool received;
+   {
+       // ----- Enter critical section
+       -----
+       interrupts::critical_section ics;
+
+       if (internal_try_receive_ (msg, nbytes, mprio))
+       {
+           instrumentation::message_queue::receive_retval (this,
+           result::ok);
+           return result::ok;
+       }
+       received = internal_try_receive_ (msg, nbytes, mprio);
+       // ----- Exit critical section
+       -----
+   }
+
+   if (received)
+   {
+       // internal_try_receive_() resumed a waiting sender.
+       Rescheduling
+       // here, with the kernel lock released, lets it run at once
+       instead
+       // of waiting for the next tick.
+       port::scheduler::reschedule ();
+
+       instrumentation::message_queue::receive_retval (this,
+       result::ok);
+       return result::ok;

```

```

+     }
+
+     thread& crt_thread = this_thread::thread ();
+
+     // Prepare a list node pointing to the current thread.
@@ -1463,25 +1494,29 @@ namespace os
+     // Don't call this from high priority interrupts.
+     assert (port::interrupts::is_priority_valid ());
+
+     bool received;
+     {
+         // ----- Enter critical section
+         -----
+         interrupts::critical_section ics;
+
+         if (internal_try_receive_ (msg, nbytes, mprio))
+         {
+             instrumentation::message_queue::try_receive_retval (this,
+
+             result::ok);
+             return result::ok;
+         }
+         else
+         {
+             instrumentation::message_queue::try_receive_retval (this,
+
+             EWOULDBLOCK);
+             return EWOULDBLOCK;
+         }
+     }
+     received = internal_try_receive_ (msg, nbytes, mprio);
+     // ----- Exit critical section
+     -----
+ }
+
+ if (received)
+ {
+     // internal_try_receive_() resumed a waiting sender.
+     Rescheduling
+     // here, with the kernel lock released, lets it run at once
+     instead
+     // of waiting for the next tick.
+     port::scheduler::reschedule ();
+
+     instrumentation::message_queue::try_receive_retval (this,
+     result::ok);
+     return result::ok;
+ }
+
+ instrumentation::message_queue::try_receive_retval (this,
+     EWOULDBLOCK);
+ return EWOULDBLOCK;
+
+ #endif
+ }
@@ -1578,19 +1613,27 @@ namespace os
+
+     // Extra test before entering the loop, with its inherent
+     weight.
+     // Trade size for speed.
+ bool received;
+ {
+     // ----- Enter critical section
+     -----
+     interrupts::critical_section ics;
+
+     if (internal_try_receive_ (msg, nbytes, mprio))
+     {
+         instrumentation::message_queue::timed_receive_retval
+         (this,

```

```

-         result::ok);
-         return result::ok;
-     }
+     received = internal_try_receive_ (msg, nbytes, mprio);
+     // ----- Exit critical section
+     -----
+ }

+ if (received)
+ {
+     // internal_try_receive_() resumed a waiting sender.
+     Rescheduling
+     // here, with the kernel lock released, lets it run at once
+     instead
+     // of waiting for the next tick.
+     port::scheduler::reschedule ();
+
+     instrumentation::message_queue::timed_receive_retval (this,
+
+     result::ok);
+     return result::ok;
+ }

+
+ thread& crt_thread = this_thread::thread ();

+
+ // Prepare a list node pointing to the current thread.

```

## PR #22: High-Resolution Monotonic Clock Port Synchronization (K + C + P)

### Why Step 13 is Needed:

- **First Cross-Repository Synchronization (Kernel + Cortex-M + POSIX-Arch):**
  - **Kernel (os-decls.h, os-clocks.cpp):** Declares and calls `port::clock_highres::has_hardware_counter()` and `hardware_counter()`.
  - **Cortex-M Port (os-inlines.h):** Returns `false/0` and fixes SysTick overflow detection by checking `(SCB->ICSR & SCB_ICSR_PENDSTSET_Msk) != 0` to adjust cycle counts during pending ticks.
  - **POSIX-Arch Port (os-inlines.h):** Implements high-resolution hardware timestamping using host `CLOCK_MONOTONIC`. Uses `timespec tp`; without struct tag to satisfy `-Werror=redundant-tags`.
- **Atomic Linking Constraint:** Declarations in the kernel and implementations in the ports must land in the same step; otherwise, uniprocessor link builds fail with undefined reference errors.

```

diff --git a/include/cmsis-plus/rtos/os-decls.h b/include/cmsis-plus/rtos/os-decls.h
index 5d625d5b..2e39d590 100644
--- a/include/cmsis-plus/rtos/os-decls.h
+++ b/include/cmsis-plus/rtos/os-decls.h
@@ -22,6 +22,11 @@
 #endif
 // Include the non-portable portable types, enums and constants
 // declarations.
 #include <cmsis-plus/rtos/port/os-decls.h>
+// Port compat (develop/xbpack-development): the C-ABI declarations
+// were split
+// out of the C++ <os.h> chain, but kernel C++ sources
+// (os_systick_handler in
+// os-clocks.cpp) and the ports (os_thread_t) still rely on them
+// being visible
+// via <os.h>, as in v7.0.1. Needed by both the SMP and single-CPU
+// builds.
+#include <cmsis-plus/rtos/os-c-decls.h>

```

```

#include <cmsis-plus/diag/trace.h>

@@ -1025,6 +1030,12 @@ namespace os
    static void
    start (void);

+    static constexpr bool
+    has_hardware_counter (void) noexcept;
+
+    static uint64_t
+    hardware_counter (void) noexcept;
+
    static uint32_t
    cycles_per_tick (void);

diff --git a/src/rtos/os-clocks.cpp b/src/rtos/os-clocks.cpp
index f91f2ae3..94b457ff 100644
--- a/src/rtos/os-clocks.cpp
+++ b/src/rtos/os-clocks.cpp
@@ -793,11 +793,19 @@ namespace os
    clock::timestamp_t
    clock_highres::now (void)
    {
-    // ----- Enter critical section
-    -----
-    interrupts::critical_section ics;
+    if constexpr (port::clock_highres::has_hardware_counter ())
+    {
+        return port::clock_highres::hardware_counter ();
+    }
+    else
+    {
+        // Prevent inconsistent values.
+        // ----- Enter critical section
+        -----
+        interrupts::critical_section ics;

-    return steady_count_ + port::clock_highres::cycles_since_tick
-    ();
+    return steady_count_ +
+    port::clock_highres::cycles_since_tick ();
+    // ----- Exit critical section
+    -----
+    }
    }

    // -----

```

---

```

diff --git a/micro-os-plus-iii-cortexm/include/cmsis-plus/rtos/port/
os-inlines.h b/micro-os-plus-iii-cortexm/include/cmsis-plus/
rtos/port/os-inlines.h
index bcf1c4..33a1a69 100644
--- a/micro-os-plus-iii-cortexm/include/cmsis-plus/rtos/port/os-
inlines.h
+++ b/micro-os-plus-iii-cortexm/include/cmsis-plus/rtos/port/os-
inlines.h
@@ -95,12 +95,14 @@ namespace os
    return locked (state::unlocked);
    }

+
+    inline bool __attribute__ ((always_inline))
+    locked (void)
+    {
+        return lock_state != state::unlocked;
+    }

```

```

+         inline void __attribute__ ((always_inline))
+         wait_for_interrupt (void)
+         {
@@ -230,6 +232,7 @@ namespace os
+
+ #endif
+
+
+ // Apparently not required by architecture, but used by
+ // FreeRTOS, with an unconvincing motivation ("... ensure
+ // the code is completely within the specified behaviour
@@ -244,6 +247,7 @@ namespace os
+         inline void __attribute__ ((always_inline))
+         critical_section::exit (rtos::interrupts::state_t state)
+         {
+
+ #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+
+ #if defined(OS_INTEGER_RTOS_CRITICAL_SECTION_INTERRUPT_PRIORITY)
@@ -362,6 +366,18 @@ namespace os
+         ;
+         }
+
+         inline constexpr bool __attribute__ ((always_inline))
+         clock_highres::has_hardware_counter (void) noexcept
+         {
+             return false;
+         }
+
+         inline uint64_t __attribute__ ((always_inline))
+         clock_highres::hardware_counter (void) noexcept
+         {
+             return 0;
+         }
+
+         inline uint32_t __attribute__ ((always_inline))
+         clock_highres::input_clock_frequency_hz (void)
+         {
@@ -391,7 +407,7 @@ namespace os
+         // yet processed, so the total cycles count in steady_count_
+         // does not yet reflect the correct value and needs to be
+         // adjusted by one full cycle length.
+         if (SysTick->CTRL & SCB_ICSR_PENDSTSET_Msk)
+         if ((SCB->ICSR & SCB_ICSR_PENDSTSET_Msk) != 0)
+         {
+             // Sample the decremting counter again to validate the
+             // initial sample.
diff --git a/micro-os-plus-iii-posix-arch/include/cmsis-plus/rtos/
port/os-inlines.h b/micro-os-plus-iii-posix-arch/include/
cmsis-plus/rtos/port/os-inlines.h
index 93944a5..0ec7ab7 100644
--- a/micro-os-plus-iii-posix-arch/include/cmsis-plus/rtos/port/os-
inlines.h
+++ b/micro-os-plus-iii-posix-arch/include/cmsis-plus/rtos/port/os-
inlines.h
@@ -129,6 +129,21 @@ namespace os
+
+ //
+ // =====
+
+         inline constexpr bool __attribute__ ((always_inline))
+         clock_highres::has_hardware_counter (void) noexcept
+         {
+             return true;
+         }
+
+

```

```

+
+     inline uint64_t __attribute__((always_inline))
+     clock_highres::hardware_counter (void) noexcept
+     {
+         timespec tp; // NB: no 'struct' – baseline gate has
+         -Werror=redundant-tags
+         ::clock_gettime (CLOCK_MONOTONIC, &tp);
+         return static_cast<uint64_t> (tp.tv_sec) * 1000000ULL
+             + static_cast<uint64_t> (tp.tv_nsec) / 1000ULL;
+     }
+
+     } /* namespace port */
+     } /* namespace rtos */
+ } /* namespace os */

```

## Appendix Section 2 — Pull Request #2: Multi-Core SMP Kernel & Port Contracts + Genuine native-smp Host Platform (Steps 14-23 collapsed)

### Why PR #2 is Needed as a Standalone Pull Request

#### 1. Atomic Integration of Closely Coupled SMP Subsystems:

- Multi-core synchronization is intrinsically interdependent: the SMP scheduler (os-core.cpp), thread lifecycle (os-thread.cpp), idle threads (os-idle.cpp), recursive kernel lock (\_smp\_klock), per-CPU current thread pointers (current\_thread\_[OS\_NCPU]), and port contracts cannot be applied as isolated bisectable steps without breaking the build. Collapsing Steps 14–23 into a single atomic commit (step/14) guarantees mathematical and compilation consistency.

#### 2. Safe 5-Stage Stack-Pointer Claim/Publish Protocol:

- In a multi-core environment, thread context switching across cores introduces severe race conditions if another CPU picks an outgoing thread before its registers are fully saved. PR #2 enforces the 5-stage protocol:
  1. *Claim*: to->context\_.port\_.stack\_ptr = nullptr marks the incoming thread as live on this CPU.
  2. *Spill*: Callee-saved registers are pushed onto the current stack.
  3. *Switch*: CPU Stack Pointer is switched to the new thread's stack.
  4. *Publish*: from->context\_.port\_.stack\_ptr = sp publishes the outgoing thread as switchable only **after** its state is completely saved.
  5. *Restore*: New thread registers are popped and execution resumes.

#### 3. Inclusion of Genuine native-smp Dual-Core Host Verification:

- Passing -D OS\_USE\_SMP\_SCHEDULER=1 on the command line only sets CMake cache variables that never reach the compiler, silently running tests in single-core mode. PR #2 includes the native-smp host platform setting defines via target\_compile\_definitions, providing immediate, genuine dual-core test execution (pthreads-as-cores, interval timers, signal IPs) and verifying 100% of RTOS APIs under real multi-core concurrency.

#### 4. Hardening Corrections (Fixes A-J):

- Resolves Clang -Weverything diagnostics (-Wc++98-compat-pedantic, -Wreserved-identifier, -Wunsafe-buffer-usage, -Wcast-function-type-strict), -Werror=redundant-tags, and wraps kernel port\_cpu\_id with -Wredundant-decls pragma.

#### 5. Verification Gate:

- undef -UOS\_USE\_SMP\_SCHEDULER projection is 100% byte-identical to Step 13.
- 72/72 baseline single-core tests PASS.

- native-smp dual-core suite (3/3: rtos-apis-test, mutex-stress-test, cmsis-os-validator-test 60/60) PASS.

## PR #23 (Kernel): Multi-Core SMP Infrastructure

### Why Part B Kernel Infrastructure is Needed:

- **Portable Multi-Core Scheduler Engine:** In `src/rtos/os-core.cpp`, replaces global scalar ready lists with per-CPU ready lists (`ready_list_[OS_NCPU]`), manages per-core current threads (`current_thread_[OS_NCPU]`), and implements the recursive kernel spinlock (`_smp_klock`).
- **Strict Invariant — No Context Switching While Holding Kernel Lock:** Prohibits context switching while `_smp_klock` is held. The context switch is scheduled, the lock is released, and the hardware PendSV / signal handler executes the actual register switch.
- **Thread Migration & CPU Affinity:** In `src/rtos/os-thread.cpp`, enforces `th_cpu_affinity` bitmasks, allowing threads to be pinned to specific cores or load-balanced dynamically across cores.
- **Dedicated Per-Core Idle Threads:** In `src/rtos/os-idle.cpp`, instantiates and registers dedicated idle threads for each physical CPU core (`os_idle_thread_core[OS_NCPU]`).
- **Redundant Declaration Wrap (Fix H):** Wraps forward declarations of `port_cpu_id()` with `#pragma GCC diagnostic ignored "-Wredundant-decls"` to avoid conflicts with port header prototypes under true SMP builds.

```
diff --git a/include/cmsis-plus/rtos/os-c-decls.h b/include/cmsis-plus/rtos/os-c-decls.h
index 86dc5429..e3cddb30 100644
--- a/include/cmsis-plus/rtos/os-c-decls.h
+++ b/include/cmsis-plus/rtos/os-c-decls.h
@@ -316,7 +316,6 @@ extern "C"
     * @brief Used to check reused threads.
     */
     os_thread_state_initialising = 6,
-
-    /**
-     * @brief In process of being destroyed.
-     */
@@ -546,6 +545,10 @@ extern "C"
     */
     bool th_enable_assert_reuse;

+
+#if defined(OS_USE_SMP_SCHEDULER)
+    uint32_t th_cpu_affinity;
+#endif
+
     } os_thread_attr_t;

     /**
@@ -592,6 +595,9 @@ extern "C"
     os_thread_state_t state;
     os_thread_prio_t prio_assigned;
     os_thread_prio_t prio_inherited;
+
+#if defined(OS_USE_SMP_SCHEDULER)
+    uint32_t cpu_affinity;
+#endif /* defined(OS_USE_SMP_SCHEDULER) */
     bool interrupted;
     os_internal_evflags_t event_flags;
     #if defined(OS_INCLUDE_RTOS_CUSTOM_THREAD_USER_STORAGE)
diff --git a/include/cmsis-plus/rtos/os-sched.h b/include/cmsis-plus/rtos/os-sched.h
index 0a2b4e78..f3b6ab60 100644
--- a/include/cmsis-plus/rtos/os-sched.h
```

```

+++ b/include/cmsis-plus/rtos/os-sched.h
@@ -55,7 +55,11 @@ namespace os

    #if !defined(OS_USE_RTOS_PORT_SCHEDULER)
        extern bool is_preemptive_;
+    #if defined(OS_USE_SMP_SCHEDULER)
+        extern thread* volatile current_thread_[OS_NCPU];
+    #else
        extern thread* volatile current_thread_;
+    #endif /* defined(OS_USE_SMP_SCHEDULER) */
        extern internal::ready_threads_list ready_threads_list_;
    #endif /* !defined(OS_USE_RTOS_PORT_SCHEDULER) */

diff --git a/include/cmsis-plus/rtos/os-thread.h b/include/cmsis-plus/
rtos/os-thread.h
index cac6777f..385e6cb2 100644
--- a/include/cmsis-plus/rtos/os-thread.h
+++ b/include/cmsis-plus/rtos/os-thread.h
@@ -745,6 +745,14 @@ namespace os
    friend port::stack::element_t*
    port::scheduler::switch_stacks (port::stack::element_t* sp);

+    #if defined(OS_USE_SMP_SCHEDULER)
+        // The SMP picker accesses state_ / context_ directly.
+        friend void
+        rtos::scheduler::internal_switch_threads (void);
+        // The idle reaper reads context_ to know a terminated
+        thread's
+        // context is no longer live on another CPU (see os-
+        idle.cpp).
+        friend void ::os_rtos_idle_actions (void);
+    #endif /* defined(OS_USE_SMP_SCHEDULER) */

    #endif

    /**
@@ -880,6 +888,9 @@ namespace os

        bool th_enable_assert_reuse = false;

+    #if defined(OS_USE_SMP_SCHEDULER)
+        uint32_t th_cpu_affinity = 0xFFFFFFFFu;
+    #endif

        // Add more attributes here.

@@ -1198,6 +1209,13 @@ namespace os
        priority_t
        priority_inherited (void);

+    #if defined(OS_USE_SMP_SCHEDULER)
+        uint32_t
+        cpu_affinity (void) const;
+
+        void
+        cpu_affinity (uint32_t mask);
+    #endif /* defined(OS_USE_SMP_SCHEDULER) */

    #if 0
        // ???
@@ -1696,6 +1714,9 @@ namespace os
        priority_t volatile prio_assigned_ = priority::none;
        priority_t volatile prio_inherited_ = priority::none;

+    #if defined(OS_USE_SMP_SCHEDULER)
+        uint32_t cpu_affinity_ = 0xFFFFFFFFu;
+    #endif /* defined(OS_USE_SMP_SCHEDULER) */

```

```

        bool volatile interrupted_ = false;

diff --git a/src/rtos/os-c-wrapper.cpp b/src/rtos/os-c-wrapper.cpp
index 4bd9e1bb..1c37050d 100644
--- a/src/rtos/os-c-wrapper.cpp
+++ b/src/rtos/os-c-wrapper.cpp
@@ -239,6 +239,8 @@ static_assert (os_thread_state_terminated ==
    thread::state::terminated,
        "adjust os_thread_state_terminated");
    static_assert (os_thread_state_destroyed == thread::state::destroyed,
        "adjust os_thread_state_destroyed");
+static_assert (os_thread_state_destroying ==
    thread::state::destroying,
+    "adjust os_thread_state_destroying");

    static_assert (os_timer_once == timer::run::once, "adjust
        os_timer_once");
    static_assert (os_timer_periodic == timer::run::periodic,
@@ -353,6 +354,12 @@ osThreadCreate (const osThreadDef_t*
    thread_def, void* args)
    thread::attributes attr;
    attr.th_priority = thread_def->tpriority;
    attr.th_stack_size_bytes = thread_def->stacksize;
+if defined(OS_USE_SMP_SCHEDULER)
+    // CMSIS-RTOS v1 assumes a single-core priority scheduling model;
+    // pin threads created through the CMSIS-RTOS v1 API to Core 0 at
+    construction.
+    attr.th_cpu_affinity = (1u << 0);
+endif
+
    // Creating thread with invalid priority should fail (validator
    requirement).
    if (thread_def->tpriority >= osPriorityError)
    {
@@ -357,6 +358,12 @@ osThreadCreate (const osThreadDef_t*
    thread_def, void* args)
        args, attr);
#pragma GCC diagnostic pop

+if defined(OS_USE_SMP_SCHEDULER)
+    // CMSIS-RTOS v1 assumes a single-core priority scheduling
+    model;
+    // pin threads created through the CMSIS-RTOS v1 API to
+    Core 0.
+    th->cpu_affinity (1u << 0);
+endif
+
    // No need to yield here, already done by constructor.
    return reinterpret_cast<osThreadId> (th);
}

diff --git a/src/rtos/os-core.cpp b/src/rtos/os-core.cpp
index c9d6859e..8b88a471 100644
--- a/src/rtos/os-core.cpp
+++ b/src/rtos/os-core.cpp
@@ -24,6 +24,11 @@
// -----

+if defined(OS_USE_SMP_SCHEDULER)
+// Provided by the SMP port (e.g. Cortex-A7); returns the current
+    core index.
+extern "C" unsigned port_cpu_id(void);
+endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+namespace
+{
+    #if defined(OS_HAS_INTERRUPTS_STACK)
@@ -130,10 +135,21 @@ namespace os

#pragma GCC diagnostic push

```

```

#pragma GCC diagnostic ignored "-Wcast-align"
+#if defined(OS_USE_SMP_SCHEDULER)
+    // One running thread per CPU, indexed by port_cpu_id().
+    thread* volatile current_thread_[OS_NCPU]
+        = {reinterpret_cast<thread*> (&tiny_thread)};
+#else
+    thread* volatile current_thread_
+        = reinterpret_cast<thread*> (&tiny_thread);
+#endif /* defined(OS_USE_SMP_SCHEDULER) */
#pragma GCC diagnostic pop

+#if defined(OS_USE_SMP_SCHEDULER)
+    // One idle thread per CPU (secondary cores register theirs at
+    // boot).
+    thread* os_idle_thread_core[OS_NCPU] = {nullptr};
+#endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+    #pragma GCC diagnostic push
+    #if defined(__clang__)
+    #pragma clang diagnostic ignored "-Wglobal-constructors"
@@ -462,6 +478,45 @@ namespace os

    #if !defined(OS_USE_RTOS_PORT_SCHEDULER)

+#if defined(OS_USE_SMP_SCHEDULER)
+    // Affinity gate: a per-core idle thread is pinned to its own
+    // core,
+    // everything else follows its cpu_affinity() bitmask.
+    //
+    // Identity against os_idle_thread_core[] is the authoritative
+    // test: it
+    // covers every core whatever the thread is named. The name
+    // rules are
+    // only the fallback for the boot window, between the moment a
+    // secondary
+    // idle thread is linked into the ready list by its constructor
+    // and the
+    // moment it is registered in os_idle_thread_core[]. Without a
+    // rule for
+    // core OS_NCPU-1 an unregistered idle thread inherits the
+    // default
+    // 0xFFFFFFFF affinity, so another core can pick it out of the
+    // ready list
+    // and run it on the same stack as its own core does.
+    static bool
+    is_thread_allowed_on_cpu (thread* th, unsigned cpu)
+    {
+        for (unsigned c = 0; c < OS_NCPU; ++c)
+        {
+            if (th == scheduler::os_idle_thread_core[c])
+            {
+                return (cpu == c);
+            }
+        }
+        const char* name = th->name ();
+        if (name != nullptr && name[0] == 'i' && name[1] == 'd'
+            && name[2] == 'l' && name[3] == 'e')
+        {
+            if (name[4] == '\0' || name[4] == '0')
+            {
+                return (cpu == 0);
+            }
+            if (name[4] >= '1' && name[4] <= '9' && name[5] == '\0')
+            {
+                return (cpu == static_cast<unsigned> (name[4] -
+'0'));
+            }
+        }
+        return (th->cpu_affinity () & (1u << cpu)) != 0;
+    }
}

```

```

+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+     void
+     internal_switch_threads (void)
+     {
+@@ -490,7 +545,11 @@ namespace os
+         scheduler::statistics::cpu_cycles_ += delta;
+
+         // Accumulate durations to old thread.
+ #if defined(OS_USE_SMP_SCHEDULER)
+ +         scheduler::current_thread_[port_cpu_id()]-
+         >statistics_.cpu_cycles_ += delta;
+ #else
+         scheduler::current_thread_>statistics_.cpu_cycles_ += delta;
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+         // Remember the timestamp for the next context switch.
+         scheduler::statistics::switch_timestamp_ = now;
+@@ -501,6 +560,77 @@ namespace os
+         // current thread and return the top priority thread.
+         if (!locked ())
+         {
+ #if defined(OS_USE_SMP_SCHEDULER)
+ +         unsigned cpu = port_cpu_id ();
+ +         thread* old_thread = scheduler::current_thread_[cpu];
+ +
+ +         instrumentation::thread::suspended (
+ +         old_thread,
+ +         OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_SWITCH);
+ +
+ +         bool is_old_idle = false;
+ +         for (unsigned c = 0; c < OS_NCPU; ++c)
+ +         {
+ +             if (old_thread == scheduler::os_idle_thread_core[c])
+ +             {
+ +                 is_old_idle = true;
+ +                 break;
+ +             }
+ +         }
+ +
+ +         if (!is_old_idle)
+ +         {
+ +             old_thread->internal_relink_running_ ();
+ +         }
+ +         else
+ +         {
+ +             old_thread->state_ = thread::state::ready;
+ +         }
+ +
+ +         // Affinity-aware pick from the ready list. Skip threads
+ +         whose
+ +         // stack_ptr is still nullptr (their context is live in
+ +         another
+ +         // CPU's registers -- publish is deferred to the asm
+ +         restore path);
+ +         // this CPU's own outgoing thread may always be re-
+ +         picked.
+ +         thread* next_thread = nullptr;
+ +         auto* sentinel =
+ +         reinterpret_cast<internal::waiting_thread_node*> (
+ +         &scheduler::ready_threads_list_);
+ +         auto* node = const_cast<internal::waiting_thread_node*> (
+ +         static_cast<const volatile
+ +         internal::waiting_thread_node*> (
+ +         scheduler::ready_threads_list_.head ()));
+ +
+ +         while (node != nullptr && node != sentinel)
+ +         {
+ +             thread* th = node->thread_;

```

```

+         if (th != nullptr && is_thread_allowed_on_cpu (th,
+             cpu)
+             && (th == old_thread
+                 || (th->state_ != thread::state::running
+                     && th->context_.port_.stack_ptr !=
+                 nullptr)))
+             {
+                 next_thread = th;
+                 node->unlink ();
+                 next_thread->state_ = thread::state::running;
+                 break;
+             }
+             node = static_cast<internal::waiting_thread_node*> (
+                 node->next ());
+         }
+
+         if (next_thread != nullptr)
+         {
+             scheduler::current_thread[cpu] = next_thread;
+         }
+         else
+         {
+             scheduler::current_thread[cpu]
+                 = scheduler::os_idle_thread_core[cpu];
+             if (scheduler::os_idle_thread_core[cpu] != nullptr)
+             {
+                 scheduler::os_idle_thread_core[cpu]->state_
+                     = thread::state::running;
+             }
+         }
+
+         instrumentation::thread::active
+             (scheduler::current_thread[cpu]);
+     #else
+         instrumentation::thread::suspended (
+             scheduler::current_thread_,
+             OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_SWITCH);
+@@ -512,6 +642,7 @@ namespace os
+             = scheduler::ready_threads_list_.unlink_head ();
+
+             instrumentation::thread::active
+                 (scheduler::current_thread_);
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+     }
+
+     // ***** Pointer switched to new thread! *****
+@@ -527,7 +658,11 @@ namespace os
+         scheduler::statistics::context_switches_++;
+
+         // Increment new thread context switches.
+ #if defined(OS_USE_SMP_SCHEDULER)
+         scheduler::current_thread[port_cpu_id()]-
+             >statistics_.context_switches_++;
+ #else
+         scheduler::current_thread->statistics_.context_switches_++;
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+ #endif /* defined(OS_INCLUDE_RTOS_STATISTICS_THREAD_CONTEXT_SWITCHES)
+     */
+ }
+
+ diff --git a/src/rtos/os-idle.cpp b/src/rtos/os-idle.cpp
+ index 0db1ab35..10272366 100644
+ --- a/src/rtos/os-idle.cpp
+ +++ b/src/rtos/os-idle.cpp
+@@ -80,6 +80,28 @@ os_rtos_idle_actions (void)
+     node = const_cast<internal::waiting_thread_node*> (
+         scheduler::terminated_threads_list_.head ());
+     thread* th = node->thread_;

```

```

+ #if defined(OS_USE_SMP_SCHEDULER)
+     // A thread links itself here in internal_exit() and only
+     then
+     // switches away, so on SMP it may still be running on
+     another CPU.
+     // Destroying it now frees the stack that CPU is executing
+     on, and
+     // lets the joiner delete the object under it. Reap it only
+     once no
+     // CPU has it current and its context has been saved -- the
+     same
+     // rule internal_switch_threads() applies to the ready list
+     -- and
+     // otherwise leave it linked for the next idle pass.
+     bool live = (__atomic_load_n (&th->context_.port_.stack_ptr,
+                                     __ATOMIC_ACQUIRE)
+                 == nullptr);
+     for (unsigned c = 0; c < OS_NCPU && !live; ++c)
+     {
+         live = (__atomic_load_n (&scheduler::current_thread_[c],
+                                     __ATOMIC_ACQUIRE)
+                 == th);
+     }
+     if (live)
+     {
+         break;
+     }
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+     if (th->state_ == thread::state::destroying
+         || th->state_ == thread::state::destroyed)
+     {
diff --git a/src/rtos/os-main.cpp b/src/rtos/os-main.cpp
index 67782880..546c709e 100644
--- a/src/rtos/os-main.cpp
+++ b/src/rtos/os-main.cpp
@@ -165,6 +165,9 @@ int

        thread::attributes attr = thread::initializer;
        attr.th_stack_size_bytes = OS_INTEGER_RTOS_MAIN_STACK_SIZE_BYTES;
+ #if defined(OS_USE_SMP_SCHEDULER)
+     attr.th_cpu_affinity = (1u << 0);
+ #endif
        #pragma GCC diagnostic push
        #if defined(__clang__)
        #pragma clang diagnostic ignored "-Wcast-function-type-strict"
@@ -176,6 +179,12 @@ int

        #endif /* defined(OS_EXCLUDE_DYNAMIC_MEMORY_ALLOCATIONS) */

+ #if defined(OS_USE_SMP_SCHEDULER)
+ // Pin main thread to core 0 so per-core peripherals (NVIC,
+ // SysTick) initialized
+ // on the boot core remain consistent for the main thread
+ // lifecycle.
+ os_main_thread->cpu_affinity (1u << 0);
+ #endif
+
+ #if !defined(OS_USE_RTOS_PORT_SCHEDULER)
+     os_startup_create_thread_idle ();
+ #endif /* !defined(OS_USE_RTOS_PORT_SCHEDULER) */
diff --git a/src/rtos/os-thread.cpp b/src/rtos/os-thread.cpp
index 79b0a71c..0aff28a9 100644
--- a/src/rtos/os-thread.cpp
+++ b/src/rtos/os-thread.cpp
@@ -19,6 +19,16 @@
+ #include <memory>
+ #include <stdexcept>

+ #if defined(OS_USE_SMP_SCHEDULER)

```

```

+extern "C" unsigned port_cpu_id(void);
+extern "C" void port_smp_ipi(unsigned cpu) __attribute__((weak));
+
+extern "C" void __attribute__((weak))
+port_smp_ipi (unsigned cpu)
+{
+ (void) cpu;
+}
+#endif /* defined(OS_USE_SMP_SCHEDULER) */

// -----

@@ -272,6 +282,9 @@ namespace os
    // Must be explicit here, since they are not done in the
    // members
    // declarations to allow th_enable_assert_reuse.
    state_ = state::initializing;
+#if defined(OS_USE_SMP_SCHEDULER)
+    cpu_affinity_ = 0xFFFFFFFFu;
+#endif /* defined(OS_USE_SMP_SCHEDULER) */

    instrumentation::thread::create_return (this);
}

@@ -286,6 +299,9 @@ namespace os
    // Must be explicit here, since they are not done in the
    // members
    // declarations to allow th_enable_assert_reuse.
    state_ = state::initializing;
+#if defined(OS_USE_SMP_SCHEDULER)
+    cpu_affinity_ = 0xFFFFFFFFu;
+#endif /* defined(OS_USE_SMP_SCHEDULER) */

    instrumentation::thread::create_return (this);
}

@@ -440,6 +456,9 @@ namespace os
    #endif /* DEBUG */

    state_ = state::initializing;
+#if defined(OS_USE_SMP_SCHEDULER)
+    cpu_affinity_ = attr.th_cpu_affinity;
+#endif /* defined(OS_USE_SMP_SCHEDULER) */

    allocator_ = &allocator;

@@ -578,7 +597,11 @@ namespace os

    if (!scheduler::started ())
    {
+#if defined(OS_USE_SMP_SCHEDULER)
+        scheduler::current_thread[port_cpu_id()] = this;
+#else
        scheduler::current_thread_ = this;
+#endif /* defined(OS_USE_SMP_SCHEDULER) */
    }

    // Add to ready list, but do not yield yet.
@@ -697,6 +720,23 @@ namespace os

    port::scheduler::reschedule ();

+#if defined(OS_INTEGER_RTOS_PORT_NCPU) && (OS_INTEGER_RTOS_PORT_NCPU
+    > 1)
+    // If the waking CPU is not in this thread's affinity mask,
+    // actively trigger a reschedule IPI to an eligible core so it
+    // doesn't wait up to 1 ms for its next local clock tick.
+    unsigned this_cpu = port_cpu_id ();
+    if ((cpu_affinity () & (1u << this_cpu)) == 0)

```

```

+         {
+             for (unsigned c = 0; c < OS_NCPU; ++c)
+             {
+                 if (c != this_cpu && (cpu_affinity () & (1u << c)) !=
+                     0)
+                 {
+                     port_smp_ipi (c);
+                     break;
+                 }
+             }
+         }
+     }
+ }
+
+ #endif
+
+ #endif
+
+ @@ -709,6 +749,19 @@ namespace os
+     *
+     * @note Can be invoked from Interrupt Service Routines.
+     */
+
+ #if defined(OS_USE_SMP_SCHEDULER)
+     uint32_t
+     thread::cpu_affinity (void) const
+     {
+         return cpu_affinity_;
+     }
+
+     void
+     thread::cpu_affinity (uint32_t mask)
+     {
+         cpu_affinity_ = mask;
+     }
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+     thread::priority_t
+     thread::priority (void)
+
+ @@ -1064,6 +1117,30 @@ namespace os
+         port::scheduler::reschedule ();
+     }
+
+
+ #if defined(OS_USE_SMP_SCHEDULER)
+     // On SMP, the joined thread may still be running on another
+     // CPU
+     // completing its final reschedule() context switch. Wait until
+     // it is
+     // no longer current on any CPU before returning, so the caller
+     // can
+     // safely reclaim the thread object and stack memory.
+     for (;;)
+     {
+         bool still_running = false;
+         for (unsigned c = 0; c < OS_NCPU; ++c)
+         {
+             if (__atomic_load_n (&scheduler::current_thread[c],
+                                   __ATOMIC_ACQUIRE) == this)
+             {
+                 still_running = true;
+                 break;
+             }
+         }
+         if (!still_running)
+         {
+             break;
+         }
+         this_thread::yield ();
+     }
+ }
+
+ #endif

```

```

    #if defined(OS_TRACE_RTOS_THREAD)
        trace::printf ("%s() @%p %s joined\n", __func__, this, name
            ());
@@ -1384,6 +1461,69 @@ namespace os
    // ----- Enter critical section
    -----
    scheduler::critical_section scs;

+ #if defined(OS_USE_SMP_SCHEDULER)
+     // On SMP the thread may be running on another CPU --
+     typically
+     // still inside its own internal_exit(), right after the
+     event that
+     // let this caller go on -- or be terminated and already
+     taken off
+     // the funeral list by an idle reaper, which destroys it
+     outside the
+     // lock. Destroying it here in either case frees the stack
+     another
+     // CPU runs on, or destroys it twice. So wait, with the lock
+     // released, until it is off every CPU and unclaimed; from
+     then on
+     // the lock held keeps it so.
+     for (;;)
+     {
+         if (__atomic_load_n (&state_, __ATOMIC_ACQUIRE) ==
+ state::destroyed)
+         {
+             break;
+         }

+         bool busy = (__atomic_load_n (&context_.port_.stack_ptr,
+                                     __ATOMIC_ACQUIRE)
+                     == nullptr);
+         unsigned busy_cpu = OS_NCPU;
+         for (unsigned c = 0; c < OS_NCPU; ++c)
+         {
+             if (__atomic_load_n (&scheduler::current_thread_[c],
+                                     __ATOMIC_ACQUIRE)
+                 == this)
+             {
+                 busy = true;
+                 busy_cpu = c;
+                 break;
+             }
+         }

+         thread::state_t st = __atomic_load_n (&state_,
+                                     __ATOMIC_ACQUIRE);
+         if (st == state::destroying)
+         {
+             busy = true;
+         }
+         else if (st == state::terminated && ready_node_.unlinked
+ ())
+         {
+             // Idle reaper claimed it and will destroy it.
+             busy = true;
+         }

+         if (!busy)
+         {
+             break;
+         }

+         // Actively trigger reschedule IPI if running on another
+         core
+         if (busy_cpu < OS_NCPU && busy_cpu != port_cpu_id ())
+         {
+             port_smp_ipi (busy_cpu);

```

```

+         }
+
+         {
+             // ----- Enter uncritical section
+             -----
+             scheduler::uncritical_section sucs;
+
+             this_thread::yield ();
+             // ----- Exit uncritical section
+             -----
+         }
+     }
+ }
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+     if (state_ == state::destroyed)
+     {
+@@ -1845,7 +1985,22 @@ namespace os
+
+     #else
+
+ #if defined(OS_USE_SMP_SCHEDULER)
+     {
+         // Read the core id and that core's current thread with
+         this core's
+         // interrupts masked. With them enabled, a preemption
+         between the
+         // two reads can move the caller to another core, and it
+         would get
+         // back the thread now running on the core it left -- as a
+         mutex
+         // owner, a waiter, or the errno it writes. The IRQ
+         critical section
+         // is the per-core mask every port provides; it also takes
+         the
+         // kernel lock, recursively, so a caller already inside one
+         pays
+         // nothing more.
+         interrupts::critical_section ics;
+         th = scheduler::current_thread_[port_cpu_id ()];
+     }
+ #else
+         th = scheduler::current_thread_;
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+ #endif
+
+     return th;

```

## PR #23 (POSIX-Arch): Host Multiprocessing Runtime & native-smp

### Why Part B POSIX Architecture Infrastructure is Needed:

- **Threads-as-Cores Multi-Processing Emulation:** In `include/host_cpu.hpp` and `src/host_cpu.cpp`, spawns `OS_NCPU` host pthreads to model physical processor cores. Each core configures a dedicated POSIX interval timer (`timer_create(CLOCK_MONOTONIC, SIGEV_THREAD_ID)`) delivering periodic tick interrupts.
- **Inter-Processor Interrupts (IPI) via POSIX Real-Time Signals:** Implements `port_smp_ipi(unsigned cpu)` using `pthread_kill(..., SIGRTMIN+1)` to signal target cores for immediate reschedule.
- **libucontext Compatibility (Fix C):** Guards `uc_sigmask` manipulation with `#if !defined(OS_INCLUDE_LIBUCONTEXT)` since standard `<ucontext.h>` has `uc_sigmask` while standalone libucontext does not.
- **Clang -Weverything Hardening (Fixes I & J):** Injects comprehensive Clang diagnostic suppression blocks covering `-Wc++98-compat-pedantic`, `-Wreserved-identifier`, `-Wunsafe-buffer-usage`, and `-Wcast-function-type-strict`.

- **Additive CMake Target Registration (Fix D):** Registers `src/host_cpu.cpp`, `src/free-store.cpp`, and `src/exception_handler.cpp` in `micro-os-plus-iii-posix-arch-interface` without altering baseline target structure.

```
diff --git a/include/cmsis-plus/rtos/port/os-inlines.h b/include/
      cmsis-plus/rtos/port/os-inlines.h
index 93944a5..0ec7ab7 100644
--- a/include/cmsis-plus/rtos/port/os-inlines.h
+++ b/include/cmsis-plus/rtos/port/os-inlines.h
@@ -129,6 +129,21 @@ namespace os

    //
    =====

+   inline constexpr bool __attribute__((always_inline))
+   clock_highres::has_hardware_counter (void) noexcept
+   {
+       return true;
+   }
+
+   inline uint64_t __attribute__((always_inline))
+   clock_highres::hardware_counter (void) noexcept
+   {
+       timespec tp; // NB: no 'struct' – baseline gate has
+       -Werror=redundant-tags
+       ::clock_gettime (CLOCK_MONOTONIC, &tp);
+       return static_cast<uint64_t> (tp.tv_sec) * 1000000ULL
+          + static_cast<uint64_t> (tp.tv_nsec) / 1000ULL;
+   }
+
    } /* namespace port */
    } /* namespace rtos */
    } /* namespace os */
diff --git a/CMakeLists.txt b/CMakeLists.txt
index ae85397..ba48588 100644
--- a/CMakeLists.txt
+++ b/CMakeLists.txt
@@ -49,6 +49,9 @@ target_include_directories(micro-os-plus-iii-posix-
      arch-interface INTERFACE
      target_sources(micro-os-plus-iii-posix-arch-interface INTERFACE
          src/diag/trace-posix.cpp
          src/rtos/os-core.cpp
+         src/host_cpu.cpp
+         src/free-store.cpp
+         src/exception_handler.cpp
      )

      target_compile_definitions(micro-os-plus-iii-posix-arch-interface
          INTERFACE
diff --git a/include/cmsis-plus/rtos/port/os-c-decls.h b/include/
      cmsis-plus/rtos/port/os-c-decls.h
index c989c07..a274f67 100644
--- a/include/cmsis-plus/rtos/port/os-c-decls.h
+++ b/include/cmsis-plus/rtos/port/os-c-decls.h
@@ -1,29 +1,15 @@
/*
 * os-c-decls.h - the width-dependent half of the POSIX port
contract.
 *
 * This file is part of the µOS++ project (https://micro-os-
plus.github.io/).
 * Copyright (c) 2016-2025 Liviu Ionescu. All rights reserved.
 * Copyright (c) 2026 Dan. All rights reserved.
 *
 * Permission to use, copy, modify, and/or distribute this software
 * for any purpose is hereby granted, under the terms of the MIT
license.
 *

```

```

- * If a copy of the license was not distributed with this file, it
- * can
- * be obtained from https://opensource.org/licenses/mit/.
- */
-
-/*
- * The initial CMSIS++ RTOS API was inspired by CMSIS RTOS API v1.x,
- * Copyright (c) 2013 ARM LIMITED
- */
-
-/*
- * This file is part of the CMSIS++ proposal, intended as a CMSIS
- * replacement for C++ applications.
- *
- * It is included in `cmsis-plus/rtos/os-c-decls.h` to customise
- * it with port specific declarations.
- *
- * These structures (which basically contain handlers)
- * are conditionally included in the system objects
- * when they are implemented using the port native objects.
+ * Derived from upstream micro-os-plus-iii-posix-arch v1.0.1. What
+ * changed and
+ * why is in os-decls.h, beside the declarations it changed.
+ */

#ifdef CMSIS_PLUS_RTOS_PORT_OS_C_DECLS_H
@@ -69,17 +55,39 @@ typedef uint32_t os_port_clock_duration_t;
// Must match port::clock::offset_t
typedef uint64_t os_port_clock_offset_t;

+typedef uint64_t os_port_thread_stack_element_t;
+typedef uint64_t os_port_thread_stack_allocation_element_t;
+
+/*
+ * The thread context.
+ *
+ * `stack_ptr` FIRST and `ucontext` second, and the order is not
+ * cosmetic.
+ *
+ * Upstream's context was the ucontext alone, because upstream had
+ * one CPU and
+ * therefore never had to ask whether a saved context was safe to
+ * resume. The
+ * SMP scheduler does ask, in the kernel, in a line no port may edit:
+ *
+ *     src/rtos/os-core.cpp, internal_switch_threads():
+ *     && (th == old_thread || th->context_.port_.stack_ptr !=
+ *         nullptr)
+ *
+ * On the ARM ports that field is the outgoing stack pointer,
+ * published by the
+ * assembly restore path only once the CPU has left the outgoing
+ * thread's
+ * stack. Here it is the same gate carrying the same meaning -- "this
+ * context
+ * is fully saved, another CPU may claim it" -- and nothing more: the
+ * register
+ * state lives in the ucontext. A CPU clears it when it claims a
+ * thread and
+ * the NEXT context to run on that CPU publishes it, which is the
+ * whole of the
+ * deferred-publish rule. See host_cpu.cpp.
+ */
typedef struct
{
- os_impl_ucontext_t ucontext; //
+ os_port_thread_stack_element_t* stack_ptr;
+ os_impl_ucontext_t ucontext;
} os_port_thread_context_t;

```

```

typedef bool os_port_scheduler_state_t;

-// Signal set (true if signal blocked)
+/* True when this CPU's interrupt signals are blocked. A per-thread
   signal
+ * mask IS per-CPU interrupt masking, because a host thread IS a CPU.
   */
typedef bool os_port_irq_state_t;

-typedef uint64_t os_port_thread_stack_element_t;
-typedef uint64_t os_port_thread_stack_allocation_element_t;
-
-#endif /* CMSIS_PLUS_RTOS_PORT_OS_C_DECLS_H_ */
diff --git a/include/cmsis-plus/rtos/port/os-decls.h b/include/cmsis-
plus/rtos/port/os-decls.h
index 4ee273a..9cc63ca 100644
--- a/include/cmsis-plus/rtos/port/os-decls.h
+++ b/include/cmsis-plus/rtos/port/os-decls.h
@@ -1,51 +1,56 @@
/*
+ * os-decls.h - the C++ half of the POSIX port contract.
+ *
+ * This file is part of the µOS++ project (https://micro-os-
+ plus.github.io/).
+ * Copyright (c) 2016-2025 Liviu Ionescu. All rights reserved.
+ * Copyright (c) 2026 Dan. All rights reserved.
+ *
+ * Permission to use, copy, modify, and/or distribute this software
+ * for any purpose is hereby granted, under the terms of the MIT
+ license.
+ *
- * If a copy of the license was not distributed with this file, it
- can
- * be obtained from https://opensource.org/licenses/mit/.
- */
-
-/*
- * This file is part of the CMSIS++ proposal, intended as a CMSIS
- * replacement for C++ applications.
- *
- * It is included in `cmsis-plus/rtos/os.h` to customise
- * it with POSIX specific declarations.
+ * This port does NOT link micro-os-plus::port-smp-decls, and that
+ target's
+ * own README says why: the shared copy is written for ports whose
+ state types
+ * are integers and whose thread context is a bare stack pointer.
+ Neither is
+ * true here -- the interrupt state is a signal-mask bit, the
+ scheduler state
+ * is a bool, and the context carries a ucontext. The SMP
+ declarations the
+ * kernel actually reads (lock_state[], _smp_klock,
+ _port_ctx_pending[]) are
+ * reproduced below with the same names, the same types and the same
+ * volatility, so the kernel cannot tell the two apart.
+ */

#ifndef CMSIS_PLUS_RTOS_PORT_OS_DECLS_H_
#define CMSIS_PLUS_RTOS_PORT_OS_DECLS_H_

-// -----
-
#if defined(OS_USE_OS_APP_CONFIG_H)
#include <cmsis-plus/os-app-config.h>
#endif

#include <cmsis-plus/rtos/port/os-c-decls.h>

-// -----

```

```

+ #if !defined(OS_NCPU)
+ #define OS_NCPU (1)
+ #endif

+ #if !defined(OS_INTEGER_SYSTICK_FREQUENCY_HZ)
+ #define OS_INTEGER_SYSTICK_FREQUENCY_HZ (1000)
+ #endif

+ /*
+  * Host stacks are large because a signal frame lands on them.
+  *
+  * Preemption here is a signal delivered on the running thread's own
+  * stack --
+  * deliberately, with no SA_ONSTACK, so that the whole signal frame
+  * migrates
+  * with the thread when another CPU resumes it (see host_cpu.cpp).
+  * That frame
+  * is several kilobytes of ucontext plus the FPU state, so upstream's
+  * 32 KiB
+  * minimum is kept rather than the 2 KiB the ARM ports use.
+  */
+ #if !defined(OS_INTEGER_RTOS_MIN_STACK_SIZE_BYTES)
+ #define OS_INTEGER_RTOS_MIN_STACK_SIZE_BYTES (32 * 1024)
+ #endif

+ #if !defined(OS_INTEGER_RTOS_DEFAULT_STACK_SIZE_BYTES)
+ #if defined(__linux__)
+ #define OS_INTEGER_RTOS_DEFAULT_STACK_SIZE_BYTES \
+ (2 * OS_INTEGER_RTOS_MIN_STACK_SIZE_BYTES)
+ #else
+ #define OS_INTEGER_RTOS_DEFAULT_STACK_SIZE_BYTES \
+ (OS_INTEGER_RTOS_MIN_STACK_SIZE_BYTES)
+ #endif /* defined(__linux__) */
+ #endif

+ #if !defined(OS_INTEGER_RTOS_MAIN_STACK_SIZE_BYTES)
@@ -58,59 +63,52 @@
+ (OS_INTEGER_RTOS_DEFAULT_STACK_SIZE_BYTES)
+ #endif

+ // -----
+ #if !defined(OS_INTEGER_RTOS_STACK_FILL_MAGIC)
+ #define OS_INTEGER_RTOS_STACK_FILL_MAGIC (0xEFBEADDEEFBEADDEULL)
+ #endif

+ #ifdef __cplusplus
+ /* Kept identical to the ARM ports so the same code reads on both. */
+ #if !defined(SMP_NO_OWNER)
+ #define SMP_NO_OWNER (0xFFFFFFFFu)
+ #endif

+ // -----
+ #ifdef __cplusplus

+ #include <signal.h>
+ // Platform definitions
+ #include <sys/time.h>

+ #include <cstdint>
+ #include <cstddef>

+ // -----
+ #pragma GCC diagnostic push

+ #if defined(__clang__)
+ #pragma clang diagnostic ignored "-Wc++98-compat"

```

```

#endif

-// -----
-
namespace os
{
    namespace rtos
    {
        namespace port
        {
-            // -----

            namespace stack
            {
-                // Assume 64-bits core.
                using element_t = os_port_thread_stack_element_t;

-                // Align stack to 8 bytes.
                using allocation_element_t =
                    os_port_thread_stack_allocation_element_t;

-                // Initial value for the minimum stack size in bytes.
                constexpr std::size_t min_size_bytes
                    = OS_INTEGER_RTOS_MIN_STACK_SIZE_BYTES;

-                // Initial value for the default stack size in bytes.
                constexpr std::size_t default_size_bytes
                    = OS_INTEGER_RTOS_DEFAULT_STACK_SIZE_BYTES;

-                constexpr element_t magic = 0xEFBEADDEEFBEADDE;
+                constexpr element_t magic = OS_INTEGER_RTOS_STACK_FILL_MAGIC;
            } /* namespace stack */

            namespace interrupts
            {
-                // True if signal blocked
                using state_t = os_port_irq_state_t;

                namespace state
@@ -118,8 +116,19 @@ namespace os
                constexpr state_t init = false;
            } /* namespace state */

-            extern sigset_t clock_set;
+            /* The signals this port treats as interrupts: the per-CPU
+            tick and
+            * the IPI. Blocking them IS masking interrupts on this CPU.
+            */
+            extern sigset_t irq_set;

+            /* Per-CPU handler-mode flag. AArch64 has the same array for
+            the same
+            * reason: there are no banked modes to ask. */
+            extern "C" volatile bool _in_isr[OS_NCPU];

+            /* The Arm CMSIS RTOS validator's host shim raises its
+            "interrupt" with
+            * kill(SIGUSR1) and increments this around the handler. The
+            port has
+            * no banked mode to ask either, so a non-zero value also
+            means handler
+            * mode (see in_handler_mode()). Defined in src/rtos/os-
+            core.cpp. */
+            extern "C" volatile uint32_t signal_nesting;
        } /* namespace interrupts */

        namespace scheduler
@@ -133,23 +142,46 @@ namespace os

```

```

    constexpr state_t init = unlocked;
} /* namespace state */

-
+ extern state_t lock_state;
+ extern volatile state_t lock_state[OS_NCPU];

+
+ /* Byte-for-byte the ARM ports' recursive kernel lock (lock
+ word,
+ * owner CPU, nesting depth). The ARM timer leaf lock
+ (smp_tlock_t,
+ * port_tmr_lock/unlock) is deliberately NOT carried here: it
+ exists
+ * for a kernel timer back-end that takes it, and this port's
+ timers
+ * are POSIX timers driven through host_cpu, so it would be
+ dead. */
+ struct smp_klock_t
+ {
+     volatile uint32_t lock;
+     volatile uint32_t owner;
+     volatile uint32_t depth;
+ };

+
+ extern smp_klock_t _smp_klock;
+ extern volatile unsigned _port_ctx_pending[OS_NCPU];
+ } /* namespace scheduler */

namespace clock
{
-
+     constexpr unsigned int signal_number = SIGALRM;
-
+ } /* namespace clock */
+ /* Not SIGALRM. ITIMER_REAL and SIGALRM are process-wide;
+ this port
+ * gives every CPU its own timer_create() timer, and a real-
+ time
+ * signal because those queue rather than coalesce.
+ *
+ * Functions, not constants: glibc's SIGRTMIN expands to a
+ call to
+ * __libc_current_sigrtmin(), so it is not a constant
+ expression. */
+ inline int
+ signal_number (void)
+ {
+     return SIGRTMIN;
+ }

-
+ using thread_context_t = struct thread_context_s
+ {
-
+     // On POSIX, the context is saved on standard (although
+ deprecated)
+ // ucontext_t structures. It requires _XOPEN_SOURCE=700L to
+ compile.
-
+     os_impl_ucontext_t ucontext; //
-
+ };
+ inline int
+ ipi_signal_number (void)
+ {
+     return SIGRTMIN + 1;
+ }
+ } /* namespace clock */

-
+ //
+ -----
+ using thread_context_t = os_port_thread_context_t;

+
+ } /* namespace port */
+ } /* namespace rtos */
@@ -157,10 +189,6 @@ namespace os

```

```

#pragma GCC diagnostic pop

-// -----
-
#endif /* __cplusplus */

-// -----
-

#endif /* CMSIS_PLUS_RTOS_PORT_OS_DECLS_H */
diff --git a/include/cmsis-plus/rtos/port/os-inlines.h b/include/
      cmsis-plus/rtos/port/os-inlines.h
index 0ec7ab7..943da20 100644
--- a/include/cmsis-plus/rtos/port/os-inlines.h
+++ b/include/cmsis-plus/rtos/port/os-inlines.h
@@ -1,53 +1,44 @@
/*
+ * os-inlines.h - the hot half of the POSIX port: masking and the
+   kernel lock.
+ *
+ * This file is part of the µOS++ project (https://micro-os-
+   plus.github.io/).
+ * Copyright (c) 2016-2025 Liviu Ionescu. All rights reserved.
+ * Copyright (c) 2026 Dan. All rights reserved.
+ *
+ * Permission to use, copy, modify, and/or distribute this software
+ * for any purpose is hereby granted, under the terms of the MIT
+   license.
+ *
- * If a copy of the license was not distributed with this file, it
-   can
- * be obtained from https://opensource.org/licenses/mit/.
- */
-
-/*
- * This file is part of the CMSIS++ proposal, intended as a CMSIS
- * replacement for C++ applications.
- *
- * If contains the scheduler implementation that uses
- * functions from the POSIX API (macOS and GNU/Linux).
+ * Read this beside micro-os-plus-iii-aarch64/include/cmsis-plus/
+   rtos/port/
+ * os-inlines.h. The two files have the same shape function for
+   function; only
+ * the mechanism differs, and that is the whole point of the port
+   seam:
+ *
- * This file is included in all src/os-*.cpp files.
+ *
+ * concern      AArch64      here
+ * -----
+ * CPU id      MRS MPIDR_EL1, & 3      thread_local, set at
+   CPU start
+ * IRQ mask    MSR DAIFSET/DAIFCLR, #2
+   pthread_sigmask(irq_set)
+ * IRQ state   MRS DAIF      sigismember(old, tick)
+ * atomics     LDAXR/STLXR + DMB ISH    __atomic_* , seq_cst
+ * in-handler  _in_isr[]      _in_isr[] (identical)
+ */

#ifndef CMSIS_PLUS_RTOS_PORT_OS_INLINE_H_
#define CMSIS_PLUS_RTOS_PORT_OS_INLINE_H_

-// -----
-

#if defined(OS_USE_OS_APP_CONFIG_H)
#include <cmsis-plus/os-app-config.h>
#endif

#include <cmsis-plus/rtos/os-c-decls.h>

-// -----

```

```

-
-#ifndef __cplusplus
-
-// -----
-
-#include <stdlib.h>
-#include <string.h>
-#include <sys/utsname.h>
-#include <sys/time.h>
-
-// For Linux
+#include <pthread.h>
+#include <signal.h>
+include <unistd.h>
+#include <time.h>

#include <cmsis-plus/diag/trace.h>

-// -----
-
#pragma GCC diagnostic push
#pragma GCC diagnostic ignored "-Wunused-parameter"

@@ -55,9 +46,9 @@
#pragma clang diagnostic ignored "-Wc++98-compat"
#endif

-// -----
-
-extern "C" uint32_t signal_nesting;
+// Out of line on purpose; see _this_cpu below.
+extern "C" unsigned
+port_cpu_id (void);

namespace os
{
@@ -65,10 +56,44 @@ namespace os
{
    namespace port
    {
        //
-
+
+/*
+ * Which CPU is running this code.
+ *
+ * This is the one deliberate use of native thread-local
+ storage in the
+ * whole port, and it is the one use that is correct by
+ construction: a
+ * host thread IS a CPU, so storage private to a host thread is
+ storage
+ * private to a CPU. It answers "where am I", never "what was I
+ doing".
+ *
+ * Everything else must obey the opposite rule. A µOS++ thread
+ migrates
+ * between CPUs, so `errno` and any `thread_local` it touches
+ belong to
+ * whichever host thread happens to be running it at that
+ instant. No
+ * port or application state may live there across a switch
+ point --
+ * read errno only inside the critical section that made the
+ call.
+ *
+ * That rule covers the ADDRESS of `_this_cpu` too, and the
+ compiler does
+ * not know it. The thread pointer is constant for the life of
+ a host
+ * thread, so clang computes `_this_cpu` once per function and
+ keeps it

```

```

+      * in a callee-saved register. A function that blocks --
+      semaphore::
+      * wait() loops around reschedule() -- then resumes on another
+      host
+      * thread and reads the CPU id of the one it left. It takes the
+      kernel
+      * lock as that CPU, skips the acquire when that CPU already
+      owns it,
+      * and two CPUs run inside the lock; the depth count strands it
+      held and
+      * every CPU spins for ever (clang 16/17 -O2: smp_test3, smp-
+      pipeline).
+      * GCC reloads the thread pointer at each access and never
+      showed it.
+      *
+      * So the read is never inlined: each call of port_cpu_id()
+      computes the
+      * address afresh, and the empty asm makes every call a side
+      effect that
+      * can be neither merged with another nor hoisted, LTO
+      included.
+      */
+      extern thread_local unsigned _this_cpu;

namespace scheduler
{
+      inline unsigned __attribute__((always_inline))
+      port_cpu_id_inline (void)
+      {
+          return ::port_cpu_id ();
+      }

      inline port::scheduler::state_t __attribute__((always_inline))
      lock (void)
@@ -85,27 +110,54 @@ namespace os
      inline bool __attribute__((always_inline))
      locked (void)
      {
-          return lock_state != state::unlocked;
+          return lock_state[port_cpu_id_inline ()] !=
+          state::unlocked;
      }

+      /* SMP: acquire the scheduler lock word.
+      *
+      * A spin, exactly as on ARM, and for the same reason: this
+      lock is
+      * held for a handful of instructions and is taken from
+      inside signal
+      * handlers, where a pthread_mutex would be a function call
+      into code
+      * that is not async-signal-safe. The pause hint keeps a
+      spinning CPU
+      * from starving the one that holds it when the host
+      oversubscribes. */
+      inline void __attribute__((always_inline))
+      wait_for_interrupt (void)
+      _smp_klock_raw_acquire (void)
+      {
-#if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
-          trace::printf ("%s() \n", __func__);
+          while (__atomic_exchange_n (&smp_klock.lock, 1u,
+          _ATOMIC_ACQUIRE)
+              != 0u)
+          {
+              #if defined(__x86_64__) || defined(__i386__)
+                  __builtin_ia32_pause ();
+              #elif defined(__aarch64__) || defined(__arm__)
+                  __asm__ volatile("yield" ::: "memory");
+              #endif
+              pause ();

```

```

+         }
+     }
+
+     inline void __attribute__ ((always_inline))
+     _smp_klock_raw_release (void)
+     {
+         __atomic_store_n (&_smp_klock.lock, 0u, __ATOMIC_RELEASE);
+     }
+
+     inline unsigned __attribute__ ((always_inline))
+     port_smp_depth (void)
+     {
+         return _smp_klock.depth;
+     }
+
+     void
+     wait_for_interrupt (void);
+
+ } /* namespace scheduler */
+
+ namespace interrupts
+ {
+
+     inline bool __attribute__ ((always_inline))
+     in_handler_mode (void)
+     {
+         return (signal_nesting > 0);
+     }
+
+     const unsigned cpu = scheduler::port_cpu_id_inline ();
+     return ((cpu < OS_NCPU) && _in_isr[cpu]) ||
+     (signal_nesting != 0);
+ }
+
+     inline bool __attribute__ ((always_inline))
@@ -114,9 +166,71 @@ namespace os
+     return true;
+ }
+
+ } /* namespace interrupts */
+
+ /*
+  * Enter an IRQ critical section: mask this CPU's interrupts
+  * and take
+  * the kernel lock. Same two steps, same order, as AArch64's.
+  *
+  * pthread_sigmask, not sigprocmask. In a multithreaded
+  * process the
+  * behaviour of sigprocmask is unspecified (POSIX.1-2017, and
+  * glibc
+  * documents it as such), and upstream's port used it
+  * throughout
+  * because upstream had exactly one thread. The replacement
+  * is not a
+  * workaround but the better abstraction: a per-thread signal
+  * mask is
+  * per-CPU interrupt masking, which is what a critical
+  * section has
+  * always meant.
+  */
+     inline rtos::interrupts::state_t __attribute__
+     ((always_inline))
+     critical_section::enter (void)
+     {
+         sigset_t old;
+         ::pthread_sigmask (SIG_BLOCK, &irq_set, &old);
+
+         const unsigned cpu = scheduler::port_cpu_id_inline ();
+         if (scheduler::_smp_klock.owner != cpu)
+         {
+             scheduler::_smp_klock_raw_acquire ();
+             scheduler::_smp_klock.owner = cpu;

```

```

+         }
+         scheduler::_smp_klock.depth = scheduler::_smp_klock.depth +
1;
+
+         return ::sigismember (&old, clock::signal_number ()) != 0;
+     }
+
+     inline void __attribute__ ((always_inline))
+     critical_section::exit (rtos::interrupts::state_t state)
+     {
+         const unsigned cpu = scheduler::port_cpu_id_inline ();
+         if (scheduler::_smp_klock.owner == cpu
+             && scheduler::_smp_klock.depth > 0)
+         {
+             const uint32_t d = scheduler::_smp_klock.depth - 1;
+             scheduler::_smp_klock.depth = d;
+             if (d == 0)
+             {
+                 scheduler::_smp_klock.owner = SMP_NO_OWNER;
+                 scheduler::_smp_klock_raw_release ();
+             }
+         }
+
+         ::pthread_sigmask (state ? SIG_BLOCK : SIG_UNBLOCK,
+                             &irq_set,
+                             nullptr);
+     }
+
+     inline rtos::interrupts::state_t __attribute__
+     ((always_inline))
+     uncritical_section::enter (void)
+     {
+         sigset_t old;
+         ::pthread_sigmask (SIG_UNBLOCK, &irq_set, &old);
+         return ::sigismember (&old, clock::signal_number ()) != 0;
+     }
+
+     inline void __attribute__ ((always_inline))
+     uncritical_section::exit (rtos::interrupts::state_t state)
+     {
+         ::pthread_sigmask (state ? SIG_BLOCK : SIG_UNBLOCK,
+                             &irq_set,
+                             nullptr);
+     }
+
+ //
+ //=====
+ } /* namespace interrupts */
+
+ namespace this_thread
+ {
@@ -124,11 +238,8 @@ namespace os
+     prepare_suspend (void)
+     {
+     }
+
+ } /* namespace this_thread */
+
+ //
+ //=====
+
+ inline constexpr bool __attribute__ ((always_inline))
+ clock_highres::has_hardware_counter (void) noexcept
+ {
@@ -138,7 +249,7 @@ namespace os
+     inline uint64_t __attribute__ ((always_inline))
+     clock_highres::hardware_counter (void) noexcept
+     {

```

```

timespec tp; // NB: no 'struct' - baseline gate has
-Werror=redundant-tags
+ timespec tp;
::clock_gettime (CLOCK_MONOTONIC, &tp);
return static_cast<uint64_t> (tp.tv_sec) * 1000000ULL
    + static_cast<uint64_t> (tp.tv_nsec) / 1000ULL;
@@ -150,10 +261,6 @@ namespace os

#pragma GCC diagnostic pop

-// -----
-
#endif /* __cplusplus */

-// -----

#endif /* CMSIS_PLUS_RTOS_PORT_OS_INLINES_H_ */
diff --git a/include/exception_handler.hpp b/include/
exception_handler.hpp
new file mode 100644
index 0000000..27423a1
--- /dev/null
+++ b/include/exception_handler.hpp
@@ -0,0 +1,29 @@
+/**
+ * @file exception_handler.hpp
+ * @brief Synchronous-fault reporting for the POSIX port.
+ *
+ * On the ARM ports a synchronous exception lands in a vector
+ * installed by
+ * startup.S and is reported by port_fatal_exception(). The host
+ * equivalent of
+ * a synchronous exception is a signal -- SIGSEGV, SIGBUS, SIGFPE,
+ * SIGILL --
+ * so exception::init() installs handlers for those and reports
+ * through the
+ * same function, with the same name, so a test that faults says so
+ * instead of
+ * dying silently with no output.
+ *
+ * This matters more here than on the silicon, not less: a host
+ * process that
+ * segfaults under a test runner produces an exit status and nothing
+ * else, and
+ * with several CPUs running there is no way afterwards to tell which
+ * one
+ * faulted or in which thread.
+ */
+#pragma once
+
+#include <cstdint>
+
+namespace exception {
+
+void init();
+
+} // namespace exception
+
+extern "C" void port_fatal_exception(std::uint64_t type,
std::uint64_t esr,
std::uint64_t elr, std::uint64_t
far,
std::uint64_t spsr);
diff --git a/include/host_cpu.hpp b/include/host_cpu.hpp
new file mode 100644
index 0000000..55861b6
--- /dev/null
+++ b/include/host_cpu.hpp
@@ -0,0 +1,129 @@
+/**
```

```

+ * host_cpu.hpp - the CPU model of the POSIX port: a host thread IS a
+ * CPU.
+ *
+ * This file is part of the µOS++ project (https://micro-os-plus.github.io/).
+ * Copyright (c) 2026 Dan. All rights reserved.
+ *
+ * Permission to use, copy, modify, and/or distribute this software
+ * for any purpose is hereby granted, under the terms of the MIT
+ * license.
+ *
+ * On the ARM ports the equivalent of this header is split between
+ * the silicon
+ * (the generic timer, the GIC/mailbox IPI, MPIDR) and the board (the
+ * spin
+ * table that releases the secondary cores). Here the silicon is the
+ * host
+ * kernel, so it is all one file, and it is port code rather than
+ * board code.
+ */
+
+
+#ifndef MICRO_OS_PLUS_III_POSIX_ARCH_HOST_CPU_HPP_
+#define MICRO_OS_PLUS_III_POSIX_ARCH_HOST_CPU_HPP_
+
+#include <cstdint>
+
+#include <cmsis-plus/rtos/os.h>
+#include <cmsis-plus/rtos/port/os-decls.h>
+
+namespace host_cpu
+{
+    using element_t = os::rtos::port::stack::element_t;
+
+    /*
+     * "Which thread is running on this CPU."
+     *
+     * The kernel spells this two different ways: under the SMP
+     * scheduler
+     * `scheduler::current_thread_` is an array indexed by CPU, and
+     * without it
+     * a single pointer (os-sched.h, guarded on OS_USE_SMP_SCHEDULER,
+     * which
+     * uos_add_app() derives from NCPU > 1).
+     *
+     * One accessor keeps the difference in one place, so the rest of
+     * the port
+     * is written once and serves both. This is the same dual-branch
+     * arrangement
+     * the cortexm port uses to run its three STM32 boards at OS_NCPU=1
+     * off the
+     * RP2350's SMP core.
+     */
+    inline os::rtos::thread* volatile&
+    current_thread (unsigned cpu)
+    {
+        #if defined(OS_USE_SMP_SCHEDULER)
+            return os::rtos::scheduler::current_thread[cpu];
+        #else
+            (void)cpu;
+            return os::rtos::scheduler::current_thread;
+        #endif
+    }
+
+    /* Install the tick and IPI signal handlers, process-wide. Called
+     * once,
+     * from port::scheduler::initialize(). */
+    void
+    install_handlers (void);
+
+    /* Give the calling CPU an alternate stack for fault reporting. Per
+     * host

```

```

+ * thread, because sigaltstack() is, and that is correct here:
+   reporting a
+ * fault does not migrate, so it may use storage the CPU owns. */
+ void
+ install_fault_stack (void);
+
+ /* Arm the calling CPU's own periodic timer, at
+ * OS_INTEGER_SYSTICK_FREQUENCY_HZ. Every CPU has one; only CPU 0
+   advances
+ * the kernel clock with it. */
+ void
+ start_this_cpu_tick (void);
+
+ /* Create the OS_NCPU-1 secondary host threads. Each sets its own
+   CPU index,
+ * arms its own tick and enters the scheduler, which is what a
+   secondary
+ * core does after the spin table releases it. */
+ void
+ start_secondary_cpus (void);
+
+ /* Ask another CPU to re-pick. pthread_kill() of the IPI signal,
+   which is
+ * this port's SGI / mailbox doorbell. */
+ void
+ send_ipi (unsigned cpu);
+
+ /* Build a never-run context's entry point. Wraps the kernel's
+   thread body
+ * in the trampoline that discharges this CPU's deferred publish
+   before the
+ * body runs -- a context that has never run still arrives on a CPU
+   that
+ * has just left another thread behind. */
+ void
+ make_entry (os_impl_ucontext_t* ctx, void* func, void* args);
+
+ /* Leave `*addr = val` for whoever runs next on this CPU. See the
+   long
+ * comment on port::scheduler::switch_stacks(). */
+ void
+ defer_publish (unsigned cpu, element_t** addr, element_t* val);
+
+ /* Discharge this CPU's deferred publish, if it has one. Called on
+   every
+ * arrival into a context: after swapcontext(), and in the
+   trampoline. */
+ void
+ publish_pending (void);
+
+ /* -----
+ * AddressSanitizer and the context switch.
+ *
+ * ASan keeps a shadow record of which stack is live so that it can
+   tell a
+ * genuine overflow from an ordinary function return. swapcontext()
+   moves
+ * the stack pointer to somewhere ASan has never seen, and without
+   being
+ * told, ASan reports the first thing the resumed thread touches as
+   a
+ * stack-buffer-overflow -- a false positive on every single
+   switch, which
+ * makes the tool useless rather than merely noisy.
+ *
+ * The fix is ASan's own fiber interface, which exists for exactly
+   this:
+ *
+ *   start_switch (&save, bottom, size)   before swapcontext /
+   setcontext
+ *   finish_switch (save)                  first thing on arrival
+ *

```

```

+ * `save` is where ASan parks the outgoing fibre's "fake stack"
+ * (its
+ * use-after-return bookkeeping). It must survive until that
+ * context is
+ * resumed, so switch_stacks() keeps it in a local -- which lives
+ * on the
+ * outgoing thread's own stack and is therefore still there, and
+ * still
+ * correct, whenever and on whichever CPU that thread comes back.
+ *
+ * Both compile to nothing when ASan is off, so the calls stay
+ * unguarded at
+ * the call sites and the switch code reads the same either way.
+ * ----- */
+ void
+ asan_start_switch (void** save, const void* bottom, std::size_t
+ size);
+
+ void
+ asan_finish_switch (void* save);
+
+} /* namespace host_cpu */
+
+#endif /* MICRO_OS_PLUS_III_POSIX_ARCH_HOST_CPU_HPP */
diff --git a/include/hw_result.hpp b/include/hw_result.hpp
new file mode 100644
index 00000000..2102b90
--- /dev/null
+++ b/include/hw_result.hpp
@@ -0,0 +1,53 @@
+/**
+ * @file hw_result.hpp
+ * @brief Bounded "done" helper, POSIX host.
+ *
+ * The sibling of micro-os-plus-iii-aarch64/include/hw_result.hpp. A
+ * test
+ * prints its RESULT line and then calls ok()/fail().
+ *
+ * On the silicon ports these compile to a semihosted SYS_EXIT under
+ * SEMIHOST
+ * and to nothing otherwise, so a plain image keeps its idle-forever
+ * behaviour. Here there is no "otherwise": the host always has an
+ * exit
+ * status, and a test process that idled for ever would have to be
+ * killed by
+ * the runner and could not report anything. So they always end the
+ * run, and
+ * the verdict reaches the shell as the exit code.
+ *
+ * _exit(), not exit(). Other CPUs are still running threads; exit()
+ * would run
+ * static destructors underneath them and turn a clean PASS into a
+ * crash in
+ * the teardown.
+ *
+ * But _exit() does not flush stdio either, so these flush first.
+ * Most tests
+ * here print through uart::uart1, which is write(2) and therefore
+ * already on
+ * its way out; a test carried from upstream's own suite prints with
+ * printf(),
+ * and to a pipe -- which is how the runner captures it -- stdout is
+ * fully
+ * buffered. Without the flush its entire output, verdict included,
+ * is
+ * discarded at the exit. (Found exactly that way: mutex-stress ran,
+ * passed and
+ * printed nothing.)
+ *
+ * Pattern in a test:
+ *
+ *     uart1 << "\nRESULT: " << (ok ? "PASS" : "FAIL") << "\n";
+ *     if (ok) hw_result::ok (); else hw_result::fail ();

```

```

+ *      for (;;) sysclock.sleep_for (1000);    // never reached
+ */
+ #pragma once
+
+ #include <cstdio>
+ #include <unistd.h>
+
+ namespace hw_result
+ {
+     [[noreturn]] inline void
+     ok () noexcept
+     {
+         std::fflush (nullptr);
+         ::_exit (0);
+     }
+
+     [[noreturn]] inline void
+     fail () noexcept
+     {
+         std::fflush (nullptr);
+         ::_exit (1);
+     }
+ } // namespace hw_result
diff --git a/src/board-contract.cpp b/src/board-contract.cpp
new file mode 100644
index 00000000..40805d4
--- /dev/null
+++ b/src/board-contract.cpp
@@ -0,0 +1,43 @@
+/*
+ * board-contract.cpp - no code. It fails the build the moment a
+ * board has
+ *
+ * not stated something the shared sources need.
+ *
+ * This file is part of the µOS++ project (https://micro-os-plus.github.io/).
+ * Copyright (c) 2026 Dan. All rights reserved.
+ *
+ * Permission to use, copy, modify, and/or distribute this software
+ * for any purpose is hereby granted, under the terms of the MIT
+ * license.
+ *
+ * The same file exists in the three cross-compiled projects and
+ * exists here
+ * for the same reason: a board that leaves something unset must
+ * produce a
+ * compile error naming the board and the thing, not a silent
+ * inheritance of
+ * another board's answer -- or, worse, a link that succeeds and a
+ * test that
+ * proves nothing.
+ */
+
+ #if defined(__APPLE__) || defined(__linux__)
+
+ #include <cmsis-plus/rtos/port/os-decls.h>
+
+ #if !defined(OS_NCPU)
+ #error "The board does not set OS_NCPU."
+ #endif
+
+ #if OS_NCPU < 1
+ #error "OS_NCPU must be at least 1."
+ #endif
+
+ #if !defined(PORT_GREETING)
+ #error "The board does not define PORT_GREETING; the test banners
+ print it."
+ #endif

```

```

+
+// A host CPU is a host thread, so OS_NCPU above the number the
+// machine can
+// actually run in parallel is not an error -- it oversubscribes,
+// which is a
+// legitimate and useful way to shake out races. It IS worth saying
+// out loud,
+// because a suite that suddenly takes ten times as long has usually
+// done this
+// by accident.
+#if OS_NCPU > 16
+#warning "OS_NCPU above 16 oversubscribes almost any host; expect the
+// tick to drift."
+#endif
+
+#endif /* defined(__APPLE__) || defined(__linux__) */
diff --git a/src/exception_handler.cpp b/src/exception_handler.cpp
new file mode 100644
index 00000000..2a09db9
--- /dev/null
+++ b/src/exception_handler.cpp
@@ -0,0 +1,224 @@
+/*
+ * exception_handler.cpp - synchronous faults, reported rather than
+ * silent.
+ *
+ * This file is part of the µOS++ project (https://micro-os-plus.github.io/).
+ * Copyright (c) 2026 Dan. All rights reserved.
+ *
+ * Permission to use, copy, modify, and/or distribute this software
+ * for any purpose is hereby granted, under the terms of the MIT
+ * license.
+ */
+
+#if defined(__APPLE__) || defined(__linux__)
+
+#include <csignal>
+#include <cstring>
+#include <initializer_list>
+#include <dlfcn.h>
+#include <execinfo.h>
+#include <ucontext.h>
+#include <unistd.h>
+
+#include <cmsis-plus/rtos/os.h>
+#include <cmsis-plus/rtos/port/os-decls.h>
+#include <cmsis-plus/rtos/port/os-inlines.h>
+#include <exception_handler.hpp>
+#include <host_cpu.hpp>
+
+namespace
+{
+    // Everything below runs in a signal handler after the process is
+    // already
+    // broken, so it uses write(2) and nothing else: no printf, no
+    // malloc, no
+    // locks. A reporting path that can itself deadlock reports
+    // nothing.
+
+    void
+    emit (const char* s)
+    {
+        std::size_t n = 0;
+        while (s[n] != '\0')
+        {
+            ++n;
+        }
+        ssize_t r = ::write (2, s, n);
+        (void)r;
+    }
+}

```

```

+
+ void
+ emit_hex (std::uint64_t v)
+ {
+     static const char digits[] = "0123456789ABCDEF";
+     char out[19] = "0x";
+     int n = 2;
+     for (int i = 15; i >= 0; --i)
+     {
+         out[n++] = digits[(v >> (i * 4)) & 0xF];
+     }
+     out[n] = '\0';
+     emit (out);
+ }
+
+ /*
+  * The program counter of the faulting instruction, from the
+  * machine context
+  * the kernel handed the handler. The ARM ports report ELR for the
+  * same
+  * reason: on an SMP fault the faulting address alone rarely
+  * identifies the
+  * bug, and `addr2line -e <image> <pc>` names the line outright.
+  */
+ /*
+  * The load bias of this image, captured in exception::init() --
+  * i.e. in a
+  * sane context, because dladdr() is not async-signal-safe.
+  * Reported
+  * alongside the raw pc so that a position-independent executable
+  * (which is
+  * the default here) can still be looked up with the static
+  * addresses in the
+  * ELF file: `addr2line -e <image> <pc-bias>`.
+  */
+ std::uint64_t g_load_bias = 0;
+
+ std::uint64_t
+ fault_pc (void* ucontext)
+ {
+     if (ucontext == nullptr)
+     {
+         return 0;
+     }
+     ucontext_t* uc = static_cast<ucontext_t*> (ucontext);
+ #if defined(__x86_64__)
+     return static_cast<std::uint64_t> (uc-
+         >uc_mcontext.gregs[REG_RIP]);
+ #elif defined(__i386__)
+     return static_cast<std::uint64_t> (uc-
+         >uc_mcontext.gregs[REG_EIP]);
+ #elif defined(__aarch64__)
+     return static_cast<std::uint64_t> (uc->uc_mcontext.pc);
+ #elif defined(__arm__)
+     return static_cast<std::uint64_t> (uc->uc_mcontext.arm_pc);
+ #else
+     (void)uc;
+     return 0;
+ #endif
+ }
+
+ void
+ fault_handler (int sig, siginfo_t* info, void* ucontext)
+ {
+     port_fatal_exception (static_cast<std::uint64_t> (sig), 0,
+         fault_pc (ucontext),
+         reinterpret_cast<std::uint64_t> (
+             info != nullptr ? info->si_addr :
+             nullptr),
+         0);
+ }

```

```

+ }
+} /* anonymous namespace */
+
+extern "C" void
+port_fatal_exception (std::uint64_t type, std::uint64_t esr,
+                      std::uint64_t elr, std::uint64_t far,
+                      std::uint64_t spsr)
+{
+ (void)esr;
+ (void)spsr;
+
+ emit ("\n!!! FATAL EXCEPTION on CPU ");
+ {
+   const unsigned cpu = port_cpu_id ();
+   char c = static_cast<char> ('0' + (cpu % 10));
+   ssize_t r = ::write (2, &c, 1);
+   (void)r;
+ }
+ emit (" -- signal ");
+ emit (::strsignal (static_cast<int> (type)));
+ emit (" at ");
+ emit_hex (far);
+ emit (" , pc ");
+ emit_hex (elr);
+ emit (" (static ");
+ emit_hex (elr - g_load_bias);
+ emit (" )\n");
+
+ /*
+  * The scheduler state, which is the whole point of reporting at
+  * all: on a
+  * machine with several CPUs, "it crashed" says nothing without
+  * knowing
+  * which thread each CPU was running and who held the kernel lock.
+  * The ARM
+  * ports print the same two facts from their claim tripwire.
+  */
+ for (unsigned c = 0; c < OS_NCPU; ++c)
+ {
+   emit ("  cpu");
+   {
+     char d = static_cast<char> ('0' + (c % 10));
+     ssize_t r = ::write (2, &d, 1);
+     (void)r;
+   }
+   emit (": ");
+   os::rtos::thread* th = host_cpu::current_thread (c);
+   emit (th == nullptr ? "(null)"
+        : (th->name () != nullptr ? th->name () :
+         "(anon)"));
+   emit ("\n");
+ }
+ emit ("  klock: owner=");
+ emit_hex (os::rtos::port::scheduler::_smp_klock.owner);
+ emit (" depth=");
+ emit_hex (os::rtos::port::scheduler::_smp_klock.depth);
+ emit (" lock=");
+ emit_hex (os::rtos::port::scheduler::_smp_klock.lock);
+ emit ("\n");
+
+ /*
+  * The call chain. Not async-signal-safe in the strict sense
+  * (backtrace()
+  * may take the loader lock on its first call), but the process is
+  * already
+  * dead and a chain of return addresses is what turns "a null
+  * pointer was
+  * dereferenced" into "this line dereferenced it".

```

```

+ */
+ {
+     void* frames[24];
+     int n = ::backtrace (frames, 24);
+     emit ("  backtrace (static):\n");
+     for (int i = 0; i < n; ++i)
+     {
+         emit ("    ");
+         emit_hex (reinterpret_cast<std::uint64_t> (frames[i]) -
+ g_load_bias);
+         emit ("\n");
+     }
+ }
+
+ // Re-raise with the handler removed, so the shell sees the real
+   cause and a
+ // core file is still produced.
+ ::signal (static_cast<int> (type), SIG_DFL);
+ ::raise (static_cast<int> (type));
+ ::_exit (128 + static_cast<int> (type));
+}
+
+namespace exception
+{
+    void
+    init ()
+    {
+        {
+            DL_info info;
+            if (::dladdr (reinterpret_cast<void*> (&port_fatal_exception),
+ &info)
+                != 0)
+            {
+                g_load_bias = reinterpret_cast<std::uint64_t>
+ (info.dli_fbase);
+            }
+        }
+
+        struct sigaction sa; // `sigaction` is also a function: the tag
+           is required
+        std::memset (&sa, 0, sizeof (sa));
+        sa.sa_flags = SA_SIGINFO | SA_ONSTACK;
+        ::sigemptyset (&sa.sa_mask);
+        sa.sa_sigaction = fault_handler;
+
+        // SA_ONSTACK here, and NOT on the tick -- the two are opposite
+           cases and
+        // the difference matters.
+        //
+        // The tick must run on the thread's own stack, so its frame
+           migrates with
+        // the thread when another CPU resumes it (host_cpu.cpp). A fault
+           handler
+        // never migrates: it reports and the process ends. And it must
+           not need
+        // the faulting stack, because a blown or wild stack is precisely
+           one of
+        // the faults worth reporting -- without an alternate stack that
+           case
+        // cannot be reported at all, which is how the first SMP fault
+           found here
+        // came out as a silent exit 139.
+        //
+        // The stack itself is per CPU and installed by host_cpu when the
+           CPU
+        // starts, because sigaltstack() is per host thread.
+        for (int sig : { SIGSEGV, SIGBUS, SIGFPE, SIGILL })
+        {
+            ::sigaction (sig, &sa, nullptr);
+        }
+    }
+}

```

```

+} /* namespace exception */
+
+#endif /* defined(__APPLE__) || defined(__linux__) */
diff --git a/src/free-store.cpp b/src/free-store.cpp
new file mode 100644
index 00000000..acc7e95
--- /dev/null
+++ b/src/free-store.cpp
@@ -0,0 +1,133 @@
+/*
+ * free-store.cpp - the application free store on the POSIX host.
+ *
+ * This file is part of the µOS++ project (https://micro-os-plus.github.io/).
+ * Copyright (c) 2016-2025 Liviu Ionescu. All rights reserved.
+ * Copyright (c) 2026 Dan. All rights reserved.
+ *
+ * Permission to use, copy, modify, and/or distribute this software
+ * for any purpose is hereby granted, under the terms of the MIT
+ * license.
+ *
+ * -----
+ * Why this file exists at all, and why it does NOT just call
+ * malloc().
+ *
+ * The kernel's own src/startup/initialise-free-store.cpp is guarded
+ * by
+ * `#if defined(__ARM_EABI__)` from top to bottom, so on a host it
+ * compiles to
+ * nothing and os_startup_initialize_free_store() is undefined --
+ * which is
+ * exactly what the carried tests call. Relaxing that guard would
+ * mean editing
+ * the kernel, which is kept merge-clean against upstream; so the
+ * port
+ * supplies the hook, which is what a port is for.
+ *
+ * Upstream's posix-arch had no such hook and let the system malloc()
+ * serve,
+ * and its NOTES.md says so. That was correct for upstream, which had
+ * ONE host
+ * thread. It is not correct here, and the difference is the whole
+ * subject of
+ * this port:
+ *
+ * A µOS++ thread may be preempted inside an allocation and resumed
+ * on a
+ * DIFFERENT CPU -- that is, a different host thread. glibc's
+ * malloc is
+ * thread-safe by taking an arena lock, and that lock would then be
+ * released
+ * by a host thread that did not take it. Nothing in the C library
+ * promises
+ * that works.
+ *
+ * Using µOS++'s own memory resource removes the question rather than
+ * betting
+ * on it: the allocator is the kernel's, its mutual exclusion is the
+ * kernel's
+ * scheduler lock, and the scheduler lock already migrates correctly
+ * because
+ * making it do so is what the rest of this port is about. It also
+ * makes the
+ * host behave like the three silicon ports, which is the point of
+ * testing on
+ * it -- a first_fit_top over a fixed block is what every board does,
+ * so an
+ * allocation pattern that exhausts the heap here exhausts it there
+ * too,
+ * instead of being quietly absorbed by a host with gigabytes to
+ * spare.
+ *
+ * -----
+ */
+

```

```

+ #if defined(__APPLE__) || defined(__linux__)
+
+ #include <cstdint>
+ #include <new>
+
+ #include <cmsis-plus/rtos/os.h>
+ #include <cmsis-plus/rtos/os-hooks.h>
+ #include <cmsis-plus/memory/first-fit-top.h>
+ #include <cmsis-plus/estd/memory_resource>
+
+ using namespace os;
+
+ #if defined(OS_TYPE_APPLICATION_MEMORY_RESOURCE)
+ using application_memory_resource =
+     OS_TYPE_APPLICATION_MEMORY_RESOURCE;
+ #else
+ using application_memory_resource = os::memory::first_fit_top;
+ #endif
+
+ namespace
+ {
+     // Same storage trick as the kernel's ARM version: the resource
+     // itself must
+     // outlive everything and must not need the free store to exist
+     // first.
+     alignas (application_memory_resource) char
+         application_free_store[sizeof (application_memory_resource)];
+ } // namespace
+
+ /*
+  * The out-of-memory hooks, for the same reason as the hook above:
+  * the
+  * kernel's definitions live in the __ARM_EABI__-guarded file. Weak,
+  * so an
+  * application that wants to reset the machine, dump the heap or
+  * coalesce
+  * before giving up replaces them by defining its own -- which is
+  * exactly the
+  * contract they have on the silicon ports.
+  */
+ void __attribute__ ((weak))
+ os_rtos_application_out_of_memory_hook (void)
+ {
+     estd::__throw_bad_alloc ();
+ }
+
+ #if defined(OS_INTEGER_RTOS_DYNAMIC_MEMORY_SIZE_BYTES)
+
+ void __attribute__ ((weak))
+ os_rtos_system_out_of_memory_hook (void)
+ {
+     estd::__throw_bad_alloc ();
+ }
+
+ #endif /* defined(OS_INTEGER_RTOS_DYNAMIC_MEMORY_SIZE_BYTES) */
+
+ /*
+  * Weak no-ops, so an application that brings up no hardware still
+  * links.
+  * Every carried test defines both, and its definitions win.
+  */
+ void __attribute__ ((weak))
+ os_startup_initialize_hardware_early (void)
+ {
+ }
+
+ void __attribute__ ((weak))
+ os_startup_initialize_hardware (void)
+ {

```

```

+}
+
+void __attribute__ ((weak))
+os_startup_initialize_free_store (void* heap_address,
+                                std::size_t heap_size_bytes)
+{
+#if !defined(OS_EXCLUDE_DYNAMIC_MEMORY_ALLOCATIONS)
+
+    new (&application_free_store)
+        application_memory_resource{ "app-heap", heap_address,
+                                     heap_size_bytes };
+
+    reinterpret_cast<rtos::memory::memory_resource*>
+        (&application_free_store)
+        ->out_of_memory_handler
+        (os_rtos_application_out_of_memory_hook);
+
+    estd::pmr::set_default_resource (
+        reinterpret_cast<estd::pmr::memory_resource*>
+        (&application_free_store));
+
+    // No sbrk() adjustment. On a bare-metal target the kernel's
+    // version pushes
+    // sbrk past the free store so newlib's malloc cannot collide with
+    // it; here
+    // the block is ordinary static storage the host already owns, so
+    // there is
+    // nothing to push it past.
+
+#else
+    (void)heap_address;
+    (void)heap_size_bytes;
+#endif /* !defined(OS_EXCLUDE_DYNAMIC_MEMORY_ALLOCATIONS) */
+}
+
+#endif /* defined(__APPLE__) || defined(__linux__) */
diff --git a/src/host_cpu.cpp b/src/host_cpu.cpp
new file mode 100644
index 00000000..87bd73f
--- /dev/null
+++ b/src/host_cpu.cpp
@@ -0,0 +1,530 @@
+/*
+ * host_cpu.cpp - host threads as CPUs: bring-up, per-CPU tick, IPI.
+ *
+ * This file is part of the µOS++ project (https://micro-os-plus.github.io/).
+ * Copyright (c) 2026 Dan. All rights reserved.
+ *
+ * Permission to use, copy, modify, and/or distribute this software
+ * for any purpose is hereby granted, under the terms of the MIT
+ * license.
+ */
+
+#if defined(__APPLE__) || defined(__linux__)
+
+#include <cerrno>
+#include <cstdio>
+#include <cstdlib>
+#include <cstring>
+
+#include <pthread.h>
+#include <signal.h>
+#include <sys/syscall.h>
+#include <time.h>
+#include <unistd.h>
+
+#include <cmsis-plus/rtos/os.h>
+#include <cmsis-plus/rtos/port/os-inlines.h>

```

```

+
+#include <host_cpu.hpp>
+
+/* Defined by the board (test/boards/<board>/src/smp.cpp), with the
+   same
+ * meaning as on the ARM boards: 0 = not started, 3 = scheduler
+   entered. The
+ * tests wait on it before they start timing anything. A board that
+   does not
+ * define it fails to link, which is the point. */
+extern "C" volatile uint32_t g_core_stage[OS_NCPU];
+
+
+// Port-side weak default so a build that does not link the SMP test
+   board still
+// resolves g_core_stage[]. The SMP test board (boards/native/src/
+   smp.cpp)
+// provides a strong definition that overrides this one when present.
+extern "C"
+{
+  __attribute__((weak)) volatile uint32_t g_core_stage[OS_NCPU] =
+    {};
+}
+
+namespace
+{
+  using namespace os::rtos::port;
+
+  /* Every CPU's host thread, so one can be signalled by another. */
+  pthread_t g_cpu_thread[OS_NCPU];
+  volatile bool g_cpu_up[OS_NCPU];
+
+  /* The deferred-publish slot, one per CPU. */
+  struct publish_slot
+  {
+    host_cpu::element_t** addr;
+    host_cpu::element_t* val;
+  };
+
+  publish_slot g_publish[OS_NCPU];
+
+  /* What a never-run context must call. Recovered by the trampoline
+     from the
+   * two pointers makecontext() carries for it. */
+  using body_t = void (*) (void*);
+
+  [[noreturn]] void
+  fatal (const char* what)
+  {
+    std::fprintf (stderr, "\n!!! posix-arch: %s: %s\n", what,
+                  std::strerror (errno));
+    std::abort ();
+  }
+
+  /*
+   * The interrupt path.
+   *
+   * Deliberately the same shape as boards/rpi-zero-2w/src/rtos/
+   port_isr.cpp:
+   * flag handler mode, do the work, mark a switch pending, clear the
+   flag,
+   * and take the switch on the way out.
+   *
+   * Two things are deliberate and neither is an oversight:
+   *
+   * - NO_SA_ONSTACK. The handler runs on the µOS++ thread's own
+   stack, so
+   * the whole signal frame is part of the context that
+   swapcontext() saves
+   * and therefore migrates with the thread when another CPU
+   resumes it. An

```

```

+ *   alternate stack belongs to the host thread and would be left
+ *   behind.
+ *   This is why OS_INTEGER_RTOS_MIN_STACK_SIZE_BYTES is 32 KiB
+ *   here.
+ *
+ * - The switch happens INSIDE the handler, not after sigreturn.
+ *   Returning
+ *   first would mean the switch only ever happened at a
+ *   cooperative point,
+ *   which is exactly the limitation upstream's NOTES.md records
+ *   ("the
+ *   scheduler runs in cooperative mode only"). swapcontext() from
+ *   a
+ *   handler is well defined here because the handler frame lives
+ *   on the
+ *   thread's stack: resuming the context resumes the handler,
+ *   which then
+ *   returns through sigreturn normally.
+ */
+ void
+ irq_epilogue (unsigned cpu)
+ {
+     if (scheduler::_port_ctx_pending[cpu] == 0)
+     {
+         return;
+     }
+
+     if (os::rtos::scheduler::locked ())
+     {
+         // Still masked by the kernel; the next tick will find it
+         pending.
+         return;
+     }
+
+     if (scheduler::_smp_klock.owner == cpu &&
+         scheduler::_smp_klock.depth > 0)
+     {
+         return;
+     }
+
+     scheduler::_port_ctx_pending[cpu] = 0;
+     scheduler::switch_stacks (nullptr);
+ }
+
+ /* Put `errno` back, out of line.
+ *
+ * Out of line on purpose. `errno` is `*_errno_location()`, which
+ * is native
+ * TLS, and the caller read it BEFORE a swapcontext() that may have
+ * resumed
+ * this thread on a different CPU -- that is, on a different host
+ * thread,
+ * with a different errno. A compiler that cached the address
+ * across the
+ * switch would write the value into the host thread the thread
+ * LEFT. This
+ * port bans native TLS across a switch point for exactly that
+ * reason, and a
+ * noinline call is how the ban is honoured here: the address is
+ * resolved
+ * inside this function, after the switch, on whichever CPU is
+ * running now.
+ */
+ [[gnu::noinline]] void
+ restore_errno (int value)
+ {
+     errno = value;
+ }
+
+ void
+ tick_handler (int, siginfo_t*, void*)
+ {

```

```

+ /* errno belongs to the THREAD, and on this port the thread is
+    the uOS++
+    * one, not the host thread it is borrowing.
+    *
+    * This handler does not, in general, return to where it was
+    raised: its
+    * epilogue switches contexts, so control leaves on one thread's
+    stack and
+    * comes back -- possibly on another CPU, possibly much later --
+    when THIS
+    * thread is resumed. Everything the handler and every thread
+    scheduled in
+    * between did to errno is therefore visible to the interrupted
+    code
+    * unless it is put back. On silicon there is no errno to spoil;
+    here
+    * there is.
+    *
+    * `saved` is a local, so like the ASan fibre save it rides the
+    interrupted thread's own stack and is still correct wherever
+    that
+    * thread comes back. */
+ const int saved = errno;
+
+ const unsigned cpu = port_cpu_id ();
+ if (cpu >= OS_NCPU)
+ {
+     restore_errno (saved);
+     return;
+ }
+
+ interrupts::_in_isr[cpu] = true;
+
+#if defined(__linux__)
+ // Only CPU 0 advances the kernel clock. Every CPU tickles itself
+ into
+ // re-picking. This is the BCM2837 arrangement exactly: four
+ cores take
+ // the 1 ms PPI, one calls os_systick_handler().
+ if (cpu == 0)
+ {
+     os_systick_handler ();
+ }
+#else
+ // On non-Linux where per-thread timers are not available
+ (SIGEV_SIGNAL
+ // process-directed delivery), guarantee os_systick_handler() is
+ called
+ // regardless of which host thread received the tick signal so
+ the clock
+ // does not stall.
+ os_systick_handler ();
+#endif
+
+ scheduler::_port_ctx_pending[cpu] = 1;
+
+ interrupts::_in_isr[cpu] = false;
+
+ irq_epilogue (cpu);
+
+ // Resumed. See the comment at the top of this function.
+ restore_errno (saved);
+ }
+
+ void
+ ipi_handler (int, siginfo_t*, void*)
+ {
+     // Same reasoning as tick_handler(); this one switches contexts
+     too.
+     const int saved = errno;
+

```

```

+     const unsigned cpu = port_cpu_id ();
+     if (cpu >= OS_NCPU)
+     {
+         restore_errno (saved);
+         return;
+     }
+
+     interrupts::_in_isr[cpu] = true;
+     scheduler::_port_ctx_pending[cpu] = 1;
+     interrupts::_in_isr[cpu] = false;
+
+     irq_epilogue (cpu);
+
+     restore_errno (saved);
+ }
+
+ /* One per CPU, never shared: a CPU reporting a fault is not going
+    anywhere
+ * else while it does so. SIGSTKSZ is a minimum, and the reporting
+    path
+ * formats numbers, so it gets room. */
+ char g_fault_stack[OS_NCPU][64 * 1024];
+
+ void
+ arm_tick (void)
+ {
+     /* struct */ sigevent sev;
+     std::memset (&sev, 0, sizeof (sev));
+
+ #if defined(__linux__)
+     // SIGEV_THREAD_ID, not ITIMER_REAL.
+     //
+     // ITIMER_REAL delivers SIGALRM to an ARBITRARY thread of the
+     // process, so
+     // with N CPUs the tick would land on whichever host thread the
+     // kernel
+     // happened to choose -- a tick is not a per-CPU event any more,
+     // and the
+     // preemption of CPU 2 could be charged to CPU 0. A per-thread
+     // timer is
+     // what a per-core timer actually is.
+     sev.sigev_notify = SIGEV_THREAD_ID;
+     sev._sigev_un._tid = static_cast<int> (::syscall (SYS_gettid));
+ #else
+     // On non-Linux (e.g. Darwin/BSD), per-thread timer creation is
+     // not available.
+     // Arm the process timer only on CPU 0 to prevent N timers firing
+     // concurrently.
+     if (port_cpu_id () != 0)
+     {
+         return;
+     }
+     sev.sigev_notify = SIGEV_SIGNAL;
+ #endif
+     sev.sigev_signo = clock::signal_number ();
+
+     timer_t timer;
+     if (::timer_create (CLOCK_MONOTONIC, &sev, &timer) != 0)
+     {
+         fatal ("timer_create");
+     }
+
+     const long period_ns = 1000000000L /
+         OS_INTEGER_SYSTICK_FREQUENCY_HZ;
+
+     /* struct */ itimerspec its;
+     its.it_value.tv_sec = 0;
+     its.it_value.tv_nsec = period_ns;
+     its.it_interval.tv_sec = 0;

```

```

+     its.it_interval.tv_nsec = period_ns;
+
+     if (::timer_settime (timer, 0, &its, nullptr) != 0)
+     {
+         fatal ("timer_settime");
+     }
+ }
+
+ void*
+ secondary_cpu_body (void* arg)
+ {
+     const unsigned cpu
+         = static_cast<unsigned> (reinterpret_cast<uintptr_t> (arg));
+
+     _this_cpu = cpu;
+
+     host_cpu::install_fault_stack ();
+
+     ::pthread_sigmask (SIG_BLOCK, &interrupts::irq_set, nullptr);
+
+     // A context to switch away from, as port::scheduler::start()
+     // builds for
+     // CPU 0. A secondary core does exactly this after the spin table
+     // releases it: give itself somewhere to save, then reschedule.
+     static os_thread_t fake_thread[OS_NCPU];
+     std::memset (&fake_thread[cpu], 0, sizeof (os_thread_t));
+     fake_thread[cpu].name = "fake_thread_sec";
+     host_cpu::current_thread (cpu)
+         = reinterpret_cast<os::rtos::thread*> (&fake_thread[cpu]);
+
+     g_cpu_up[cpu] = true;
+
+     host_cpu::start_this_cpu_tick ();
+
+     g_core_stage[cpu] = 3;
+
+     scheduler::reschedule ();
+
+     // Only if this CPU never found anything to run.
+     ::pthread_sigmask (SIG_UNBLOCK, &interrupts::irq_set, nullptr);
+     for (;;)
+     {
+         ::pause ();
+     }
+
+     return nullptr;
+ }
+
+ [[noreturn]] void
+ trampoline (void* func, void* args)
+ {
+     // libucontext contexts carry no uc_sigmask, so guarantee the
+     // masked
+     // start here, before the deferred publish (unmasked again
+     // below).
+     ::pthread_sigmask (SIG_BLOCK, &interrupts::irq_set, nullptr);
+     // Arrival. The switch that brought us here started on another
+     // stack; this
+     // tells ASan the move is complete and that THIS stack is the
+     // live one.
+     // nullptr because a context that has never run has no outgoing
+     // fibre
+     // bookkeeping of its own to restore.
+     host_cpu::asan_finish_switch (nullptr);
+
+     // A context that has never run still arrives on a CPU that has
+     // just left
+     // another thread behind, so the publish is owed here too -- and
+     // it is

```

```

+ // owed BEFORE this thread can be interrupted, or the tick's own
+ // switch
+ // would overwrite the slot and strand the outgoing thread.
+ // context::create
+ // starts every new context with this CPU's interrupts masked for
+ // exactly
+ // this window; it ends here.
+ host_cpu::publish_pending ();
+
+ ::pthread_sigmask (SIG_UNBLOCK, &interrupts::irq_set, nullptr);
+
+ reinterpret_cast<body_t> (func) (args);
+
+ // The kernel's thread body does not return.
+ std::fprintf (stderr, "\n!!! posix-arch: thread body
+ returned\n");
+ std::abort ();
+ }
+
+} /* anonymous namespace */
+
+namespace host_cpu
+{
+ void
+ install_handlers (void)
+ {
+ struct sigaction sa; // `sigaction` is also a function: the tag
+ // is required
+ std::memset (&sa, 0, sizeof (sa));
+
+ // SA_RESTART so a preempted read()/write() resumes rather than
+ // failing
+ // with EINTR; the tests print through write(), a thousand times
+ // a second.
+ // No SA_ONSTACK -- see irq_epilogue() above.
+ sa.sa_flags = SA_SIGINFO | SA_RESTART;
+ ::sigemptyset (&sa.sa_mask);
+ ::sigaddset (&sa.sa_mask, clock::signal_number ());
+ ::sigaddset (&sa.sa_mask, clock::ipi_signal_number ());
+
+ sa.sa_sigaction = tick_handler;
+ if (::sigaction (clock::signal_number (), &sa, nullptr) != 0)
+ {
+ fatal ("sigaction(tick)");
+ }
+
+ sa.sa_sigaction = ipi_handler;
+ if (::sigaction (clock::ipi_signal_number (), &sa, nullptr) != 0)
+ {
+ fatal ("sigaction(ipi)");
+ }
+
+ g_cpu_thread[0] = ::pthread_self ();
+ g_cpu_up[0] = true;
+ g_core_stage[0] = 3;
+
+ install_fault_stack ();
+ }
+
+ void
+ install_fault_stack (void)
+ {
+ const unsigned cpu = port_cpu_id ();
+ if (cpu >= OS_NCPU)
+ {
+ return;
+ }
+
+

```

```

+   stack_t ss;
+   std::memset (&ss, 0, sizeof (ss));
+   ss.ss_sp = g_fault_stack[cpu];
+   ss.ss_size = sizeof (g_fault_stack[cpu]);
+   ss.ss_flags = 0;
+   ::sigaltstack (&ss, nullptr);
+ }
+
+ void
+ start_this_cpu_tick (void)
+ {
+   arm_tick ();
+ }
+
+ void
+ start_secondary_cpus (void)
+ {
+   for (unsigned cpu = 1; cpu < OS_NCPU; ++cpu)
+   {
+     pthread_attr_t attr;
+     ::pthread_attr_init (&attr);
+
+     if (::pthread_create (
+         &g_cpu_thread[cpu], &attr, secondary_cpu_body,
+         reinterpret_cast<void*> (static_cast<uintptr_t>
+ (cpu)))
+         != 0)
+     {
+       fatal ("pthread_create(cpu)");
+     }
+
+     ::pthread_attr_destroy (&attr);
+   }
+
+   // Wait for every CPU to have an index and a context to save
+   // into, the
+   // way the ARM ports wait on the per-core stage flags before
+   // proceeding.
+   for (unsigned cpu = 1; cpu < OS_NCPU; ++cpu)
+   {
+     while (!g_cpu_up[cpu])
+     {
+       ::usleep (100);
+     }
+   }
+ }
+
+ void
+ send_ipi (unsigned cpu)
+ {
+   if (cpu < OS_NCPU && g_cpu_up[cpu])
+   {
+     ::pthread_kill (g_cpu_thread[cpu], clock::ipi_signal_number
+ ());
+   }
+ }
+
+ void
+ make_entry (os_impl_ucontext_t* ctx, void* func, void* args)
+ {
+   #pragma GCC diagnostic push
+   #if defined(__clang__)
+   #pragma clang diagnostic ignored "-Wc++98-compat-pedantic"
+   #endif
+   // Two pointer arguments. glibc carries them whole -- verified on
+   // this
+   // host before it was relied on, because the interface is
+   // declared with

```

```

+ // int arguments and a truncated `args` would be a pointer to
+ // nothing.
+ os_impl_makecontext (
+     ctx, reinterpret_cast<void (*) (void)> (trampoline), 2, func,
+     args);
+ #pragma GCC diagnostic pop
+ }
+
+ void
+ defer_publish (unsigned cpu, element_t** addr, element_t* val)
+ {
+     g_publish[cpu].addr = addr;
+     g_publish[cpu].val = val;
+ }
+
+ /* See the long comment in host_cpu.hpp. Declared by hand rather
+    than through
+    * <sanitizer/asan_interface.h>, so that a toolchain without the
+    header still
+    * builds the port -- the symbols come from the ASan runtime, which
+    is only
+    * linked when -fsanitize=address is on, and the calls are compiled
+    out
+    * otherwise.
+    *
+    * __has_feature is clang's (and GCC >= 14's): it is tested in its
+    own #if,
+    * because an older GCC rejects `__has_feature (x)` even behind a
+    * `defined (__has_feature) &&`. */
+ #if defined(__SANITIZE_ADDRESS__)
+ #define UOS_HAVE_ASAN 1
+ #elif defined(__has_feature)
+ #if __has_feature(address_sanitizer)
+ #define UOS_HAVE_ASAN 1
+ #endif
+ #endif
+ #if defined(UOS_HAVE_ASAN)
+ extern "C" void
+ __sanitizer_start_switch_fiber (void** fake_stack_save, const void*
+     bottom,
+
+     std::size_t size);
+ extern "C" void
+ __sanitizer_finish_switch_fiber (void* fake_stack_save, const void**
+     bottom_old,
+
+     std::size_t* size_old);
+ #endif
+
+ void
+ asan_start_switch (void** save, const void* bottom, std::size_t
+     size)
+ {
+     #if defined(UOS_HAVE_ASAN)
+     __sanitizer_start_switch_fiber (save, bottom, size);
+     #else
+     (void)save;
+     (void)bottom;
+     (void)size;
+     #endif
+ }
+
+ void
+ asan_finish_switch (void* save)
+ {
+     #if defined(UOS_HAVE_ASAN)
+     __sanitizer_finish_switch_fiber (save, nullptr, nullptr);
+     #else
+     (void)save;
+     #endif
+ }
+
+

```

```

+ void
+ publish_pending (void)
+ {
+     // The CPU index must be re-read: this runs after a swapcontext()
+     // that
+     // may well have been performed by a different CPU.
+     const unsigned cpu = port_cpu_id ();
+     if (cpu >= OS_NCPU)
+     {
+         return;
+     }
+
+     element_t** addr = g_publish[cpu].addr;
+     if (addr == nullptr)
+     {
+         return;
+     }
+
+     g_publish[cpu].addr = nullptr;
+     __atomic_store_n (addr, g_publish[cpu].val, __ATOMIC_RELEASE);
+ }
+
+} /* namespace host_cpu */
+
+extern "C" void port_smp_ipi (unsigned cpu);
+
+extern "C" void
+port_smp_ipi (unsigned cpu)
+{
+    host_cpu::send_ipi (cpu);
+}
+
+#endif /* defined(__APPLE__) || defined(__linux__) */
diff --git a/src/rtos/os-core.cpp b/src/rtos/os-core.cpp
index 91689ca..c073489 100644
--- a/src/rtos/os-core.cpp
+++ b/src/rtos/os-core.cpp
@@ -1,34 +1,67 @@
/*
 * os-core.cpp - the port's half of the µOS++ III SMP scheduler,
 *               POSIX host.
 *
 * This file is part of the µOS++ project (https://micro-os-plus.github.io/).
 * Copyright (c) 2016-2025 Liviu Ionescu. All rights reserved.
 * Copyright (c) 2026 Dan. All rights reserved.
 *
 * Permission to use, copy, modify, and/or distribute this software
 * for any purpose is hereby granted, under the terms of the MIT
 * license.
 *
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 * can
 * be obtained from https://opensource.org/licenses/mit/.
 * -----
 * The model: a host thread IS a CPU.
 *
 * OS_NCPU host threads are created at startup and never destroyed.
 * Each one
 * runs the scheduler and is, for every purpose the kernel can
 * observe, a
 * core: it has its own interrupt mask (its signal mask), its own
 * tick
 * (its own timer_create timer), its own handler-mode flag and its
 * own entry
 * in lock_state[] and _port_ctx_pending[].
 *
 * µOS++ threads remain ucontext contexts switched WITHIN a CPU,
 * which is

```

```

+ * upstream's machinery, preserved deliberately. What is new is that
+ *   a context
+ * saved by one CPU may be resumed by another -- and that is the only
+ *   genuinely
+ * hard part of this port, because it is a data race unless the
+ *   handover is
+ * ordered. See switch_stacks() below.
+ *
+ * Read this file beside micro-os-plus-iii-aarch64/src/rtos/os-
+ *   core.cpp. The
+ * two have the same functions doing the same things in the same
+ *   order.
+ * -----
+ */

#if defined(__APPLE__) || defined(__linux__)

-// -----
-
#include <cassert>
+#include <cstdlib>
+#include <cstring>

#include <cmsis-plus/rtos/os.h>
+#include <cmsis-plus/rtos/os-hooks.h>
#include <cmsis-plus/rtos/port/os-inlines.h>

-#include <sys/time.h>
+#include <host_cpu.hpp>

-// -----
+#include <sys/utsname.h>
+#include <sys/time.h>
+#include <unistd.h>

#if defined(__clang__)
#pragma clang diagnostic ignored "-Wc++98-compat"
#endif

-// -----
+extern os::rtos::thread* os_idle_thread;

-uint32_t signal_nesting;
+#if defined(OS_USE_SMP_SCHEDULER)
+namespace os
+{
+  namespace rtos
+  {
+    {
+      namespace scheduler
+      {
+        {
+          extern thread* os_idle_thread_core[OS_NCPU];
+        }
+      }
+    }
+  }
+}
+#endif /* defined(OS_USE_SMP_SCHEDULER) */

namespace os
{
@@ -38,138 +71,26 @@ namespace os
{
//
-----

-#pragma GCC diagnostic push
-#pragma GCC diagnostic ignored "-Wdeprecated-declarations"
+  thread_local unsigned _this_cpu = 0;

-  void

```

```

-     context::create (void* context, void* func, void* args)
-     {
-         /* class */ rtos::thread::context* th_ctx
-         = static_cast< /* class */ rtos::thread::context*>
-         (context);
-         memset (&th_ctx->port_, 0, sizeof (th_ctx->port_));
-
- #pragma GCC diagnostic push
- #if defined(__clang__)
- #elif defined(__GNUC__)
- #pragma GCC diagnostic ignored "-Wuseless-cast"
- #endif
-         os_impl_ucontext_t* ctx =
-         reinterpret_cast<os_impl_ucontext_t*> (
-             &(th_ctx->port_.ucontext));
- #pragma GCC diagnostic pop
-
- #if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
-         trace::printf ("port::context::%s() getcontext %p\n",
-             __func__, ctx);
- #endif
-
-         if (os_impl_getcontext (ctx) != 0)
-         {
-             trace::printf ("port::context::%s() getcontext failed
- with %s\n",
-                 __func__, strerror (errno));
-             abort ();
-         }
-
-         // The context in itself is not needed, but makecontext()
-         // requires a context obtained by getcontext().
-
-         // Remove the parent link.
-         // TODO: maybe use this to link to exit code.
-         ctx->uc_link = nullptr;
-
-         // Configure the new stack to default values.
-         ctx->uc_stack.ss_sp = th_ctx->stack ().bottom ();
-         ctx->uc_stack.ss_size = th_ctx->stack ().size ();
-         ctx->uc_stack.ss_flags = 0;
-
- #if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
-         trace::printf ("port::context::%s() makecontext %p\n",
-             __func__, ctx);
- #endif
-
- #pragma GCC diagnostic push
- #if defined(__clang__)
- #pragma clang diagnostic ignored "-Wc++98-compat-pedantic"
- #endif
-         os_impl_makecontext (ctx, reinterpret_cast<func_t> (func), 1,
-             args);
- #pragma GCC diagnostic pop
-
-         // context->port_.saved = false;
-     }
-
- #pragma GCC diagnostic pop
-
-     //
-     -----
-
-     namespace interrupts
-     {
-         //
-         -----
-
-         sigset_t clock_set;

```

```

+     sigset_t irq_set;
-
-     //
-     -----
-
-     // Enter an IRQ critical section
-     rtos::interrupts::state_t
-     critical_section::enter (void)
-     {
-#if defined(OS_TRACE_RTOS_SCHEDULER)
-         trace::printf ("c ");
-#endif
-         sigset_t old;
-         sigprocmask (SIG_BLOCK, &clock_set, &old);
-
-#pragma GCC diagnostic push
-#pragma GCC diagnostic ignored "-Wsign-conversion"
-         return sigismember (&old, clock::signal_number);
-#pragma GCC diagnostic pop
-     }
-
-     // Exit an IRQ critical section
-     void
-     critical_section::exit (rtos::interrupts::state_t state)
-     {
-#if defined(OS_TRACE_RTOS_SCHEDULER)
-         trace::printf (" c");
-#endif
-         sigprocmask (state ? SIG_BLOCK : SIG_UNBLOCK, &clock_set,
-         nullptr);
-     }
-
-     //
-     =====
-
-     // Enter an IRQ uncritical section
-     rtos::interrupts::state_t
-     uncritical_section::enter (void)
-     {
-#if defined(OS_TRACE_RTOS_SCHEDULER)
-         trace::printf ("u ");
-#endif
-         sigset_t old;
-         sigprocmask (SIG_UNBLOCK, &clock_set, &old);
-
-#pragma GCC diagnostic push
-#pragma GCC diagnostic ignored "-Wsign-conversion"
-         return sigismember (&old, clock::signal_number);
-#pragma GCC diagnostic pop
-     }
-
-     // Exit an IRQ critical section
-     void
-     uncritical_section::exit (rtos::interrupts::state_t state)
-     {
-#if defined(OS_TRACE_RTOS_SCHEDULER)
-         trace::printf (" u");
-#endif
-         sigprocmask (state ? SIG_BLOCK : SIG_UNBLOCK, &clock_set,
-         nullptr);
-     }
+     extern "C" volatile bool _in_isr[OS_NCPU];
+     volatile bool _in_isr[OS_NCPU] = {};
+
+     extern "C" volatile uint32_t signal_nesting;
+     volatile uint32_t signal_nesting = 0;
+ } /* namespace interrupts */

```

```

- //
- -----
-
namespace scheduler
{
+ volatile state_t lock_state[OS_NCPU] = {};
-
- //
- -----
-
state_t lock_state;
+ smp_klock_t _smp_klock = { 0, SMP_NO_OWNER, 0 };
+ volatile unsigned _port_ctx_pending[OS_NCPU] = {};

//
- -----

@@ -179,96 +100,301 @@ namespace os
/* struct */ utsname name;
if (::uname (&name) != -1)
{
- trace::printf ("POSIX synthetic, running on %s %s %s",
+ trace::printf ("POSIX synthetic SMP, running on %s %s
%s",
name.machine, name.sysname,
name.release);
}
else
{
- trace::printf ("POSIX synthetic");
+ trace::printf ("POSIX synthetic SMP");
}

- trace::puts ("; non-preemptive");
+ trace::printf ("; %d CPU%s, %d Hz tick, preemptive\n",
OS_NCPU,
(OS_NCPU == 1) ? "" : "s",
OS_INTEGER_SYSTICK_FREQUENCY_HZ);
}

result_t
initialize (void)
{
- signal_nesting = 0;
+ // Must be done before the first critical section: every
mask
+ // operation in this port names this set.
+ ::sigemptyset (&interrupts::irq_set);
+ ::sigaddset (&interrupts::irq_set, clock::signal_number
());
+ ::sigaddset (&interrupts::irq_set, clock::ipi_signal_number
());

- // Must be done before the first critical section.
- sigemptyset (&interrupts::clock_set);
+ for (unsigned c = 0; c < OS_NCPU; ++c)
+ {
+ lock_state[c] = state::init;
+ _port_ctx_pending[c] = 0;
+ interrupts::_in_isr[c] = false;
+ }

-#pragma GCC diagnostic push
-#if defined(__clang__)
-#elif defined(__GNUC__)
-#pragma GCC diagnostic ignored "-Wsign-conversion"
-#endif
- sigaddset (&interrupts::clock_set, clock::signal_number);

```

```

-#pragma GCC diagnostic pop
+     host_cpu::install_handlers ();
+
+     /*
+      * The application's hardware hooks.
+      *
+      * On a bare-metal target the kernel's own src/startup/
+      startup.cpp
+      * calls these two before main(), and every carried test
+      relies on
+      * that: os_startup_initialize_hardware() is where a test
+      brings up
+      * its console, prints its banner, installs the free store
+      and calls
+      * exception::init().
+      *
+      * Here there is no such startup. Upstream's NOTES.md
+      states the
+      * rule -- "For portability reasons, execution starts in
+      the main()
+      * function" -- and startup.cpp is __ARM_EABI__-guarded
+      from end to
+      * end, so nothing calls them at all. The first thing every
+      test's
+      * main() calls is scheduler::initialize(), so this is
+      where that
+      * startup belongs, and the tests need no edit.
+      *
+      * Found by running: without this the banner never printed,
+      the
+      * application free store was never installed (the kernel's
+      * malloc resource stayed in place), and exception::init()
+      never
+      * ran -- so the first SMP fault reported nothing at all.
+      *
+      * After install_handlers(), deliberately:
+      exception::init() wants
+      * the alternate signal stack that call sets up.
+      */
+     os_startup_initialize_hardware_early ();
+     os_startup_initialize_hardware ();
+
+     // Interrupts masked until start(), as every port does.
+     :pthread_sigmask (SIG_BLOCK, &interrupts::irq_set,
+     nullptr);
+
+     return result::ok;
+ }
+
+ //
+ -----

```

```

-#pragma GCC diagnostic push
-#pragma GCC diagnostic ignored "-Wdeprecated-declarations"
-
-     void
-     start (void)
+     state_t
+     locked (state_t state)
+     {
+     {
-         rtos::interrupts::critical_section ics;
-
-         // Determine the next thread.
-         rtos::scheduler::current_thread_
-         = rtos::scheduler::ready_threads_list_.unlink_head
+         ();
+     }
+     os_assert_throw (!interrupts::in_handler_mode (), EPERM);
-#pragma GCC diagnostic push

```

```

-#if defined(__clang__)
-#elif defined(__GNUC__)
-#pragma GCC diagnostic ignored "-Wuseless-cast"
-#endif
-
-    os_impl_ucontext_t* new_context
-    = reinterpret_cast<os_impl_ucontext_t*> (&(
-        rtos::scheduler::current_thread_
-        >context_.port_.ucontext));
-#pragma GCC diagnostic pop
-
-#if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
-    trace::printf ("port::scheduler::%s() ctx %p %s\n",
-        __func__,
-
-        new_context,
-        rtos::scheduler::current_thread_>name ());
-#endif
+
+    if (state == state::locked)
+    {
+        // Block the tick BEFORE reading the CPU id. The tick
+        handler
+        // swapcontext()s, and the thread can resume on another
+        host
+        // thread: read before the block, the id could name the
+        CPU it
+        // left, and lock_state[] and the kernel lock would be
+        taken for
+        // that CPU, never to be released by the matching
+        unlock.
+        ::pthread_sigmask (SIG_BLOCK, &interrupts::irq_set,
+        nullptr);
+        const unsigned cpu = port_cpu_id ();
+        state_t tmp = lock_state[cpu];
+        if (tmp != state::locked)
+        {
+            if (_smp_klock.owner != cpu)
+            {
+                _smp_klock_raw_acquire ();
+                _smp_klock.owner = cpu;
+            }
+            _smp_klock.depth = _smp_klock.depth + 1;
+            lock_state[cpu] = state::locked;
+        }
+        return tmp;
+    }
+    else
+    {
+        // Holding the lock, the tick is blocked and the thread
+        cannot
+        // move; not holding it, both CPUs read "unlocked"
+        anyway.
+        const unsigned cpu = port_cpu_id ();
+        state_t tmp = lock_state[cpu];
+        if (tmp != state::unlocked)
+        {
+            lock_state[cpu] = state::unlocked;
+            if (_smp_klock.owner == cpu && _smp_klock.depth >
+0)
+            {
+                const uint32_t d = _smp_klock.depth - 1;
+                _smp_klock.depth = d;
+                if (d == 0)
+                {
+                    _smp_klock.owner = SMP_NO_OWNER;
+                    _smp_klock_raw_release ();
+                }
+            }
+            ::pthread_sigmask (SIG_UNBLOCK,
+            &interrupts::irq_set,
+
+            nullptr);
+        }
+    }

```

```

+         return tmp;
+     }
+ }

-     lock_state = state::init;
+ //
+ -----

-#if defined NDEBUG
-     os_impl_setcontext (new_context);
-#else
-     int res = os_impl_setcontext (new_context);
-     assert (res == 0);
-#endif
-     abort ();
+ void
+ wait_for_interrupt (void)
+ {
+     // The idle thread's "WFI". sigsuspend() unblocks this
+     // CPU's tick
+     // and IPI and sleeps until one of them is delivered AND
+     // handled, so
+     // an idle CPU costs nothing while still being preemptible.
+     sigset_t mask;
+     ::pthread_sigmask (SIG_SETMASK, nullptr, &mask);
+     ::sigdelset (&mask, clock::signal_number ());
+     ::sigdelset (&mask, clock::ipi_signal_number ());
+     ::sigsuspend (&mask);
+ }

+ //
+ -----

- state_t
- locked (state_t state)
+ /*
+  * The context switch.
+  *
+  * Named switch_stacks() because that is what the kernel
+  * calls this
+  * seam on every port, and because it is a friend of
+  * rtos::thread --
+  * which is how it may touch context_ at all. Unlike the ARM
+  * ports it
+  * performs the switch itself rather than returning a stack
+  * pointer to
+  * an assembly restore path; there is no assembly here to
+  * return to.
+  *
+  *
+  * THE DEFERRED PUBLISH.
+  *
+  * The SMP picker in the kernel skips any thread whose
+  * stack_ptr is
+  * null, meaning "not safe to claim". The window that rule
+  * exists for
+  * is real here too: between the moment this CPU decides to
+  * leave
+  * old_thread and the moment swapcontext() has finished
+  * writing
+  * old_thread's registers into its ucontext, another CPU must
+  * not
+  * resume it -- it would run a half-saved context.
+  *
+  * So the publish cannot be done by the CPU that is leaving:
+  * once
+  * swapcontext() returns, this CPU is already executing the
+  * INCOMING
+  * thread. It is done by whoever arrives next on this CPU
+  * instead. The
+  * outgoing thread's address and value are left in this CPU's
+  * slot,
+  * and the first thing any resumed context does -- here,
+  * after

```

```

+      * swapcontext(), and in host_cpu's trampoline for a context
+      that has
+      * never run -- is publish it.
+      *
+      * This is the same mechanism as AArch64's _smp_pub_addr/
+      _smp_pub_val
+      * pair, which its assembly restore path applies only after
+      SP has
+      * left the outgoing stack. Same hazard, same answer,
+      different
+      * machine.
+      */
+      stack::element_t*
+      switch_stacks (stack::element_t* sp)
+      {
-         os_assert_throw (!interrupts::in_handler_mode (), EPERM);
+         (void)sp;
+
+         /*
+          * MASK THIS CPU FIRST. The switch must be atomic with
+          respect to
+          * this CPU's own interrupts, and on the ARM ports it is
+          atomic for
+          * free: the whole of switch_stacks() runs inside the IRQ
+          path, with
+          * interrupts already masked by the exception entry.
+          *
+          * Here it is not free. reschedule() reaches this function
+          directly
+          * from thread mode, where this CPU's tick is unmasked --
+          so the
+          * timer could land between clearing old_thread's publish
+          flag and
+          * swapcontext() finishing the save, and the handler would
+          re-enter
+          * this same function on a half-performed switch. That is a
+          * genuinely reentrant context switch, and it was the cause
+          of the
+          * intermittent SIGSEGV this port showed on smp_test2
+          (roughly three
+          * runs in five) the first time it ran multi-core.
+          *
+          * The mask is saved by swapcontext() into the outgoing
+          context and
+          * restored from the incoming one, so it follows the thread
+          across
+          * CPUs, which is exactly what interrupt state should do.
+          */
+          sigset_t saved_mask;
+          ::pthread_sigmask (SIG_BLOCK, &interrupts::irq_set,
+          &saved_mask);
+
+          const unsigned cpu = port_cpu_id ();
+
+          // Take the kernel lock outright. reschedule() has already
+          returned
+          // early if this CPU still owns it, so it is not held here.
+          _smp_klock_raw_acquire ();
+          _smp_klock.owner = cpu;
+          _smp_klock.depth = 1;
+
+          rtos::thread* old_thread = host_cpu::current_thread (cpu);
+
+          stack::element_t** pub_addr
+          = &old_thread->context_.port_.stack_ptr;
+          // Any stable non-null value: to the kernel this field is a
+          flag,
+          // and the only thing it ever asks is whether it is null.
+          stack::element_t* pub_val =
+          reinterpret_cast<stack::element_t*> (
+          &old_thread->context_.port_.ucontext);
+

```

```

+ // Park the outgoing thread before the picker can see it.
+ It may
+ // still re-pick it -- the kernel's test is
+ // `th == old_thread || stack_ptr != nullptr` -- which is
+ correct,
+ // because this CPU never left it.
+ __atomic_store_n (pub_addr, static_cast<stack::element_t*>
+ (nullptr),
+                 __ATOMIC_RELEASE);
+
+ rtos::scheduler::internal_switch_threads ();
+
+ rtos::thread* new_thread = host_cpu::current_thread (cpu);
+
+ if (new_thread == nullptr)
+ {
+     trace::printf (
+         "\n!!! no ready thread and no idle thread on CPU
+ %u !!!\n",
+         cpu);
+     ::abort ();
+ }
+
- state_t tmp;
+ if (new_thread == old_thread)
+ {
+     // Nothing to do; republish immediately and let go.
+     __atomic_store_n (pub_addr, pub_val, __ATOMIC_RELEASE);
+     _smp_klock.depth = 0;
+     _smp_klock.owner = SMP_NO_OWNER;
+     _smp_klock_raw_release ();
+     ::pthread_sigmask (SIG_SETMASK, &saved_mask, nullptr);
+     return nullptr;
+ }
+
- {
-     rtos::interrupts::critical_section ics;
+ os_impl_ucontext_t* new_uc = &new_thread-
+ >context_.port_.ucontext;
+
+ // Claim the incoming context: from here no other CPU may
+ take it.
+ __atomic_store_n (&new_thread->context_.port_.stack_ptr,
+                 static_cast<stack::element_t*> (nullptr),
+                 __ATOMIC_RELEASE);
+
+ host_cpu::defer_publish (cpu, pub_addr, pub_val);
+
+ /* Release the kernel lock. THE ORDER MATTERS, and it is
+ the same
+ * order and the same reason as the AArch64 port documents
+ at
+ * length: owner and depth are cleared BEFORE the lock
+ word.
+ *
+ * Storing the lock word first opens a window in which
+ another CPU
+ * wins the lock and installs its own owner/depth, which
+ the two
+ * stores below then wipe. Its critical_section::exit() is
+ guarded
+ * by (owner == cpu && depth > 0), so it never clears the
+ lock word
+ * again, and every CPU spins for ever -- a silent, total
+ freeze.
+ * On the ARM ports that was the root cause of the
+ smp_test4
+ * hardware deadlock. */
+ _smp_klock.depth = 0;
+ _smp_klock.owner = SMP_NO_OWNER;
+ _smp_klock_raw_release ();

```

```

+
+     os_impl_ucontext_t* old_uc = &old_thread-
+>context_.port_.ucontext;
+
+     /* Tell AddressSanitizer the stack is about to move, and
+     where to.
+     * `asan_save` lives on the OUTGOING thread's stack, so it
+     is still
+     * there -- and still this thread's -- whenever and on
+     whichever CPU
+     * that thread is resumed. Compiles to nothing without
+     -fsanitize=
+     * address; see host_cpu.hpp. */
+     void* asan_save = nullptr;
+     host_cpu::asan_start_switch (&asan_save,
+     new_thread->stack ().bottom
+     (),
+     new_thread->stack ().size ());
+
+     if (os_impl_swapcontext (old_uc, new_uc) != 0)
+     {
+         trace::printf ("port::scheduler::%s() swapcontext
+         failed: %s\n",
+         __func__, strerror (errno));
+         ::abort ();
+     }
+
+     tmp = lock_state;
+     lock_state = state;
+ }
+ // Resumed -- and NOT necessarily on the CPU that left.
+ Everything
+ // below must re-read the CPU index; nothing captured above
+ is
+ // valid any more.
+ host_cpu::asan_finish_switch (asan_save);
+ host_cpu::publish_pending ();
+
+     return tmp;
+ // Unmask last, and only now: everything above this line is
+ the
+ // switch, and the switch is not interruptible.
+ ::pthread_sigmask (SIG_SETMASK, &saved_mask, nullptr);
+
+     return nullptr;
+ }
+
+ //
+ -----

```

@@ -276,145 +402,141 @@ namespace os

```

void
reschedule (void)
{
+     const unsigned cpu = port_cpu_id ();
+
+     if (rtos::scheduler::locked ()
+         || rtos::interrupts::in_handler_mode ()
+         || (rtos::interrupts::in_handler_mode ()
+             && !rtos::scheduler::preemptive ()))
+     {
+         -#if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
+         -         // trace::printf ("port::scheduler::%s() deny\n",
+         -         __func__);
+         -#endif
+         return;
+     }
+
+     -#if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
+     -         trace::printf ("port::scheduler::%s()\n", __func__);
+     -#endif

```

```

+         // If this CPU still owns the kernel lock -- resume_one()
+         called
+         // from inside an interrupts::critical_section, as
+         message_queue's
+         // send/receive do -- switching here would strand owner and
+         depth on
+         // the outgoing thread's context and deadlock the next
+         acquirer.
+         // Defer to the next tick, which runs once that section has
+         exited.
+         if (_smp_klock.owner == cpu && _smp_klock.depth > 0)
+         {
+             _port_ctx_pending[cpu] = 1;
+             return;
+         }
+
-         // For some complicated reasons, the context save/restore
-         // functions must be called in the same the function.
-         // The idea to inline functions does not work, since
-         // the compiler does not inline functions with context
+         calls.
+         if (rtos::interrupts::in_handler_mode ())
+         {
+             // Taken on the way out of the handler, where the mask
+             is right.
+             _port_ctx_pending[cpu] = 1;
+             return;
+         }
+
-         bool save = false;
-         rtos::thread* old_thread;
-         os_impl_ucontext_t* old_ctx;
-         os_impl_ucontext_t* new_ctx;
+         switch_stacks (nullptr);
+     }
+
-     {
-         rtos::interrupts::critical_section ics;
-
-         old_thread = rtos::scheduler::current_thread_;
-         if ((old_thread->state_ == rtos::thread::state::running)
-             || (old_thread->state_ ==
-                 rtos::thread::state::suspended)
-             || (old_thread->state_ ==
-                 rtos::thread::state::ready))
-         {
-             save = true;
-         }
-     }
-#if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
-     trace::printf ("port::scheduler::%s() from %s %d %d\n",
+     __func__,
-                     old_thread->name (), old_thread->state_,
+     save);
-#endif
+     //
+     -----
+
-#pragma GCC diagnostic push
-#if defined(__clang__)
-#elif defined(__GNUC__)
-#pragma GCC diagnostic ignored "-Wuseless-cast"
-#endif
-
-         old_ctx = reinterpret_cast<os_impl_ucontext_t*> (
-             &old_thread->context_.port_.ucontext);
-#pragma GCC diagnostic pop
+     [[noreturn]] void
+     start (void)
+     {
+         ::pthread_sigmask (SIG_BLOCK, &interrupts::irq_set,
+                             nullptr);
+     }

```

```

+         // A context to switch AWAY from. The very first switch has
+         to save
+         // something, and this CPU's host-thread context is not a
+         μOS++
+         // thread. The ARM ports use a zeroed fake_thread for the
+         same
+         // reason; this one is never scheduled again, so it needs
+         no stack.
+         static os_thread_t fake_thread[OS_NCPU];
+         const unsigned cpu = port_cpu_id ();
+         memset (&fake_thread[cpu], 0, sizeof (os_thread_t));
+         fake_thread[cpu].name = "fake_thread";
+         host_cpu::current_thread (cpu)
+             = reinterpret_cast<rtos::thread*> (&fake_thread[cpu]);
+
+         for (unsigned c = 0; c < OS_NCPU; ++c)
+         {
+             lock_state[c] = state::init;
+         }
+
+         rtos::scheduler::internal_switch_threads ();
+ #if defined(OS_USE_SMP_SCHEDULER)
+         // The SMP picker asks each CPU for its own idle thread;
+         CPU 0's is
+         // the kernel's. The secondaries' are installed by the
+         board, from
+         // test-smp-boot.cpp, exactly as on the ARM boards.
+         rtos::scheduler::os_idle_thread_core[0] = ::os_idle_thread;
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+ #pragma GCC diagnostic push
+ #if defined(__clang__)
+ #elif defined(__GNUC__)
+ #pragma GCC diagnostic ignored "-Wuseless-cast"
+ #endif
+         new_ctx = reinterpret_cast<os_impl_ucontext_t*> (
+             &rtos::scheduler::current_thread_
+             >context_.port_.ucontext);
+ #pragma GCC diagnostic pop
+         }
+         host_cpu::start_this_cpu_tick ();
+
+         if (old_ctx != new_ctx)
+         {
+             if (save)
+             {
+ #if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
+                 trace::printf (
+                     "port::scheduler::%s() swapcontext %s -> %s
+                     \n",
+                     __func__, old_thread->name (),
+                     rtos::scheduler::current_thread_>name ());
+ #endif
+                 if (os_impl_swapcontext (old_ctx, new_ctx) != 0)
+                 {
+                     trace::printf (
+                         "port::scheduler::%s() swapcontext failed
+                         with %s\n",
+                         __func__, strerror (errno));
+                     abort ();
+                 }
+             }
+             else
+             {
+ #if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
+                 trace::printf ("port::scheduler::%s() setcontext
+                 %s\n",
+                     __func__,
+                     rtos::scheduler::current_thread_
+                     >name ());

```

```

-#endif
-
-        // context->port_.saved = false;
-        if (os_impl_setcontext (new_ctx) != 0)
-        {
-            trace::printf (
-                "port::scheduler::%s() setcontext failed
with %s\n",
-                __func__, strerror (errno));
-            abort ();
-        }
-    }
-    }
-    else
+    reschedule ();
+
+    // Reached only if reschedule() found nothing to run at
+    all.
+    ::pthread_sigmask (SIG_UNBLOCK, &interrupts::irq_set,
+        nullptr);
+    for (;;)
+    {
-#if defined(OS_TRACE_RTOS_THREAD_CONTEXT)
-        trace::printf ("port::scheduler::%s() same %s\n",
+        __func__,
+
+        old_thread->name ());
-#endif
+
+        ::pause ();
+    }
+
+}

-#pragma GCC diagnostic pop
-
-    //
-    -----
-
-} /* namespace scheduler */
-
-    //
-    -----
-
-    static void
-    systick_clock_signal_handler (int signum)
+    void
+    context::create (void* context, void* func, void* args)
+    {
-#if defined(OS_TRACE_RTOS_SYSCLOCK_TICK)
-        trace::printf ("{i ");
-#endif
-
-        if (signum != clock::signal_number)
-        {
-            char ce = '?';
-
-#pragma GCC diagnostic push
-#pragma GCC diagnostic ignored "-Wunused-result"
+        /* class */ rtos::thread::context* th_ctx
+        = static_cast< /* class */ rtos::thread::context*>
+        (context);
-
-            write (1, &ce, 1);
+
+        memset (&th_ctx->port_, 0, sizeof (th_ctx->port_));
-#pragma GCC diagnostic pop
+
+        os_impl_ucontext_t* ctx = &th_ctx->port_.ucontext;
-
-        return;
+
+        if (os_impl_getcontext (ctx) != 0)
+        {

```

```

+         trace::printf ("port::context::%s() getcontext failed:
%s\n",
+             __func__, strerror (errno));
+         ::abort ();
+     }

-     signal_nesting++;
-     // Call the ticks timer ISR.
-     os_systick_handler ();
-     signal_nesting--;
-#if defined(OS_TRACE_RTOS_SYSCLOCK_TICK)
-     trace::printf (" i}");
+     // The context itself is not wanted; makecontext() merely
+     requires one
+     // obtained from getcontext().
+     ctx->uc_link = nullptr;
+     ctx->uc_stack.ss_sp = th_ctx->stack ().bottom ();
+     ctx->uc_stack.ss_size = th_ctx->stack ().size ();
+     ctx->uc_stack.ss_flags = 0;

+     /*
+      * The starting signal mask: THIS CPU'S INTERRUPTS MASKED.
+      *
+      * Two wrong answers were tried before this one, which is why
+      the
+      * reasoning is written down.
+      *
+      * getcontext() snapshots the CALLER's mask, and a thread is
+      very
+      * often created from inside a critical section -- so
+      inheriting it
+      * would start the thread with interrupts masked FOR EVER,
+      because
+      * nothing would ever unmask them. Clearing the mask entirely
+      is
+      * wrong in the opposite direction: a context resumed by
+      * switch_stacks() comes back with interrupts masked and
+      unmask only
+      * after it has discharged this CPU's deferred publish, and a
+      context
+      * that has never run arrives on a CPU owing exactly the same
+      publish.
+      * Starting it unmasked lets a tick land inside the
+      trampoline before
+      * that publish happens -- and that tick's own switch
+      overwrites the
+      * pending slot, so the thread it was owed to is never
+      republished and
+      * is lost from the ready list for good.
+      *
+      * So a new context begins the way a resumed one does:
+      masked. The
+      * trampoline publishes, then unmask, and from that point
+      the thread
+      * is preemptible like any other.
+      */
+#if !defined(OS_INCLUDE_LIBUCONTEXT)
+     ::sigemptyset (&ctx->uc_sigmask);
+     ::sigaddset (&ctx->uc_sigmask, clock::signal_number ());
+     ::sigaddset (&ctx->uc_sigmask, clock::ipi_signal_number ());
+#endif
+
+     host_cpu::make_entry (ctx, func, args);

+     // Published: this context has never run, so it is complete
+     by
+     // definition and any CPU may claim it.
+     th_ctx->port_.stack_ptr
+     = reinterpret_cast<stack::element_t*> (&th_ctx-
+>port_.ucontext);
+ }

```

```

//
=====
@@ -422,75 +544,11 @@ namespace os
void
clock_systick::start (void)
{
-    // set handler
-    struct sigaction sa;
-#if defined(__APPLE__)
-    sa.__sigaction_u.__sa_handler = systick_clock_signal_handler;
-#elif defined(__linux__)
-#pragma GCC diagnostic push
-#if defined(__clang__)
-#pragma clang diagnostic ignored "-Wdisabled-macro-expansion"
-#endif
-    sa.sa_handler = systick_clock_signal_handler;
-#pragma GCC diagnostic pop
-#else
-#error Platform unsupported
-#endif
-    sigemptyset (&sa.sa_mask);
-    sa.sa_flags = SA_RESTART;
-
-    if (sigaction (clock::signal_number, &sa, nullptr) != 0)
-    {
-        trace::printf ("port::clock_systick::%s() sigaction()
failed\n",
-            __func__);
-        abort ();
-    }
-
-    // set timer
-    /* struct */ itimerval tv;
-    // first clear all fields
-#if defined(__APPLE__)
-    memset (&tv, 0, sizeof (tv));
-#else
-    timerclear (&tv.it_value);
-#endif
-    // then set the required ones
-
-#if 1
-    tv.it_value.tv_sec = 0;
-    tv.it_value.tv_usec = 1000000 /
rtos::clock_systick::frequency_hz;
-    tv.it_interval.tv_sec = 0;
-    tv.it_interval.tv_usec = 1000000 /
rtos::clock_systick::frequency_hz;
-#else
-    tv.it_value.tv_sec = 1;
-    tv.it_value.tv_usec = 0; // 1000000 /
-    //
rtos::clock_systick::frequency_hz;
-    tv.it_interval.tv_sec = 1;
-    tv.it_interval.tv_usec
= 0; // 1000000 / rtos::clock_systick::frequency_hz;
-#endif
-
-    if (setitimer (ITIMER_REAL, &tv, nullptr) != 0)
-    {
-        trace::printf ("port::clock_systick::%s() setitimer()
failed\n",
-            __func__);
-        abort ();
-    }
-#if 0
-    // Used for initial debugging, to see the signals arriving

```

```

-         pause ();
-         for (int i = 50; i > 0; --i)
-         {
-             for (int j = 100; j > 0; --j)
-             {
-                 char c = '.';
-                 write (1, &c, 1);
-             }
-             char cn = '\n';
-             write (1, &cn, 1);
-         }
-#endif
+ // Each CPU arms its own timer when it starts; this is CPU
+ // 0's, and
+ // the secondaries' are armed by
+ host_cpu::start_secondary_cpus().
+ // Only CPU 0 advances the kernel clock -- exactly as on the
+ BCM2837,
+ // where all four cores take a 1 ms PPI but only core 0 calls
+ // os_systick_handler(). The others use theirs to preempt
+ themselves.
+ }

+ //
+ =====
@@ -498,15 +556,13 @@ namespace os
static uint64_t previous_timestamp;

static uint64_t
- get_current_micros (void);
-
- uint64_t
get_current_micros (void)
{
-     /* struct */ timeval tp;
-     gettimeofday (&tp, nullptr);
+     /* struct */ timespec tp;
+     ::clock_gettime (CLOCK_MONOTONIC, &tp);

-     return static_cast<uint64_t> (tp.tv_sec * 1000000 +
+     return static_cast<uint64_t> (tp.tv_sec) * 1000000ULL
+         + static_cast<uint64_t> (tp.tv_nsec) / 1000ULL;
+ }

void
@@ -518,8 +574,8 @@ namespace os
uint32_t
clock_highres::input_clock_frequency_hz (void)
{
-     // The posix system clock resolution is 1 us, so it makes no
-     // sense to assume a frequency higher than 1 MHz.
+     // CLOCK_MONOTONIC is read here at microsecond resolution, so
+     // a
+     // higher frequency would be a claim the source cannot
+     // support.
    return 1000000;
}

@@ -538,9 +594,8 @@ namespace os
clock_highres::cycles_since_tick (void)
{
    uint64_t ts = get_current_micros ();
-     uint32_t delta = static_cast<uint32_t> (ts -
-         previous_timestamp);
-
-     return delta;
+     return static_cast<uint32_t> (ts - previous_timestamp);
}

```

```

    } /* namespace port */
@@ -549,4 +604,12 @@ namespace os

// -----

+// Never inlined, and never merged or hoisted: see _this_cpu in os-
+    inlines.h.
+extern "C" __attribute__((noinline)) unsigned
+port_cpu_id(void)
+{
+    __asm__ volatile("" ::: "memory");
+    return os::rtos::port::_this_cpu;
+}
+
+    #endif /* defined(__APPLE__) || defined(__linux__) */
diff --git a/include/cmsis-plus/rtos/port/os-decls.h b/include/cmsis-
plus/rtos/port/os-decls.h
index 9cc63ca..825f285 100644
--- a/include/cmsis-plus/rtos/port/os-decls.h
+++ b/include/cmsis-plus/rtos/port/os-decls.h
@@ -83,6 +83,8 @@

    #if defined(__clang__)
    #pragma clang diagnostic ignored "-Wc++98-compat"
+    #pragma clang diagnostic ignored "-Wc++98-compat-pedantic"
+    #pragma clang diagnostic ignored "-Wreserved-identifier"
    #endif

    namespace os
diff --git a/include/cmsis-plus/rtos/port/os-inlines.h b/include/
cmsis-plus/rtos/port/os-inlines.h
index 943da20..e9e0e69 100644
--- a/include/cmsis-plus/rtos/port/os-inlines.h
+++ b/include/cmsis-plus/rtos/port/os-inlines.h
@@ -44,6 +44,8 @@

    #if defined(__clang__)
    #pragma clang diagnostic ignored "-Wc++98-compat"
+    #pragma clang diagnostic ignored "-Wc++98-compat-pedantic"
+    #pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
    #endif

    // Out of line on purpose; see _this_cpu below.
diff --git a/src/exception_handler.cpp b/src/exception_handler.cpp
index 2a09db9..e607d03 100644
--- a/src/exception_handler.cpp
+++ b/src/exception_handler.cpp
@@ -10,6 +10,21 @@

    #if defined(__APPLE__) || defined(__linux__)

+// Clang -Weverything hardening for the SMP host port (the
+    destination harness is
+// stricter than the smp branch's own build): nullptr/alias/range-
+    for, per-CPU
+// array indexing, signal macros, fatal hooks and the ucontext
+    makecontext cast.
+// Placed before the includes so the port headers are covered too.
+    #if defined(__clang__)
+    #pragma clang diagnostic ignored "-Wc++98-compat"
+    #pragma clang diagnostic ignored "-Wc++98-compat-pedantic"
+    #pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
+    #pragma clang diagnostic ignored "-Wdisabled-macro-expansion"
+    #pragma clang diagnostic ignored "-Wmissing-noreturn"
+    #pragma clang diagnostic ignored "-Wunreachable-code"
+    #pragma clang diagnostic ignored "-Wunreachable-code-return"
+    #pragma clang diagnostic ignored "-Wcast-function-type-strict"

```

```
+#endif
+
#include <csignal>
#include <cstring>
#include <initializer_list>
@@ -24,6 +39,7 @@
#include <exception_handler.hpp>
#include <host_cpu.hpp>

+
namespace
{
    // Everything below runs in a signal handler after the process is
    // already
diff --git a/src/free-store.cpp b/src/free-store.cpp
index acc7e95..eb743b7 100644
--- a/src/free-store.cpp
+++ b/src/free-store.cpp
@@ -42,6 +42,21 @@

    #if defined(__APPLE__) || defined(__linux__)

// Clang -Weverything hardening for the SMP host port (the
// destination harness is
// stricter than the smp branch's own build): nullptr/alias/range-
// for, per-CPU
// array indexing, signal macros, fatal hooks and the ucontext
// makecontext cast.
// Placed before the includes so the port headers are covered too.
#if defined(__clang__)
#pragma clang diagnostic ignored "-Wc++98-compat"
#pragma clang diagnostic ignored "-Wc++98-compat-pedantic"
#pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
#pragma clang diagnostic ignored "-Wdisabled-macro-expansion"
#pragma clang diagnostic ignored "-Wmissing-noreturn"
#pragma clang diagnostic ignored "-Wunreachable-code"
#pragma clang diagnostic ignored "-Wunreachable-code-return"
#pragma clang diagnostic ignored "-Wcast-function-type-strict"
#endif
+
#include <cstddef>
#include <new>

@@ -50,6 +65,7 @@
#include <cmsis-plus/memory/first-fit-top.h>
#include <cmsis-plus/estd/memory_resource>

+
using namespace os;

    #if defined(OS_TYPE_APPLICATION_MEMORY_RESOURCE)
diff --git a/src/host_cpu.cpp b/src/host_cpu.cpp
index 87bd73f..9eeb32b 100644
--- a/src/host_cpu.cpp
+++ b/src/host_cpu.cpp
@@ -10,6 +10,21 @@

    #if defined(__APPLE__) || defined(__linux__)

// Clang -Weverything hardening for the SMP host port (the
// destination harness is
// stricter than the smp branch's own build): nullptr/alias/range-
// for, per-CPU
// array indexing, signal macros, fatal hooks and the ucontext
// makecontext cast.
// Placed before the includes so the port headers are covered too.
#if defined(__clang__)
#pragma clang diagnostic ignored "-Wc++98-compat"

```

```

#pragma clang diagnostic ignored "-Wc++98-compat-pedantic"
#pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
#pragma clang diagnostic ignored "-Wdisabled-macro-expansion"
#pragma clang diagnostic ignored "-Wmissing-noreturn"
#pragma clang diagnostic ignored "-Wunreachable-code"
#pragma clang diagnostic ignored "-Wunreachable-code-return"
#pragma clang diagnostic ignored "-Wcast-function-type-strict"
#endif
+
#include <cerrno>
#include <cstdio>
#include <cstdlib>
@@ -26,6 +41,7 @@

#include <host_cpu.hpp>

+
/* Defined by the board (test/boards/<board>/src/smp.cpp), with the
   same
   * meaning as on the ARM boards: 0 = not started, 3 = scheduler
   entered. The
   * tests wait on it before they start timing anything. A board that
   does not
diff --git a/src/rtos/os-core.cpp b/src/rtos/os-core.cpp
index c073489..f59b35a 100644
--- a/src/rtos/os-core.cpp
+++ b/src/rtos/os-core.cpp
@@ -30,6 +30,21 @@

#if defined(__APPLE__) || defined(__linux__)

// Clang -Weverything hardening for the SMP host port (the
  destination harness is
// stricter than the smp branch's own build): nullptr/alias/range-
  for, per-CPU
// array indexing, signal macros, fatal hooks and the ucontext
  makecontext cast.
// Placed before the includes so the port headers are covered too.
#if defined(__clang__)
#pragma clang diagnostic ignored "-Wc++98-compat"
#pragma clang diagnostic ignored "-Wc++98-compat-pedantic"
#pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
#pragma clang diagnostic ignored "-Wdisabled-macro-expansion"
#pragma clang diagnostic ignored "-Wmissing-noreturn"
#pragma clang diagnostic ignored "-Wunreachable-code"
#pragma clang diagnostic ignored "-Wunreachable-code-return"
#pragma clang diagnostic ignored "-Wcast-function-type-strict"
#endif
+
#include <cassert>
#include <cstdlib>
#include <cstring>

```

## PR #23 (Cortex-M): Multi-Core SMP Port Contract

### Why Part B Cortex-M Architecture Infrastructure is Needed:

- **Hardware Abstraction for Multi-Core Cortex-M:** In `include/cmsis-plus/rtos/port/os-inlines.h`, defines per-core `lock_state[OS_NCPU]`, recursive `_smp_klock_enter()` / `_smp_klock_exit()` with PRIMASK interrupt masking.
- **Atomic 5-Stage PendSV Context Switch:** In `src/rtos/os-core.cpp`, implements PendSV context switch with the 5-stage stack pointer claim/publish handshake (`old_thread->context_.port_.stack_ptr = sp; internal_switch_threads(); new_thread->context_.port_.stack_ptr = nullptr;`). This guarantees

that a thread's context cannot be claimed by another core while its registers are being saved.

- **Multi-Core Greeting & Verification:** Logs active core count (trace::printf(", SMP %u CPU", OS\_NCPU)) during scheduler initialization.

```
diff --git a/include/cmsis-plus/rtos/port/os-inlines.h b/include/
cmsis-plus/rtos/port/os-inlines.h
index bcf1c44..33a1a69 100644
--- a/include/cmsis-plus/rtos/port/os-inlines.h
+++ b/include/cmsis-plus/rtos/port/os-inlines.h
@@ -95,12 +95,14 @@ namespace os
    return locked (state::unlocked);
}

+
+   inline bool __attribute__ ((always_inline))
+   locked (void)
+   {
+       return lock_state != state::unlocked;
+   }
+
+
+   inline void __attribute__ ((always_inline))
+   wait_for_interrupt (void)
+   {
@@ -230,6 +232,7 @@ namespace os

#endif

+
+       // Apparently not required by architecture, but used by
+       // FreeRTOS, with an unconvincing motivation ("... ensure
+       // the code is completely within the specified behaviour
@@ -244,6 +247,7 @@ namespace os
    inline void __attribute__ ((always_inline))
    critical_section::exit (rtos::interrupts::state_t state)
    {
+
+   #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)

    #if defined(OS_INTEGER_RTOS_CRITICAL_SECTION_INTERRUPT_PRIORITY)
@@ -362,6 +366,18 @@ namespace os
    ;
    }

+
+   inline constexpr bool __attribute__ ((always_inline))
+   clock_highres::has_hardware_counter (void) noexcept
+   {
+       return false;
+   }
+
+
+   inline uint64_t __attribute__ ((always_inline))
+   clock_highres::hardware_counter (void) noexcept
+   {
+       return 0;
+   }
+
+
+   inline uint32_t __attribute__ ((always_inline))
+   clock_highres::input_clock_frequency_hz (void)
+   {
@@ -391,7 +407,7 @@ namespace os
    // yet processed, so the total cycles count in steady_count_
    // does not yet reflect the correct value and needs to be
    // adjusted by one full cycle length.
-   if (SysTick->CTRL & SCB_ICSR_PENDSTSET_Msk)
+   if ((SCB->ICSR & SCB_ICSR_PENDSTSET_Msk) != 0)
```

```

        {
            // Sample the decrementing counter again to validate the
            // initial sample.
diff --git a/include/cmsis-plus/rtos/port/os-c-decls.h b/include/
        cmsis-plus/rtos/port/os-c-decls.h
index 2eacaf3..348df94 100644
--- a/include/cmsis-plus/rtos/port/os-c-decls.h
+++ b/include/cmsis-plus/rtos/port/os-c-decls.h
@@ -48,6 +48,13 @@ typedef struct
    os_port_thread_stack_element_t* stack_ptr;
} os_port_thread_context_t;

+#if defined(OS_USE_SMP_SCHEDULER)
+
+/* The kernel lock's owner when nobody holds it (os-decls.h,
+   smp_klock_t). */
+#define SMP_NO_OWNER 0xFFFFFFFFu
+
+#endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+#define OS_INTEGER_RTOS_STACK_FILL_MAGIC (0xEFBEADDE)

+#define OS_HAS_INTERRUPTS_STACK
diff --git a/include/cmsis-plus/rtos/port/os-decls.h b/include/cmsis-
plus/rtos/port/os-decls.h
index 8036342..1747acc 100644
--- a/include/cmsis-plus/rtos/port/os-decls.h
+++ b/include/cmsis-plus/rtos/port/os-decls.h
@@ -116,8 +116,34 @@ namespace os
    constexpr state_t init = unlocked;
} /* namespace state */

+#if defined(OS_USE_SMP_SCHEDULER)
+
+    // The kernel's SMP scheduler. Both cores declare it here:
+    // upstream's
+    // at OS_NCPU=1 (os-core.cpp), and the RP2350's at OS_NCPU=1
+    // or 2
+    // (os-core-rp2350.cpp, whose include-rp2350/ has no os-
+    // decls.h of its
+    // own). One lock state per CPU, as the SMP kernel indexes
+    // it.
+    extern state_t lock_state[OS_NCPU];
+    extern uint32_t lock_primask[OS_NCPU];
+
+    // The kernel lock, recursive. owner/depth record who holds
+    // it and how
+    // deeply, the bookkeeping the SMP ports keep. What excludes
+    // the other
+    // core is the core's business: with one CPU, masking
+    // interrupts IS the
+    // mutual exclusion, so there is no lock word and no LDREX/
+    // STREX; on
+    // the RP2350 it is SIO hardware spinlock 0 (include-
+    // rp2350/.../os-inlines.h).
+    struct smp_klock_t
+    {
+        volatile uint32_t owner;
+        volatile uint32_t depth;
+    };
+
+    extern smp_klock_t _smp_klock;
+
+#else
+
+    extern state_t lock_state;
+
+#endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+} /* namespace scheduler */

```

```

        using thread_context_t = struct context_s
diff --git a/include/cmsis-plus/rtos/port/os-inlines.h b/include/
        cmsis-plus/rtos/port/os-inlines.h
index 33a1a69..813d114 100644
--- a/include/cmsis-plus/rtos/port/os-inlines.h
+++ b/include/cmsis-plus/rtos/port/os-inlines.h
@@ -64,6 +64,9 @@ namespace os
        trace::printf ("0/0+");
    #endif

        trace::printf ("", preemptive");
+    #if defined(OS_USE_SMP_SCHEDULER)
+    +        trace::printf ("", SMP %u CPU", static_cast<unsigned>
+        (OS_NCPU));
+    #endif
    #if (defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)) \
        && defined(OS_INTEGER_RTOS_CRITICAL_SECTION_INTERRUPT_PRIORITY)
        trace::printf ("", BASEPRI(%u)",
@@ -95,6 +98,47 @@ namespace os
        return locked (state::unlocked);
    }

+    #if defined(OS_USE_SMP_SCHEDULER)
+    +
+    +        // One CPU: it is always CPU 0 (os-core.cpp asserts OS_NCPU
+    == 1).
+    +        inline unsigned __attribute__ ((always_inline))
+    port_cpu_id (void)
+    +        {
+    +            return 0;
+    +        }
+
+    +        inline bool __attribute__ ((always_inline))
+    locked (void)
+    +        {
+    +            return lock_state[port_cpu_id ()] != state::unlocked;
+    +        }
+
+    +        // Enter/leave the kernel lock. The caller has already masked
+    +        // interrupts (critical_section, switch_stacks); with one CPU
+    that is
+    +        // the exclusion, and this only records the owner and the
+    nesting
+    +        // depth. Nothing spins: there is no other CPU to wait for.
+    +        inline void __attribute__ ((always_inline))
+    _smp_klock_enter (void)
+    +        {
+    +            _smp_klock.owner = port_cpu_id ();
+    +            _smp_klock.depth = _smp_klock.depth + 1;
+    +        }
+
+    +        inline void __attribute__ ((always_inline))
+    _smp_klock_exit (void)
+    +        {
+    +            if (_smp_klock.depth > 0)
+    +            {
+    +                uint32_t d = _smp_klock.depth - 1;
+    +                _smp_klock.depth = d;
+    +                if (d == 0)
+    +                {
+    +                    _smp_klock.owner = SMP_NO_OWNER;
+    +                }
+    +            }
+    +        }
+    +    }
+
+    #else

        inline bool __attribute__ ((always_inline))

```

```

        locked (void)
@@ -102,6 +146,7 @@ namespace os
        return lock_state != state::unlocked;
    }

+endif /* defined(OS_USE_SMP_SCHEDULER) */

        inline void __attribute__ ((always_inline))
        wait_for_interrupt (void)
@@ -232,6 +277,10 @@ namespace os

    #endif

+if defined(OS_USE_SMP_SCHEDULER)
+    // Interrupts are masked: take the kernel lock.
+    scheduler::_smp_klock_enter ();
+endif

        // Apparently not required by architecture, but used by
        // FreeRTOS, with an unconvincing motivation ("... ensure
@@ -247,6 +296,10 @@ namespace os
        inline void __attribute__ ((always_inline))
        critical_section::exit (rtos::interrupts::state_t state)
        {
+if defined(OS_USE_SMP_SCHEDULER)
+    // Release the kernel lock while interrupts are still
+    masked.
+    scheduler::_smp_klock_exit ();
+endif

    #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)

diff --git a/src/rtos/os-core.cpp b/src/rtos/os-core.cpp
index d88d5ec..ad9b65a 100644
--- a/src/rtos/os-core.cpp
+++ b/src/rtos/os-core.cpp
@@ -62,12 +62,40 @@ static_assert
    (OS_INTEGER_RTOS_CRITICAL_SECTION_INTERRUPT_PRIORITY

    #endif /* defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__) */

+// The kernel's SMP scheduler on ONE CPU. This core has no second
+// core to
+// start, no cross-core lock word and no IPI: the SMP branch below
+// exists so
+// the kernel's SMP code paths (per-CPU current thread and lock
+// state, the
+// affinity-aware picker, the stack_ptr liveness rule) run on a
+// single Cortex-M.
+// More CPUs need a board that supplies a lock and an IPI (see
+// CMakeLists.txt).
+if defined(OS_USE_SMP_SCHEDULER)
+static_assert (OS_NCPU == 1,
+    "the upstream Cortex-M core runs the SMP scheduler on
+    1 CPU "
+    "only");
+endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+// -----

    #if defined(__ARM_ARCH_6M__)
    extern uint32_t __vectors_start;
    #endif

+if defined(OS_USE_SMP_SCHEDULER)
+// The kernel's idle thread, registered as CPU 0's idle in start().
+extern os::rtos::thread* os_idle_thread;
+
+namespace os

```

```

+{
+ namespace rtos
+ {
+   namespace scheduler
+   {
+     // The kernel's per-CPU idle threads (kernel os-core.cpp).
+     extern thread* os_idle_thread_core[OS_NCPU];
+   } /* namespace scheduler */
+ } /* namespace rtos */
+} /* namespace os */
+#endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+ // -----

extern "C"
@@ -358,7 +386,12 @@ namespace os

    namespace scheduler
    {
+#if defined(OS_USE_SMP_SCHEDULER)
+      state_t lock_state[OS_NCPU];
+      smp_klock_t _smp_klock = { SMP_NO_OWNER, 0 };
+#else
+      state_t lock_state;
+#endif /* defined(OS_USE_SMP_SCHEDULER) */

      result_t
      initialize (void)
@@ -494,7 +527,15 @@ namespace os
      rtos::thread* pth = reinterpret_cast<rtos::thread*>
        (&fake_thread);

      // Make the fake thread look like the current thread.
+#if defined(OS_USE_SMP_SCHEDULER)
+      rtos::scheduler::current_thread_[0] = pth;
+
+      // The picker falls back to the CPU's idle thread; CPU 0's
+      is the
+      // kernel's.
+      rtos::scheduler::os_idle_thread_core[0] = ::os_idle_thread;
+#else
+      rtos::scheduler::current_thread_ = pth;
+#endif /* defined(OS_USE_SMP_SCHEDULER) */

      // Trigger the PendSV; the exception will happen a bit
      later,
      // after re-enabling the interrupts.
@@ -507,7 +548,11 @@ namespace os

    #endif

+#if defined(OS_USE_SMP_SCHEDULER)
+    lock_state[0] = state::init;
+#else
+    lock_state = state::init;
+#endif /* defined(OS_USE_SMP_SCHEDULER) */

    // Enable all interrupts; allow PendSV to occur.
    __enable_irq ();
@@ -531,8 +576,17 @@ namespace os
    {
      rtos::interrupts::critical_section ics;

+#if defined(OS_USE_SMP_SCHEDULER)
+      // With one CPU the scheduler lock stays what it is
+      upstream: a
+      // flag the picker reads. It holds no interrupt mask; no
+      other

```

```

+         // core can run the picker meanwhile.
+         unsigned cpu = port_cpu_id ();
+         tmp = lock_state[cpu];
+         lock_state[cpu] = state;
+     #else
+         tmp = lock_state;
+         lock_state = state;
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+ }

+     return tmp;
@@ -830,6 +884,44 @@ namespace os
+     SCB->ICSR = SCB_ICSR_PENDSVCLR_Msk;
+     #pragma GCC diagnostic pop

+ #if defined(OS_USE_SMP_SCHEDULER)
+ +
+     // Interrupts are masked: take the kernel lock for the
+     picker.
+     _smp_klock_enter ();
+
+     unsigned cpu = port_cpu_id ();
+     rtos::thread* old_thread =
+     rtos::scheduler::current_thread_[cpu];
+
+     // Save the current SP in the initial context. This also
+     publishes
+     // the outgoing thread as switchable again: the SMP picker
+     skips a
+     // thread whose stack_ptr is null (its context is live on a
+     CPU).
+     old_thread->context_.port_.stack_ptr = sp;
+
+     rtos::scheduler::internal_switch_threads ();
+
+     rtos::thread* new_thread =
+     rtos::scheduler::current_thread_[cpu];
+
+     if (new_thread == nullptr)
+     {
+         // No ready thread and no idle thread -- fatal. Release
+         the
+         // kernel lock first so the picker cannot be entered by
+         anything
+         // else, then halt with interrupts masked (do not
+         restore the
+         // saved mask: the scheduler is already torn).
+         _smp_klock_exit ();
+         __asm__ volatile ("cpsid if" ::: "memory");
+         for (;;)
+             __asm__ volatile ("wfi");
+     }
+
+     // Prepare a local copy of the new thread SP, and mark its
+     context
+     // live on this CPU (the same thread, when it was re-
+     picked).
+     stack::element_t* out_sp = new_thread-
+     >context_.port_.stack_ptr;
+     new_thread->context_.port_.stack_ptr = nullptr;
+
+     _smp_klock_exit ();
+ #else
+ +
+     rtos::thread* old_thread =
+     rtos::scheduler::current_thread_;
+
+     // Save the current SP in the initial context.
@@ -851,6 +943,8 @@ namespace os
+     stack::element_t* out_sp

```

```

        = rtos::scheduler::current_thread_
        >context_.port_.stack_ptr;

+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+ #if defined(__ARM_ARCH_7M__) || defined(__ARM_ARCH_7EM__)
+
+ #if defined(OS_INTEGER_RTOS_CRITICAL_SECTION_INTERRUPT_PRIORITY)
+ @@ -886,6 +980,17 @@ namespace os
+
+ // -----
+
+ #if defined(OS_USE_SMP_SCHEDULER)
+ // The kernel's CPU index (declared extern "C" by the kernel).
+ extern "C" unsigned
+ port_cpu_id (void)
+ {
+   return os::rtos::port::scheduler::port_cpu_id ();
+ }
+ #endif /* defined(OS_USE_SMP_SCHEDULER) */
+
+ // -----
+
+ using namespace os::rtos;
+
+ /**

```

## Appendix Section 3 — Pull Request #3: Port Package Semantic Releases (Step 24 / Part C1)

### Why PR #3 is Needed as a Standalone Pull Request

1. **Formalizing and Publishing the SMP Port ABI Contract:**
  - PR #2 established and verified the SMP hardware abstraction contract across Cortex-M and POSIX architectures. PR #3 bumps posix-arch to v1.1.0 and cortexm to v1.2.0 in package.json, creating the official semantic release points required by downstream silicon and tests in PR #5 and PR #6.
2. **Strict Release Safety Automation:**
  - Standard npm version invokes the upstream postversion: "git push origin --all && git push origin --tags" hook, which attempts unauthorized remote pushes. PR #3 utilizes scripts/smp/release-port.sh to update package.json directly via Python and generate local git tags only.
3. **Decoupled Review:**
  - Allows package versioning to be reviewed and verified independently of functional code changes.

## PR #24: Port Package Semantic Version Releases

### Why Step 24 (Port Releases) is Needed:

- **Formalizing the SMP Port ABI Contract:** Releases posix-arch v1.1.0 and cortexm v1.2.0 in package.json, publishing the multi-core HAL interfaces required by downstream silicon and tests.
- **Release Safety Discipline:** Modifies package.json directly via Python script (release-port.sh), preventing npm version from executing its postversion hook and triggering unauthorized remote git pushes.

```

# No source diff exists in the `smp` branch: it keeps the baseline
# versions
# (posix-arch 1.0.1, cortexm 1.1.0). The release is cut by the
# integration tool:

```

```
scripts/smp/release-port.sh "$P" 1.1.0 "SMP host CPU model + highres
clock"
scripts/smp/release-port.sh "$C" 1.2.0
"SMP Cortex-M port lock + IPI + highres ICSR"
# -> micro-os-plus-iii-posix-arch/package.json: "version": "1.0.1" ->
"1.1.0"
# -> micro-os-plus-iii-cortexm/package.json: "version": "1.1.0" ->
"1.2.0"
```

## Appendix Section 4 — Pull Request #4: History-Preserving Dissolution of devices Repository (Part 0)

### Why PR #4 is Needed as a Standalone Pull Request

#### 1. Preserving Upstream Six-Repository Architecture:

- Upstream xpack-development has no micro-os-plus-iii-devices repository. PR #4 dissolves this repository by folding device drivers, FatFs, USB, and SoC silicon files directly into the architecture repositories that own them (posix-arch, cortexm, aarch32, aarch64).

#### 2. 100% Git Commit History Preservation:

- Using git subtree split and git subtree add, all commit histories, author attributions, and timestamps are preserved in the destination architecture repositories.

#### 3. Backwards Compatibility for the Legacy Target:

- Existing board configurations keep linking micro-os-plus::devices. After the fold it resolves to the architecture repository's own drivers; until then the AArch ports resolve it by conditionally consuming a sibling micro-os-plus-iii-devices working copy (see the CMake excerpt in GITHUB-PROGRESSIVE-PR-GUIDE.md §PR #25).

## PR #25: History-Preserving Devices Repository Subtree Dissolution (diffstat only)

### Why Part 0 (devices Dissolution) is Needed:

- **Preserving Upstream Six-Repository Architecture:** Folds device drivers, FatFs, USB, and SoC silicon files from micro-os-plus-iii-devices directly into the architecture repositories that own them (posix-arch, cortexm, aarch32, aarch64).
- **100% History Preservation:** Executed via git subtree split and git subtree add, preserving all author commits, dates, and commit messages.
- **Zero Breaking Changes:** Existing boards keep linking the legacy micro-os-plus::devices target; it resolves to the folded content, or to a sibling devices working copy consumed conditionally by the AArch ports.

```
# posix-arch
4514a37 migrate(devices): soc/native -> soc/native (history preserved)
2 files changed, 282 insertions(+)
f0bae5d migrate(devices): fatfs -> drivers/fatfs (history preserved)
7 files changed, 8394 insertions(+)
ac8d5d6 migrate(devices): src -> drivers/src (history preserved)
3 files changed, 2988 insertions(+)
52a32a5 migrate(devices): include -> drivers/include (history
preserved)
4 files changed, 704 insertions(+)
# cortexm
b484b99 migrate(devices): soc/stm32f4xx -> soc/stm32f4xx (history
preserved)
36 files changed, 63693 insertions(+)
```

```
3195eed migrate(devices): soc/rp2350 -> soc/rp2350 (history preserved)
12 files changed, 9556 insertions(+)
```

## Appendix Section 5 — Pull Request #5: New Silicon Targets & Modular Add-Only CMake Architecture (Steps 25-26 / Part C2)

### Why PR #5 is Needed as a Standalone Pull Request

- 1. Production Support for Modern Multi-Core Silicon:**
  - **ARM Cortex-M33 (ARMv8-M Mainline):** Dual-core SSE-200 / MPS2 AN521 with Message Handling Unit (MHU) IPI signaling and per-core SysTick.
  - **Raspberry Pi RP2350:** Dual Cortex-M33 microcontroller with recursive `_smp_klock` backed by SIO Hardware Spinlock 0 (`rp2350::reg (rp2350::SIO_SPINLOCK0)`) and inter-core FIFO IRQ 25.
- 2. Modular Add-Only Build System:**
  - Introduces standardized toolchains (`cmake/toolchains/arm-none-eabi.cmake`, `aarch64-none-elf.cmake`) and the `cmake/uos-app.cmake` helper macro.
- 3. Preserving Fat Kernel Architecture:**
  - Appends `micro-os-plus::cortexm-qemu-m33` to Cortex-M `CMakeLists.txt` and links the fat `micro-os-plus::iii` library. Avoids disruptive decomposition into granular sub-libraries and guarantees zero breakage for existing single-core Cortex-M platforms.
- 4. Verification Gate:**
  - Dual-core Cortex-M33 QEMU platform (`2xcortex-m33`) passes all 4 test suites (`rtos-apis`, `mutex-stress`, `cmsis-os-validator 60/60`, and `fp-switch`).

## PR #26: Modern Multi-Core Silicon Ports (ARM Cortex-M33 & RP2350)

### Why Step 25 (New Silicon Cores) is Needed:

- **ARM Cortex-M33 (ARMv8-M Mainline) Multi-Core Port:** Implements `include-m33/` and `src/rtos/os-core-m33.cpp` for dual-core SSE-200 / MPS2 AN521 with hardware Message Handling Unit (MHU) IPI signaling and per-core SysTick.
- **Raspberry Pi RP2350 (Dual Cortex-M33) Silicon Port:** Implements `include-rp2350/` and `src/rtos/os-core-rp2350.cpp`, backing recursive `_smp_klock` with SIO Hardware Spinlock 0 (`rp2350::reg (rp2350::SIO_SPINLOCK0)`) and cross-core wakeups with inter-core FIFO IRQ 25.
- **Kernel IPI Wake-Up Dispatch:** In `src/rtos/os-thread.cpp`, adds IPI dispatching in `thread::resume()` when `OS_INTEGER_RTOS_PORT_NCPU > 1`.

```
// cortexm/include-rp2350/cmsis-plus/rtos/port/os-inlines.h
// ---- SMP kernel lock (SIO hardware spinlock 0)
// -----

// Claim the hardware spinlock (reading it returns 0 while another
// core holds it; any write releases it).
inline void __attribute__((always_inline))
_smp_klock_raw_acquire(void)
{
    while (rp2350::reg (rp2350::SIO_SPINLOCK0) == 0)
    {
    }
    __asm__ volatile ("dmb" ::: "memory");
}
```

```

inline void __attribute__ ((always_inline))
_smp_klock_raw_release (void)
{
    __asm__ volatile ("dmb" ::: "memory");
    rp2350::reg (rp2350::SIO_SPINLOCK0) = 1;
    __asm__ volatile ("dsb" ::: "memory");
}

// Mask IRQs on this core and enter the kernel lock (recursive, with
// owner tracking so nested critical sections only bump the depth).
inline void __attribute__ ((always_inline))
port_set_lock (void)
{
    __asm__ volatile ("cpsid i" ::: "memory");
    unsigned cpu = port_cpu_id ();
    if (_smp_klock.owner != cpu)
    {
        _smp_klock_raw_acquire ();
        _smp_klock.owner = cpu;
    }
    _smp_klock.depth = _smp_klock.depth + 1;
}

```

## PR #27: Modular Add-Only CMake Build Architecture & Cross Toolchains

### Why Step 26 (Modular CMake & 2xcortex-m33 Target) is Needed:

- **Modular Add-Only Build System:** Adds cmake/uos-app.cmake and standardized cross-compilation toolchains (arm-none-eabi.cmake, aarch64-none-elf.cmake).
- **Additive Port Target Registration:** Appends micro-os-plus::cortexm-qemu-m33 to Cortex-M CMakeLists.txt and links the fat micro-os-plus::iii library, completely preserving baseline uniprocessor stability.
- **Dual-Core QEMU M33 Test Platform:** Adds tests/platforms/2xcortex-m33/ (reusing the generic tests/device-qemu-cortexm device) running on QEMU mps2-an521 --cpu cortex-m33 --smp 2.

```

# cortexm/CMakeLists.txt (additive target registration)
add_library (micro-os-plus-cortexm-qemu-m33-interface INTERFACE
    EXCLUDE_FROM_ALL)

target_include_directories (micro-os-plus-cortexm-qemu-m33-interface
    INTERFACE
    "include-m33"
    "include"
    "include/cmsis-plus/legacy"
)

target_compile_options (micro-os-plus-cortexm-qemu-m33-interface
    INTERFACE
    -mcpu=cortex-m33
    -mthumb
    -mfloat-abi=hard
    -mfpu=fpv5-sp-d16
)

target_sources (micro-os-plus-cortexm-qemu-m33-interface INTERFACE
    src/rtos/os-core-m33.cpp
    src/diag/trace-cortexm.cpp
)

target_link_libraries (micro-os-plus-cortexm-qemu-m33-interface
    INTERFACE

```

```

micro-os-plus::iii
)

add_library (micro-os-plus::cortexm-qemu-m33 ALIAS micro-os-plus-
cortexm-qemu-m33-interface)

```

## Appendix Section 6 — Pull Request #6: New Test Suites & Multi-Architecture Add-Only Platforms (Steps 27-28 / Part C3)

### Why PR #6 is Needed as a Standalone Pull Request

1. **Hardware FPU Context Switch Stress Testing:**
  - Adds tests/sources/fp-switch/ to stress-test hardware floating-point register preservation across multi-core context switches under aggressive preemption (6 threads, 3000 ticks).
2. **Test Scaffolding Relocation:**
  - Absorbs the non-standard root test\_smpl/ directory into tests/smp-support/ (hw\_result.hpp, board-contract.cpp, run-qemu.sh, run-host.sh, run-hw.sh, OpenOCD configs).
3. **Multi-Architecture 4-Core QEMU Platforms:**
  - Integrates 4-core QEMU raspi3b platforms for AArch32 (ARMv7-A) and AArch64 (ARMv8-A), exercising secondary core boot via spin-tables and cross-core synchronization.
4. **Composite CI Action (test-smp-cmake):**
  - Appends test-smp-cmake to tests/package.json, executing test-native-smp, test-2xcortex-m33-cmake, and AArch platforms in a single automated step.
5. **Verification Gate:**
  - 72/72 baseline tests pass + test-smp-cmake 100% green across all architectures.

### PR #28: Multi-Core FPU Stress Testing & Scaffolding Relocation

#### Why Step 27 (New Test Suites & test\_smpl Absorption) is Needed:

- **Hardware FPU Context Switch Stress Testing:** Adds tests/sources/fp-switch/ (6 threads, 3000 ticks) verifying floating-point register preservation across multi-core context switches under aggressive preemption.
- **Scaffolding Relocation:** Absorbs non-standard root test\_smpl/ directory into tests/smp-support/ (hw\_result.hpp, board-contract.cpp, run-qemu.sh, run-host.sh, run-hw.sh, OpenOCD configs).

```

diff --git a/tests/cmake/tests-main.cmake b/tests/cmake/tests-
main.cmake
index eecebed0..586148af 100644
--- a/tests/cmake/tests-main.cmake
+++ b/tests/cmake/tests-main.cmake
@@ -53,9 +53,73 @@ xpack_add_dependencies_subdirectories (

# -----

-# Add the project library, defined one level above.
-message (VERBOSE "Adding top library...")
-add_subdirectory (".." "top-bin")
+# The library under test is the architecture port, which itself adds
+the SMP
+# kernel (micro-os-plus::iii) and the devices package. Do NOT add the
+kernel
+# separately: both the kernel and the port define micro-os-plus::iii,
+and

```

```

+## adding it twice fails on the duplicate alias. The port is chosen
    from the
+## platform name.
+message (VERBOSE "Selecting the library under test for $
    {PLATFORM_NAME}...")
+get_filename_component (_uos_siblings "${CMAKE_SOURCE_DIR}/../.."
    ABSOLUTE)
+
+set (UOS_SMP_DIR "${CMAKE_SOURCE_DIR}/.."
+     CACHE PATH "µOS++ III SMP kernel working copy" FORCE)
+set (UOS_AARCH32_DIR "${_uos_siblings}/micro-os-plus-iii-aarch32"
+     CACHE PATH "µOS++ III AArch32 port working copy")
+set (UOS_AARCH64_DIR "${_uos_siblings}/micro-os-plus-iii-aarch64"
+     CACHE PATH "µOS++ III AArch64 port working copy")
+set (UOS_POSIX_ARCH_DIR "${_uos_siblings}/micro-os-plus-iii-posix-
    arch"
+     CACHE PATH "µOS++ III POSIX-arch port working copy")
+set (UOS_CORTEXM_DIR "${_uos_siblings}/micro-os-plus-iii-cortexm"
+     CACHE PATH "µOS++ III Cortex-M port working copy")
+
+foreach (_dep_var IN ITEMS UOS_AARCH32_DIR UOS_AARCH64_DIR
    UOS_POSIX_ARCH_DIR UOS_CORTEXM_DIR)
+  if (NOT EXISTS "${${_dep_var}}/CMakeLists.txt"
+      AND EXISTS "${${_dep_var}}.git/CMakeLists.txt")
+    set (${_dep_var} "${${_dep_var}}.git" CACHE PATH
+        "µOS++ III sibling working copy" FORCE)
+  endif ()
+endforeach ()
+
+if (PLATFORM_NAME MATCHES "^aarch32")
+  message (VERBOSE "Adding the AArch32 port (brings iii +
    devices)...")
+  if (NOT EXISTS "${UOS_AARCH32_DIR}/CMakeLists.txt")
+    message (FATAL_ERROR "Cannot find the AArch32 port at $
    {UOS_AARCH32_DIR}")
+  endif ()
+  add_subdirectory ("${UOS_AARCH32_DIR}" "port-bin")
+elseif (PLATFORM_NAME MATCHES "^aarch64")
+  message (VERBOSE "Adding the AArch64 port (brings iii +
    devices)...")
+  if (NOT EXISTS "${UOS_AARCH64_DIR}/CMakeLists.txt")
+    message (FATAL_ERROR "Cannot find the AArch64 port at $
    {UOS_AARCH64_DIR}")
+  endif ()
+  add_subdirectory ("${UOS_AARCH64_DIR}" "port-bin")
+elseif (PLATFORM_NAME MATCHES "^native")
+  # The POSIX-arch port: the same kernel and devices, on the host
    compiler.
+  message (VERBOSE "Adding the POSIX-arch port (brings iii +
    devices)...")
+  if (NOT EXISTS "${UOS_POSIX_ARCH_DIR}/CMakeLists.txt")
+    message (FATAL_ERROR "Cannot find the POSIX-arch port at $
    {UOS_POSIX_ARCH_DIR}")
+  endif ()
+  add_subdirectory ("${UOS_POSIX_ARCH_DIR}" "port-bin")
+elseif (PLATFORM_NAME MATCHES "^cortexm" OR PLATFORM_NAME MATCHES
    "^qemu-cortex")
+  # The Cortex-M port (RP2040/RP2350, STM32F4...) and the generic
    QEMU
+  # Cortex-M machines (qemu-cortex-m0/m3/m4f/m7f): brings iii +
    devices.
+  message (VERBOSE "Adding the Cortex-M port (brings iii +
    devices)...")
+  if (NOT EXISTS "${UOS_CORTEXM_DIR}/CMakeLists.txt")
+    message (FATAL_ERROR "Cannot find the Cortex-M port at $
    {UOS_CORTEXM_DIR}")
+  endif ()
+  add_subdirectory ("${UOS_CORTEXM_DIR}" "port-bin")
+elseif (PLATFORM_NAME MATCHES "^pico2" OR PLATFORM_NAME MATCHES
    "^2xcortex")
+  # The pico2's Cortex-M33 run emulated (pico2-1cpu, 2xcortex-m33).
    Same port,

```

```
+ # but the platform links its GENERIC M33 core; no board sources are
+   used.
+ message (VERBOSE "Adding the Cortex-M port for the emulated
+   pico2...")
+ if (NOT EXISTS "${UOS_CORTEXM_DIR}/CMakeLists.txt")
+   message (FATAL_ERROR "Cannot find the Cortex-M port at $
+     ${UOS_CORTEXM_DIR}")
+ endif ()
+ add_subdirectory ("${UOS_CORTEXM_DIR}" "port-bin")
+else ()
+ # Fallback: the plain kernel, one level above (upstream behaviour).
+ message (VERBOSE "Adding the top library...")
+ add_subdirectory (".." "top-bin")
+endif ()

# -----
# Platform specifics.
```

```
# Relocation of shared test scaffolding via absorb-test-smpl.sh
git mv test_smpl/include/hw_result.hpp      tests/smp-support/
      include/hw_result.hpp
git mv test_smpl/src/board-contract.cpp     tests/smp-support/src/
      board-contract.cpp
git mv test_smpl/run-qemu.sh                tests/smp-support/run-
      qemu.sh
git mv test_smpl/run-host.sh                tests/smp-support/run-
      host.sh
git mv test_smpl/run-hw.sh                  tests/smp-support/run-
      hw.sh
rm -rf test_smpl
```

## PR #29: Multi-Architecture 4-Core Platforms & Composite CI Action

### Why Step 28 (Add-Only Platforms & test-smp-cmake) is Needed:

- **Multi-Architecture 4-Core QEMU Platforms:** Integrates 4-core QEMU raspibb platforms for AArch32 (ARMv7-A) and AArch64 (ARMv8-A) with spin-table secondary core boot.
- **Thin-Core Sub-Targets:** Injects micro-os-plus::iii-posix-io, ::iii-semihosting, ::iii-newlib-reent, and ::test-support into Kernel CMakeLists.txt.
- **Composite CI Action (test-smp-cmake):** Adds a single command to execute all multi-core platforms across all architectures.

```
// tests/package.json - the genuine aggregate action (excerpt)
"test-smp-cmake": [
  "xpm run test-aarch32-rpi-zero-2w-cmake",
  "xpm run test-aarch32-rpi3b-cmake",
  "xpm run test-aarch64-rpi-zero-2w-cmake",
  "xpm run test-aarch64-rpi3b-cmake",
  "xpm run test-2xcortex-m33-cmake",
  "xpm run test-pico2-lcpu-cmake",
  "xpm run test-cortexm-pico2-cmake",
  "xpm run test-cortexm-pico2-rp2350b-psram-cmake"
]
```

## Appendix Section 7 — Pull Request #7: Documentation Suite & Final Merge Reconciliation (Steps 29-30 / Part D)

### Why PR #7 is Needed as a Standalone Pull Request

- 1. Complete Architectural Documentation:**
  - Synchronizes and publishes all Markdown specifications, flowcharts, and compiled PDF runbooks (MICRO-OS-PLUS-SMP-VS-SINGLECORE-ANALYSIS.pdf, SMP-UPSTREAM-INTEGRATION-PLAN.pdf, Implementation-SMP-Integration.pdf, pull-request.pdf, files-modif-by-step.pdf).
- 2. Pristine CI Metadata Restoration:**
  - Reconciles repository histories with upstream origin/xpack-development, restoring .github/workflows/ci.yml, README.md, LICENSE, and Doxygen templates to be 100% byte-identical to upstream.
- 3. Cleansing of Developer Tooling:**
  - Removes transient scripts/smp/ directory and executes finalize.sh to confirm full ecosystem green status before manual PR opening.

### PR #30: Comprehensive Architectural Documentation & Specification Runbooks Suite

#### Why Step 29 (Documentation Synchronization) is Needed:

- **Complete Architectural Documentation:** Synchronizes and compiles all Markdown specifications, flowcharts, and PDF runbooks (MICRO-OS-PLUS-SMP-VS-SINGLECORE-ANALYSIS.pdf, SMP-UPSTREAM-INTEGRATION-PLAN.pdf, Implementation-SMP-Integration.pdf, pull-request.pdf, files-modif-by-step.pdf).

```
# Compilation of synchronized PDF documentation suite
cd docs && ./render-pdfs.sh Implementation-SMP-Integration \
  --title "SMP Integration – Implementation Runbook" \
  --subtitle "Concrete, verifiable execution of the 30-step plan" \
  --footer "micro-os-plus-iii-smp"
```

### PR #31: Final Upstream Master Reconciliation & Repository Cleansing

#### Why Step 30 (Final Merge Reconciliation) is Needed:

- **Pristine CI Metadata Restoration:** Reconciles the accumulated step branch with origin/xpack-development, restoring .github/workflows/ci.yml, README.md, LICENSE, and Doxygen templates to be 100% byte-identical to upstream.
- **Tooling Cleansing:** Removes transient scripts/smp/ directory and executes finalize.sh for final verification.

```
# finalize.sh execution summary
git -C "$K" merge origin/xpack-development --ff-only || true
git -C "$C" merge origin/xpack-development --ff-only || true
git -C "$P" merge origin/xpack-development --ff-only || true

# Restore untouched upstream CI configurations
git checkout origin/xpack-development -- .github/ README.md LICENSE

# Remove development-only helper tooling
rm -rf scripts/smp/
```

```
# Final validation gate
xpm run test-all && xpm run test-smp-cmake
```

## Appendix Section 8 — PR #32: Dedicated Physical Silicon Board Wiring (Raspberry Pi Pico 2)

### Why Step 31 is Needed

#### 1. Hardware Verification on Physical Silicon:

- Provides standalone hardware test configuration and OpenOCD SWD scripts for the Raspberry Pi Pico 2 (RP2350 dual Cortex-M33).

#### 2. Secondary Core Boot Sequence:

- Releases Core 1 through the RP2350 bootrom FIFO (multicore::launch\_core1()), using core 1's handler-mode stack top from port\_core1\_stack\_top(). Core 1 then enters port\_smp\_secondary\_start(), builds a fake thread context, starts its own SysTick/PendSV and reschedules.

## PR #32: Dedicated Physical Silicon Board Wiring (Raspberry Pi Pico 2)

### Why Step 31 (Dedicated Pico 2 Hardware Platform) is Needed:

- **Hardware Verification on Physical Silicon:** Provides standalone hardware test configuration and OpenOCD SWD scripts for physical RP2350 silicon.
- **Secondary Core Boot Sequence:** Releases Core 1 through the RP2350 bootrom FIFO (multicore::launch\_core1() / port\_core1\_stack\_top()) and enables the core-0 SIO FIFO IPI only after the launch has completed.

```
// cortexm/src/rtos/os-core-rp2350.cpp
// Cross-core IPI: wake another core over the SIO inter-core FIFO.
void
port_smp_ipi (unsigned cpu)
{
    if (cpu == port_cpu_id ())
    {
        return;
    }
    if (multicore::fifo_write_ready ())
    {
        multicore::fifo_push (0x4D53u); // "SM"
    }
}

// Enable core 0's cross-core IPI channel. MUST be called only AFTER
// the
// core-1 bootrom launch has completed: while the launch is in
// progress the
// FIFO carries the bootrom handshake words, and a FIFO IRQ handler
// would
// drain them away (stalling the launch forever).
void
port_smp_core0_ipi_enable (void)
{
    while (rp2350::reg (rp2350::SIO_FIFO_ST) & rp2350::FIFO_ST_VLD)
    {
        volatile uint32_t d = rp2350::reg (rp2350::SIO_FIFO_RD);
        (void) d;
    }
}
```

```
rp2350::reg (rp2350::SIO_FIFO_ST) = 0x0C; // write-1-to-clear WOF|
    ROE
NVIC_EnableIRQ (SIO_IRQ_FIFO_IRQn);
}
```

## Appendix — Why the C/D actions cannot be reduced to PR diffs

The C/D PRs *can* be — and partly are — represented as pull requests: PR #24-#32 carry unified diffs for the smaller code files. The remaining operations cannot be reduced to a PR for four concrete reasons:

| Why                                                 | Example                                                                                                                                                                        | A PR diff cannot express it because                                                                                                                                                                          |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>History/ref operations, not content</b>          | migrate-devices.sh<br>(git subtree split + subtree add),<br>release-port.sh<br>(tags)                                                                                          | A PR is a content delta between two branch tips; subtree ingestion creates <b>new commits with preserved authorship</b> and merge structure, and releases create <b>tags</b> — metadata/history, not a diff. |
| <b>Base- and order-dependent programmatic edits</b> | integrate-aarch-harness.sh (asserts the ::test-support anchor, inserts sub-targets only if absent, rewrites iii→iii-core, injects JSON configs),<br>normalize-tests-package.sh | The scripts edit <b>whatever the current file is</b> ; a fixed hunk with fixed context cannot be robust to a different base or order (it fails). The scripts are the order-aware reference.                  |
| <b>Derived / binary artifacts</b>                   | docs/**/* .pdf,<br>render-pdfs.sh                                                                                                                                              | PDFs are rendered binaries; there is no meaningful text patch, and the result depends on the render toolchain.                                                                                               |
| <b>Wholesale / huge trees</b>                       | test/pico2/** ≈ 34k lines, include-m33/,<br>include-rp2350/                                                                                                                    | A new-file diff is <i>possible</i> but would be tens of thousands of lines; the scripts lift them verbatim from the canonical origin/smp, which is equivalent and compact.                                   |

It is not that a PR *fundamentally* cannot hold them — full new-file diffs could be inlined — but then you would still have to handle separately: subtree history, tags, merges/reconciliation (*finalize.sh*), binary PDFs, and the order- and base-sensitive config edits.

Practically, a PR diff is the right tool for **code content**, and an action script is the right tool for **history, metadata, generated artifacts, and order-dependent edits**. That is why the playbook splits the work into: applicable **PR diffs** for Parts A/B (and the smaller C/D files) plus **action scripts** for the rest — and the two paths were verified to produce a **byte-identical tree (0 differences)**.